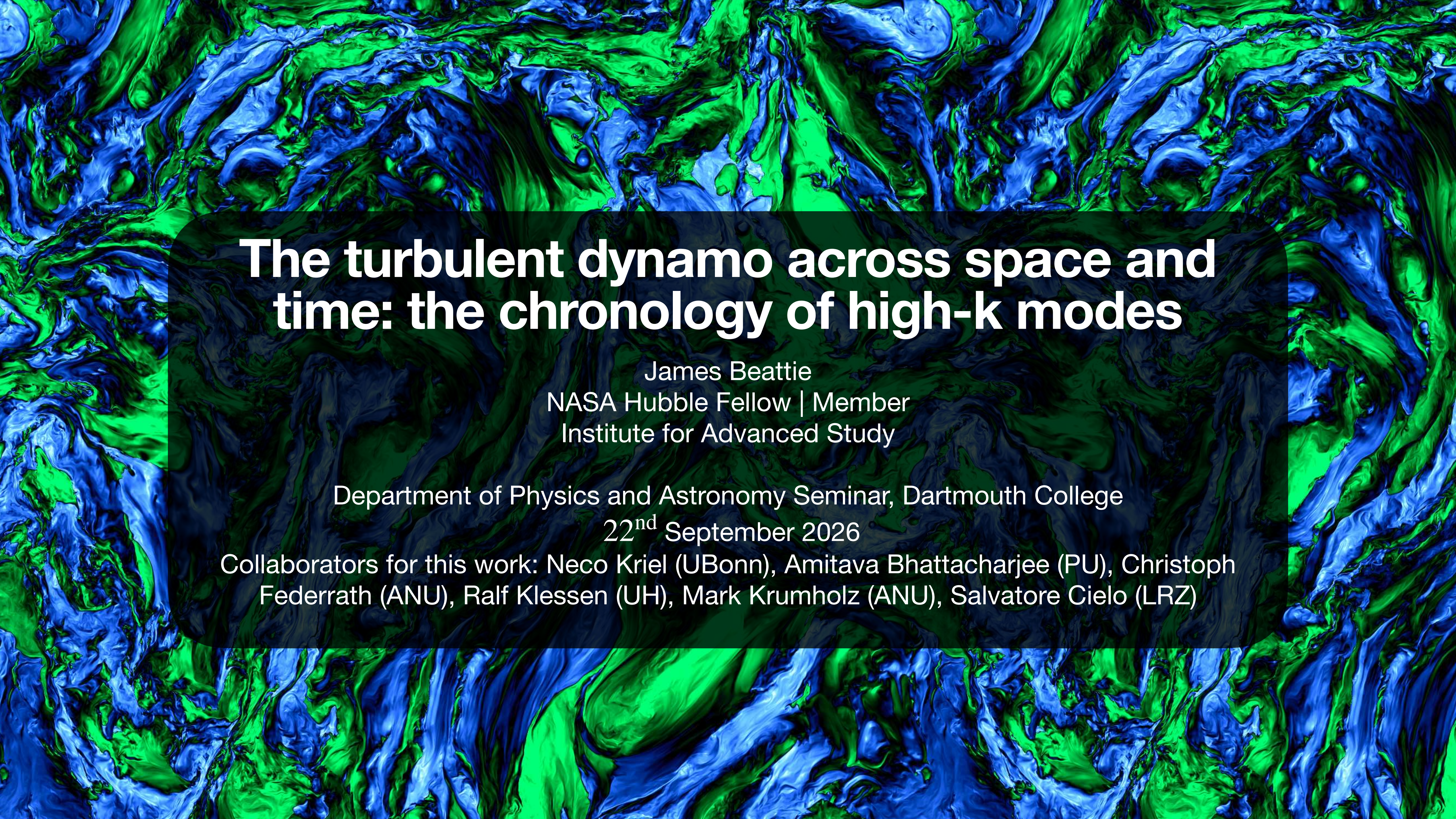




The turbulent dynamo across space and time: the chronology of high- k modes

James Beattie
NASA Hubble Fellow | Member
Institute for Advanced Study

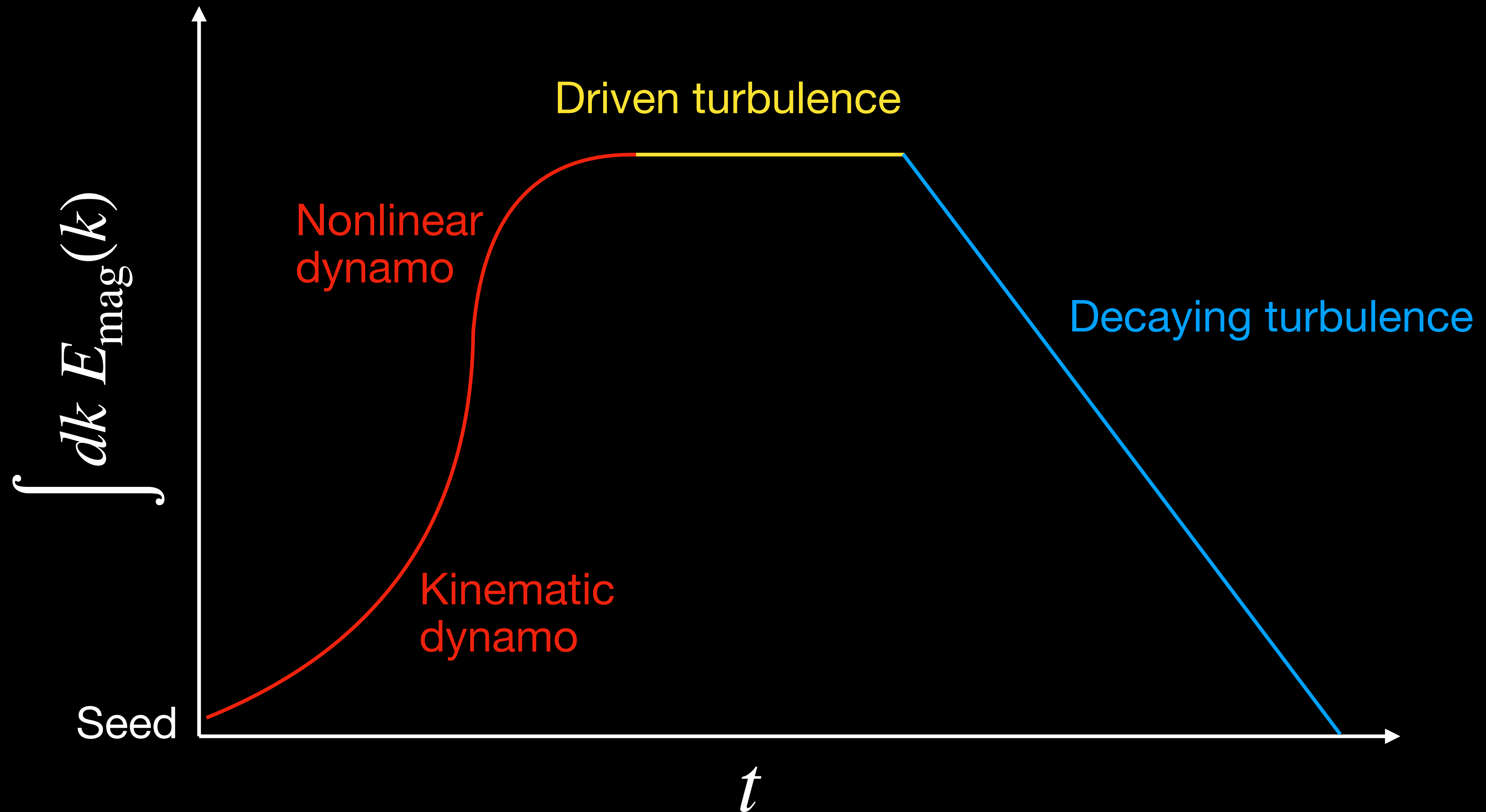
Department of Physics and Astronomy Seminar, Dartmouth College
22nd September 2026

Collaborators for this work: Neco Kriel (UBonn), Amitava Bhattacharjee (PU), Christoph Federrath (ANU), Ralf Klessen (UH), Mark Krumholz (ANU), Salvatore Cielo (LRZ)

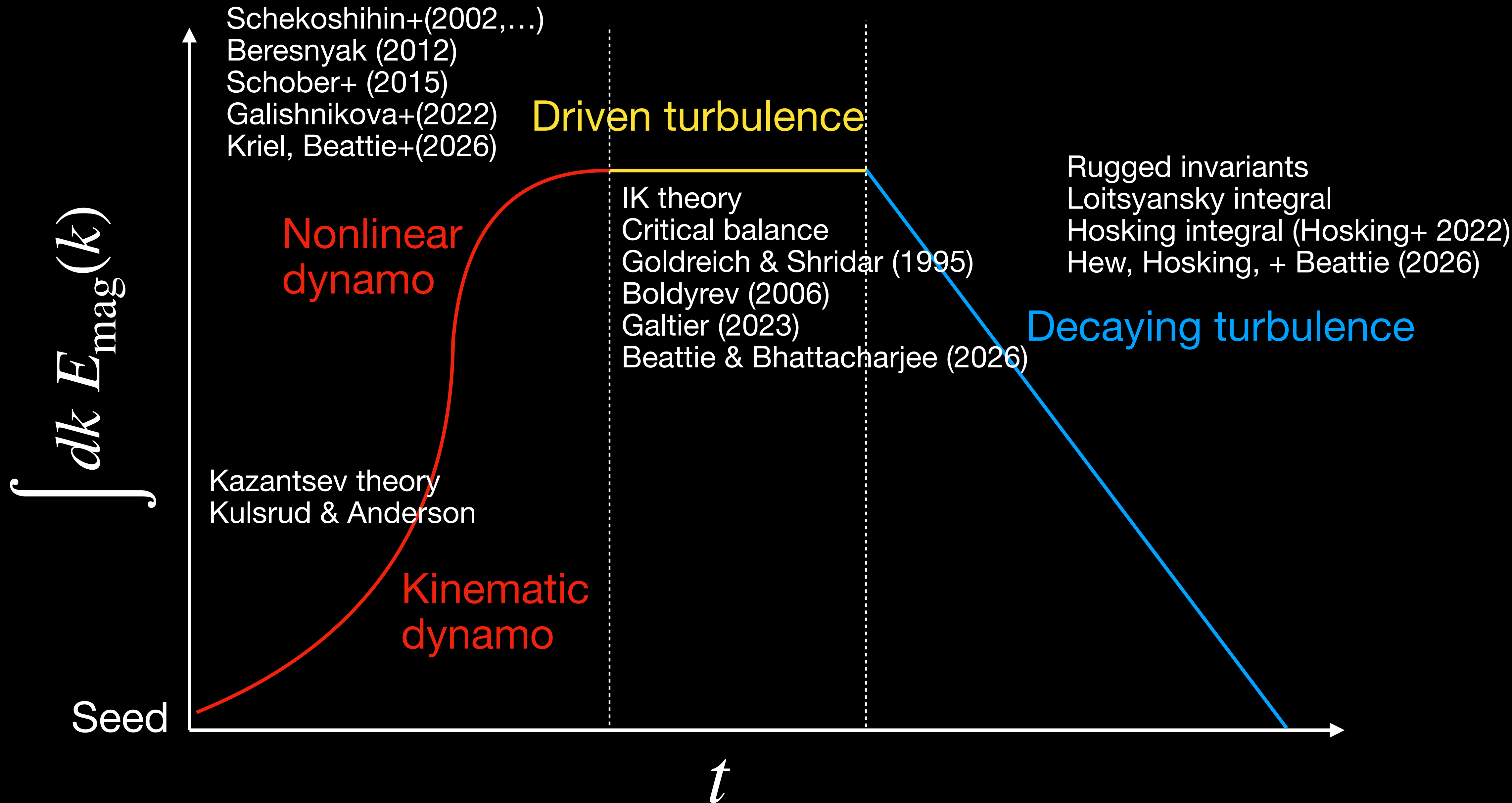
Goals for this afternoon seminar

1. Get folks excited about the universality of turbulent dynamos
2. Grow a small-scale (kinematic + nonlinear) field in the Dartmouth College seminar.
3. Venture into the dynamo saturation with $10,080^3$ domain that can resolve inertial ranges, presumably in the MHD asymptotic state, focussing on the role of alignment.

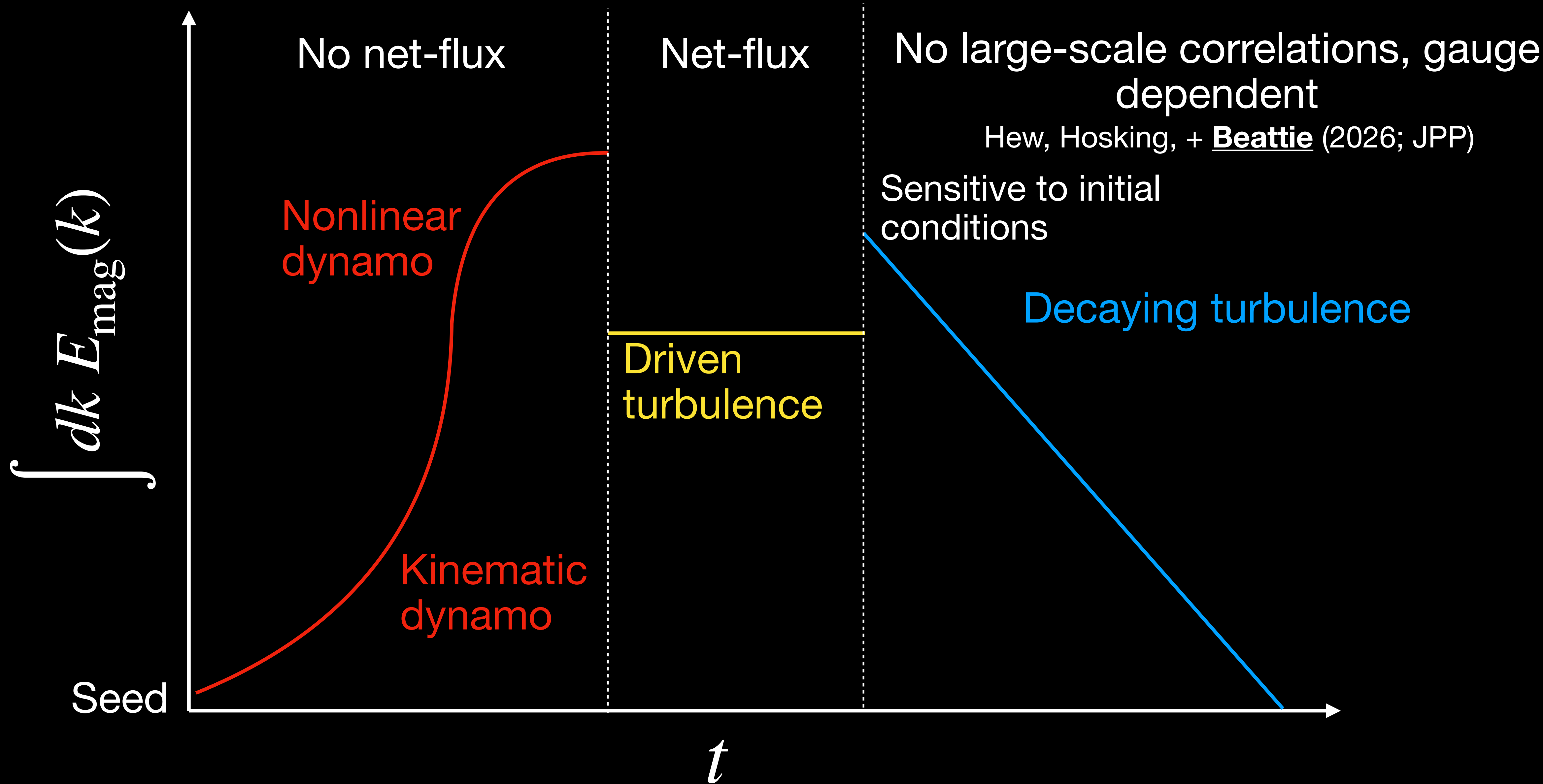
The lifecycle of a high-k mode spans across a patchwork quilt of theories



The lifecycle of a high-k mode spans across a patchwork quilt of theories

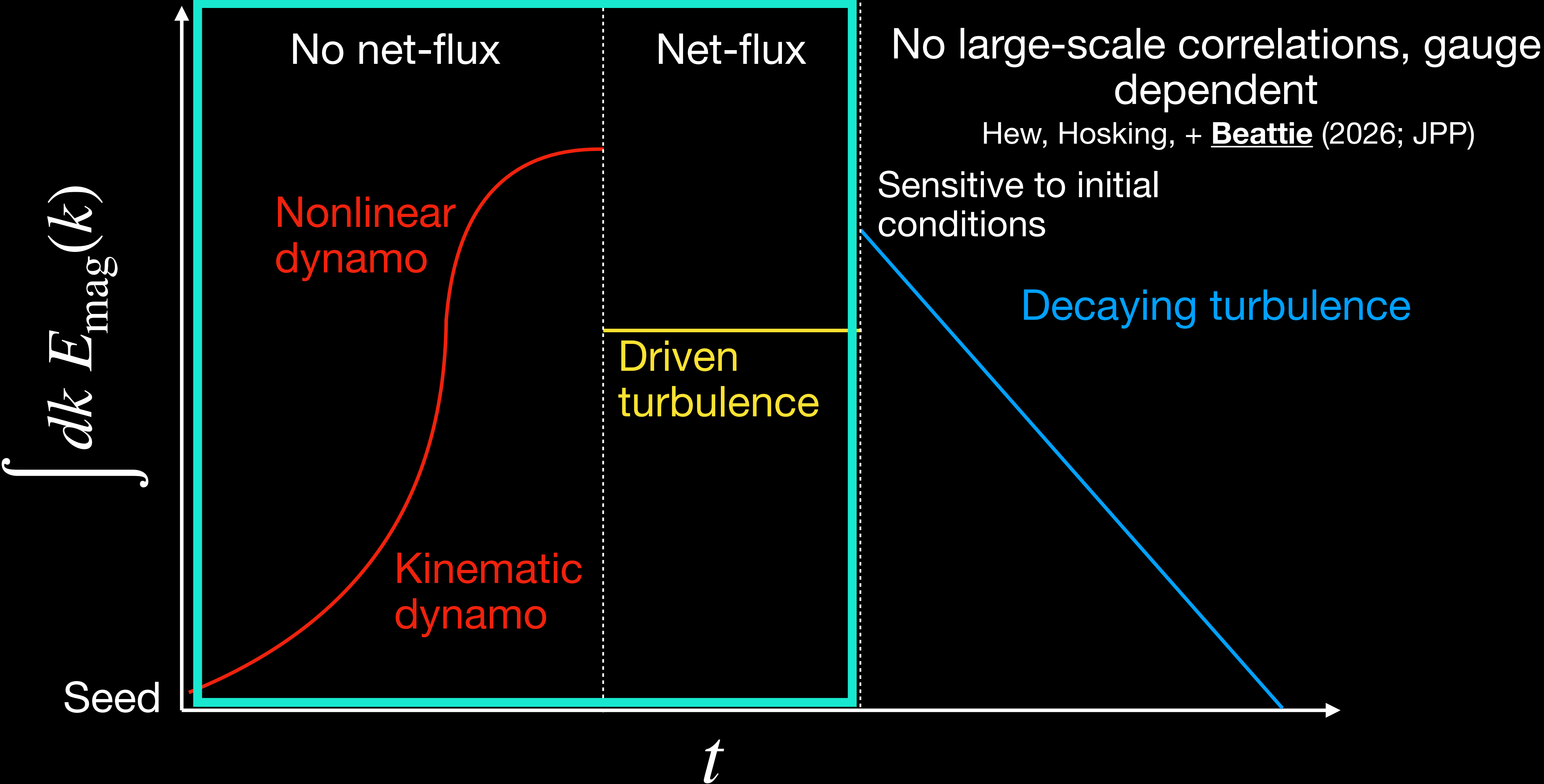


The lifecycle of a high-k mode spans across a patchwork quilt of theories



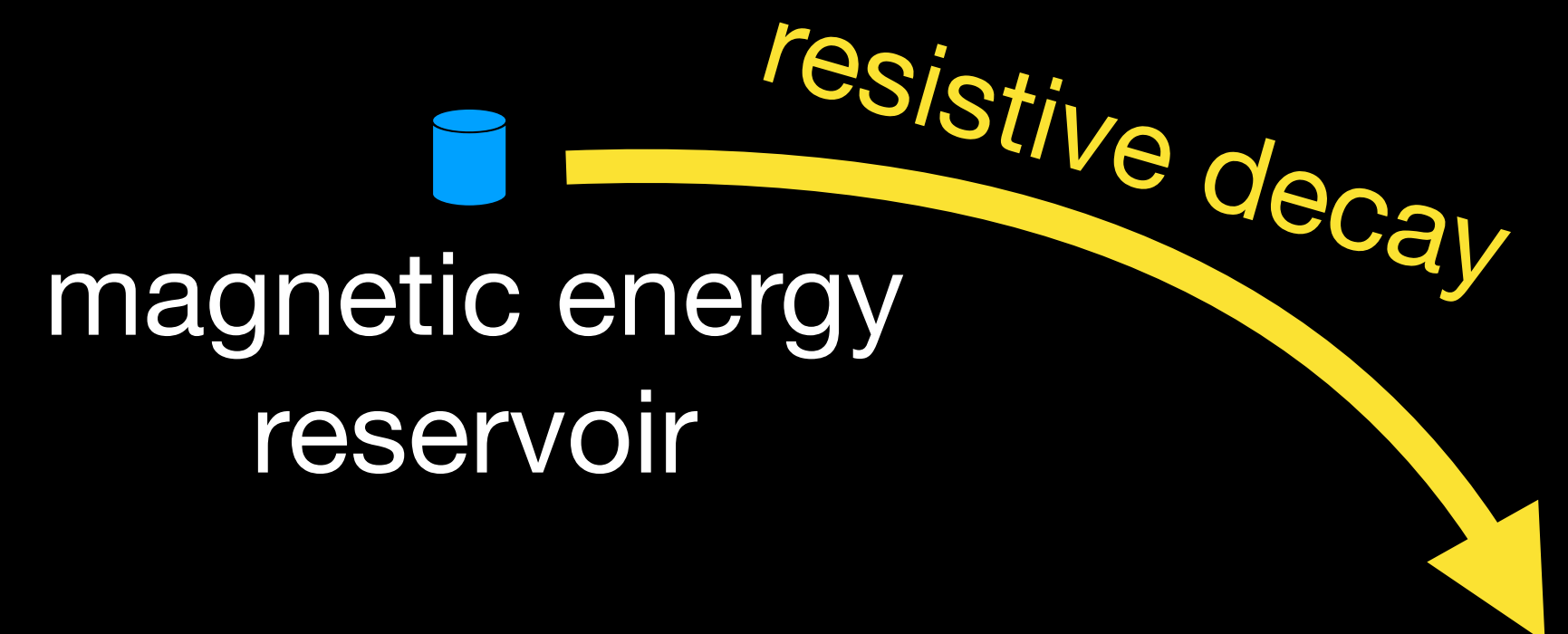
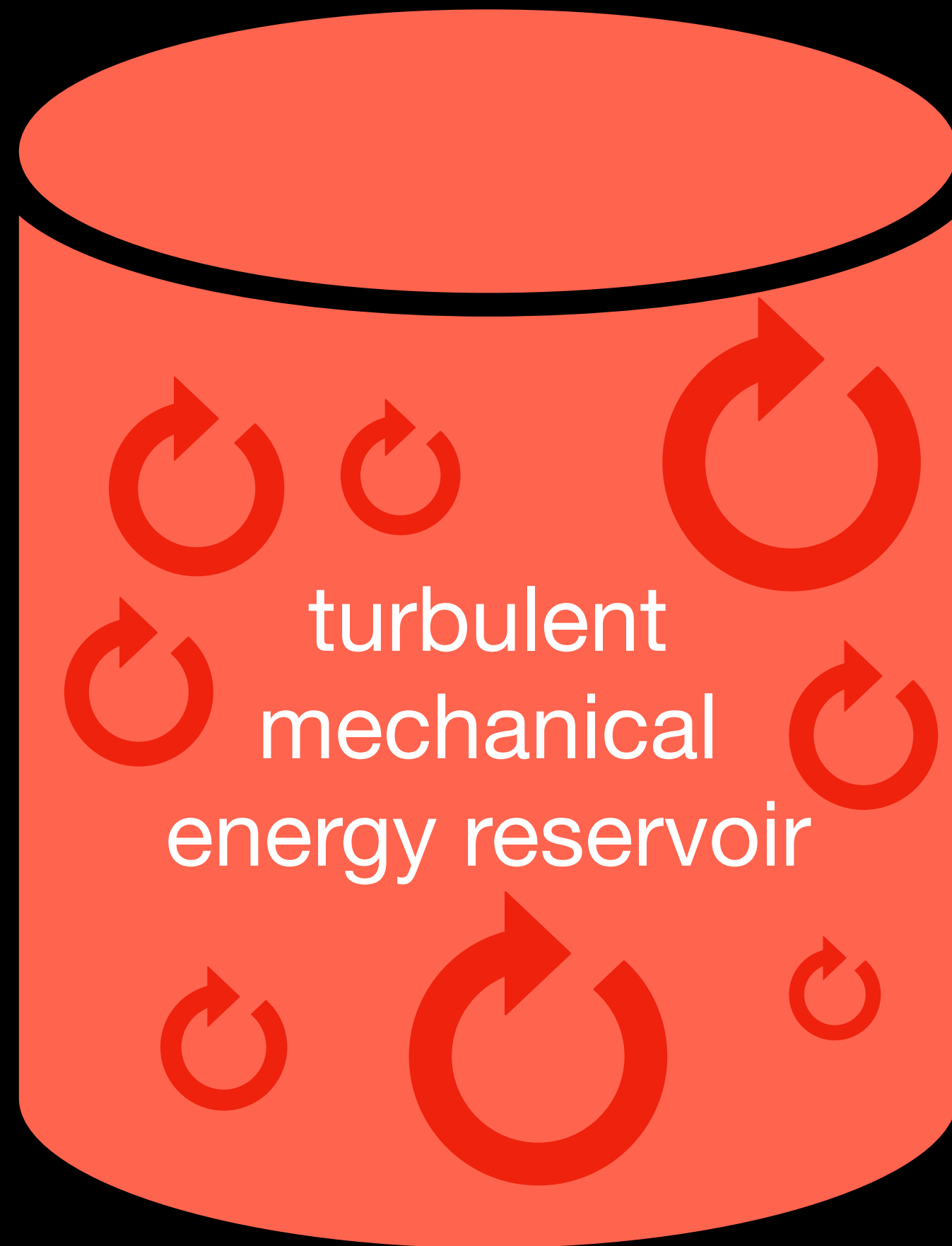
The lifecycle of a high-k mode spans across a patchwork quilt of theories

This afternoon we're here



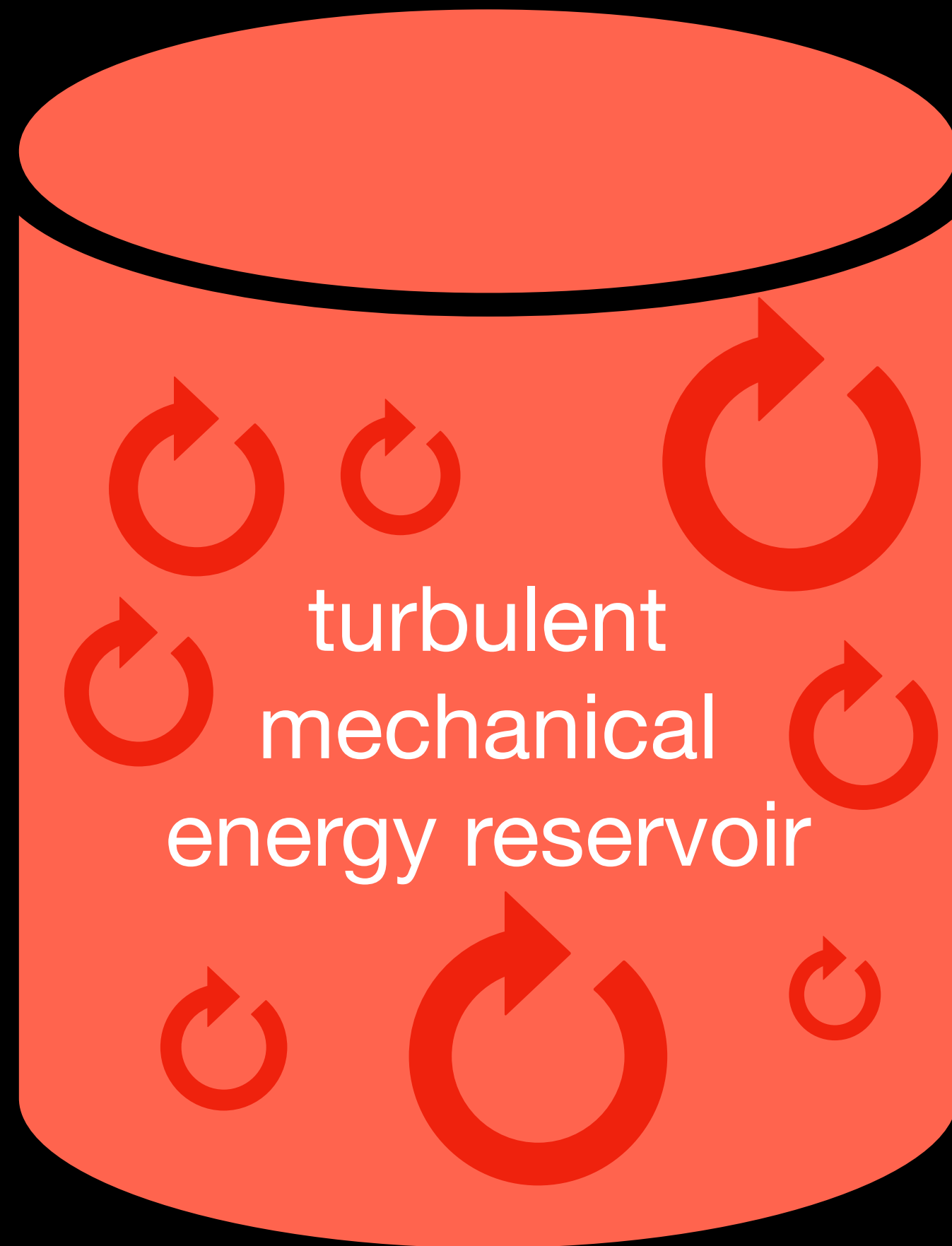
What is a turbulence, magnetic dynamo?

Starting with a seed magnetic field



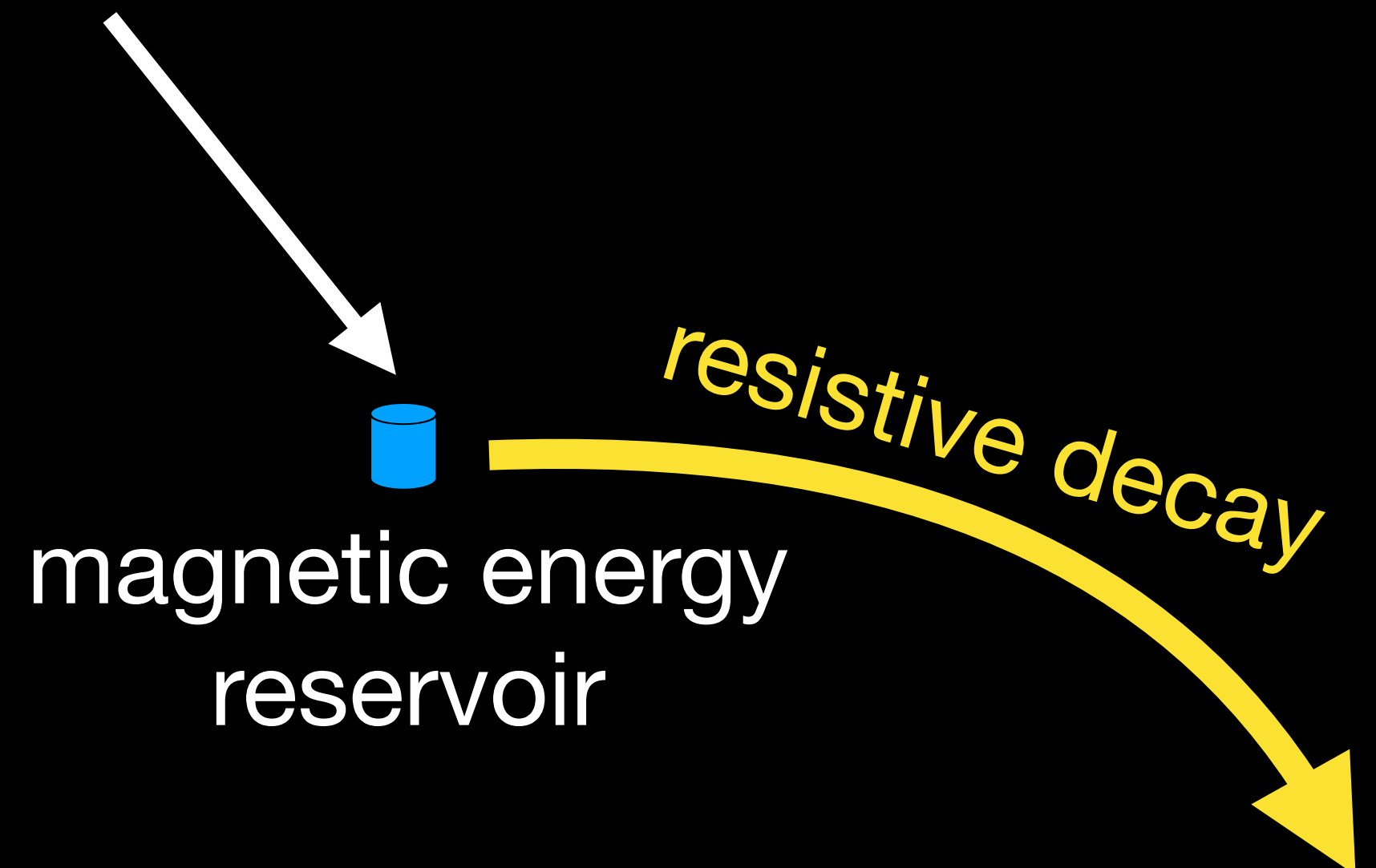
What is a turbulence, magnetic dynamo?

Starting with a seed magnetic field



Weibel instability, (Zhou + 2023,
Biermann battery, 2024, etc.)

...
come to IAS and discuss ;)

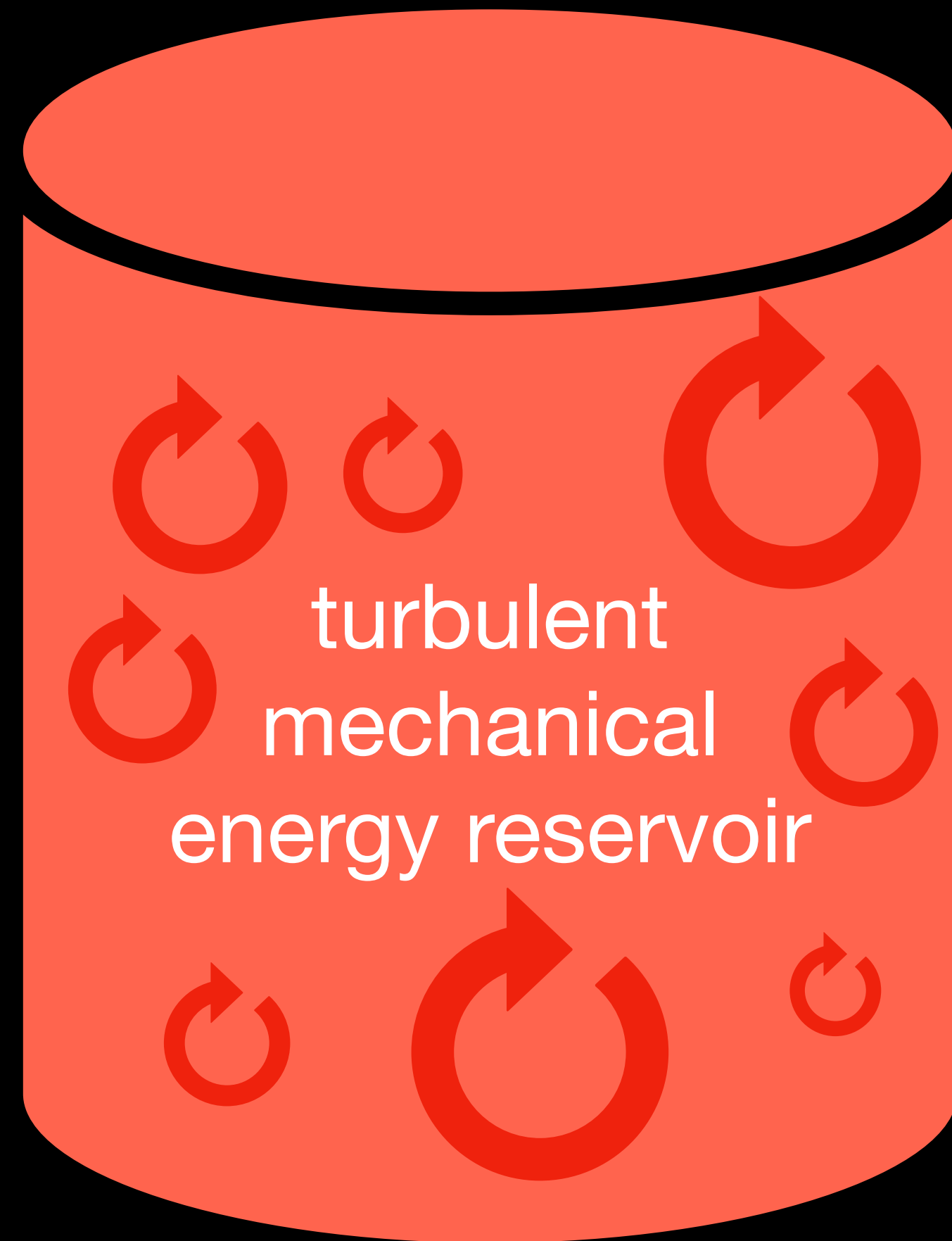


What is a turbulence, magnetic dynamo?

Kinematic growth

$$Rm \sim \frac{U_0 L}{\eta} > Rm_{\text{crit}}$$

can't be too resistive



energy flux in

$$B_i B_j \partial_j u_i$$

$$\eta B_i \partial_j \partial_j B_i$$

resistive decay

magnetic energy
reservoir

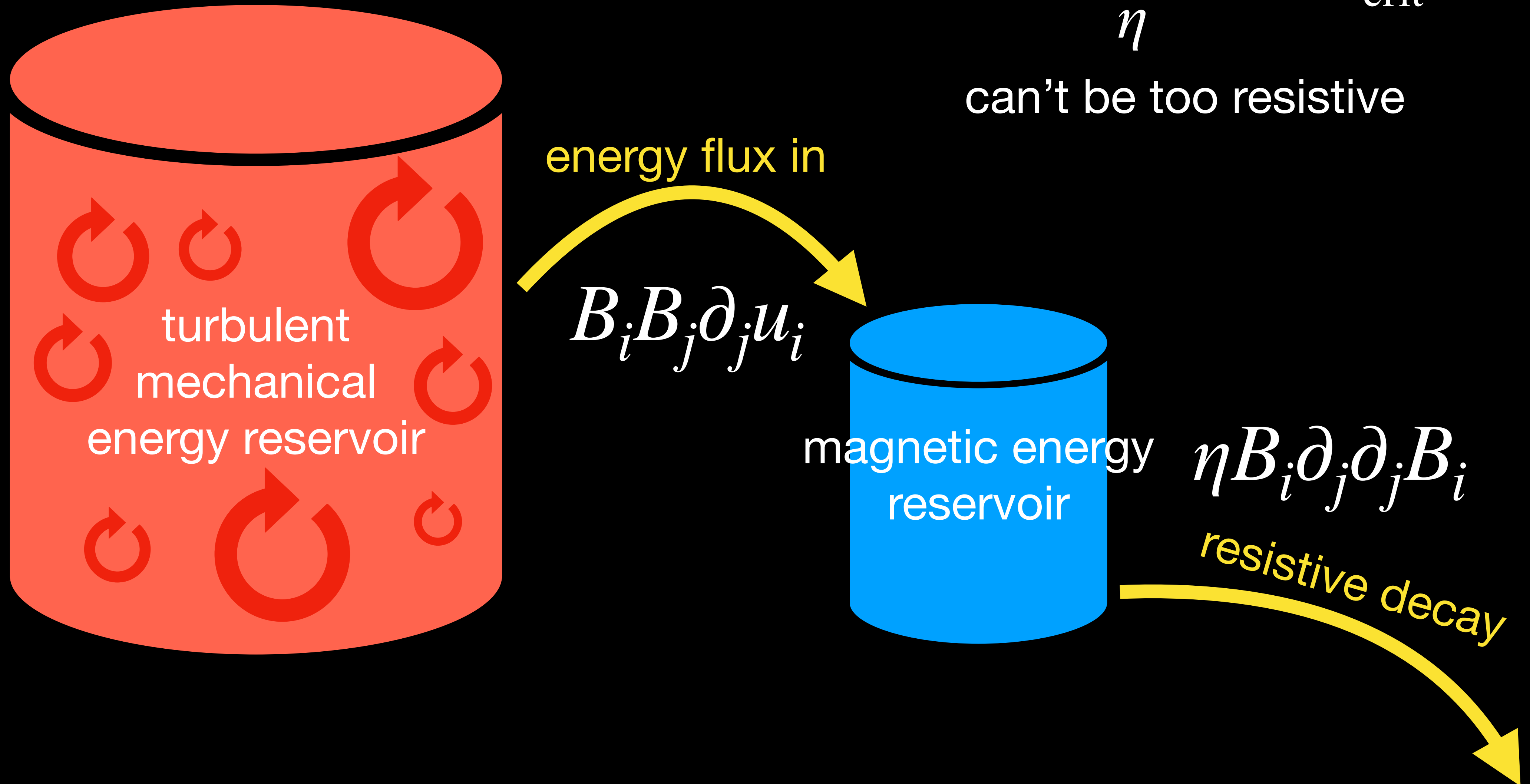


What is a turbulence, magnetic dynamo?

Kinematic growth

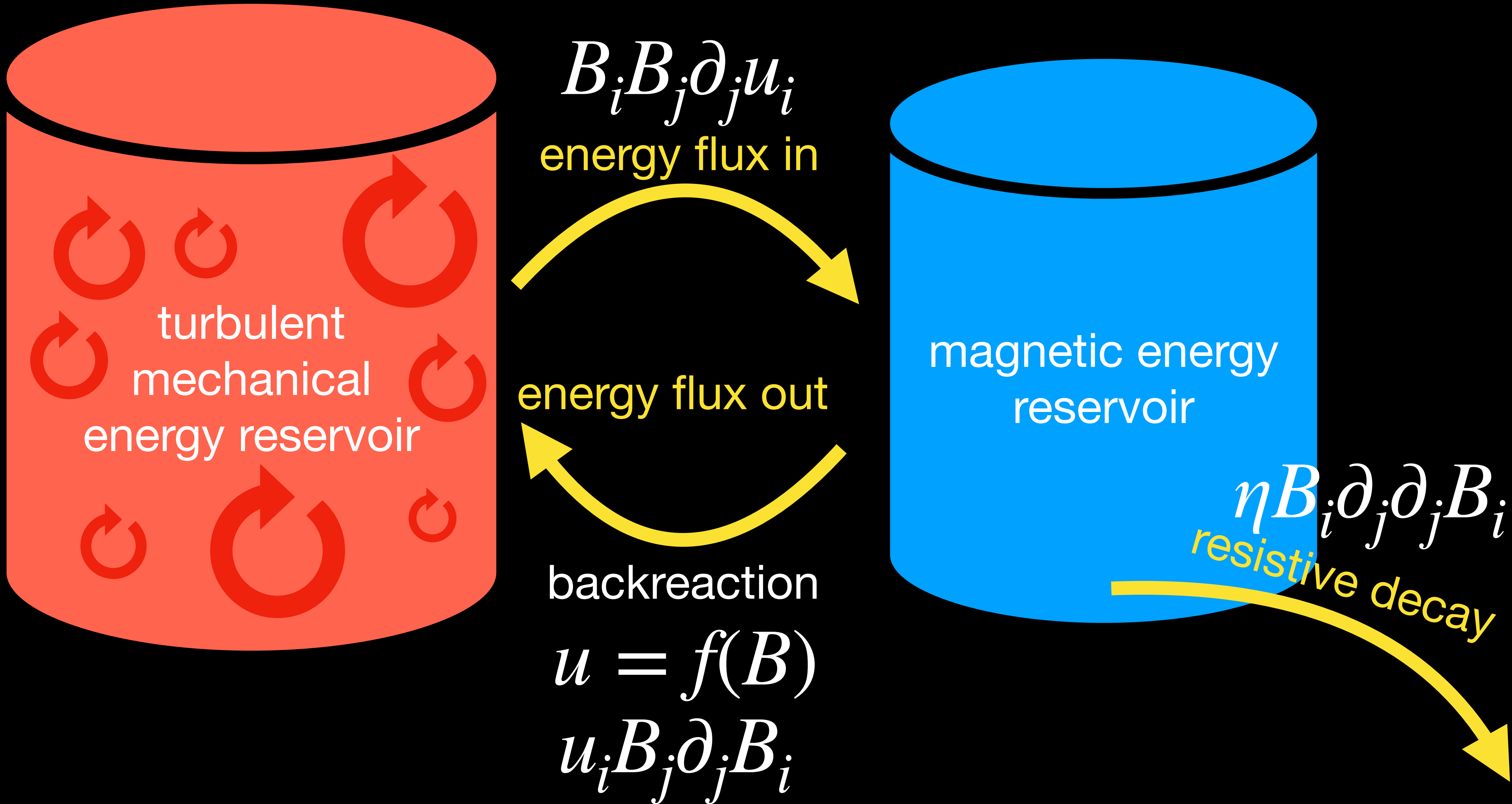
$$Rm \sim \frac{U_0 L}{\eta} > Rm_{\text{crit}}$$

can't be too resistive



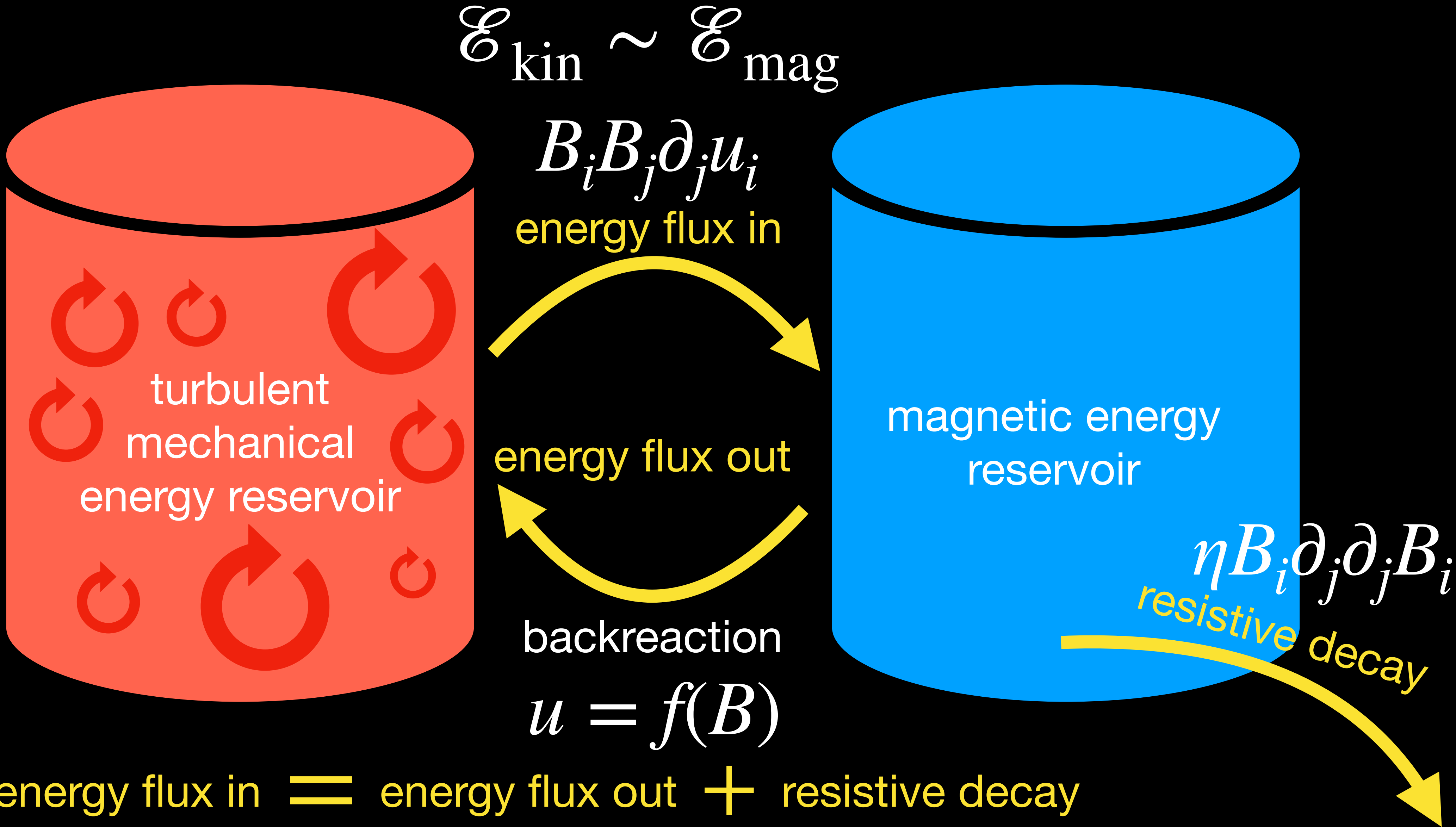
What is a turbulence, magnetic dynamo?

Nonlinearities and backreaction



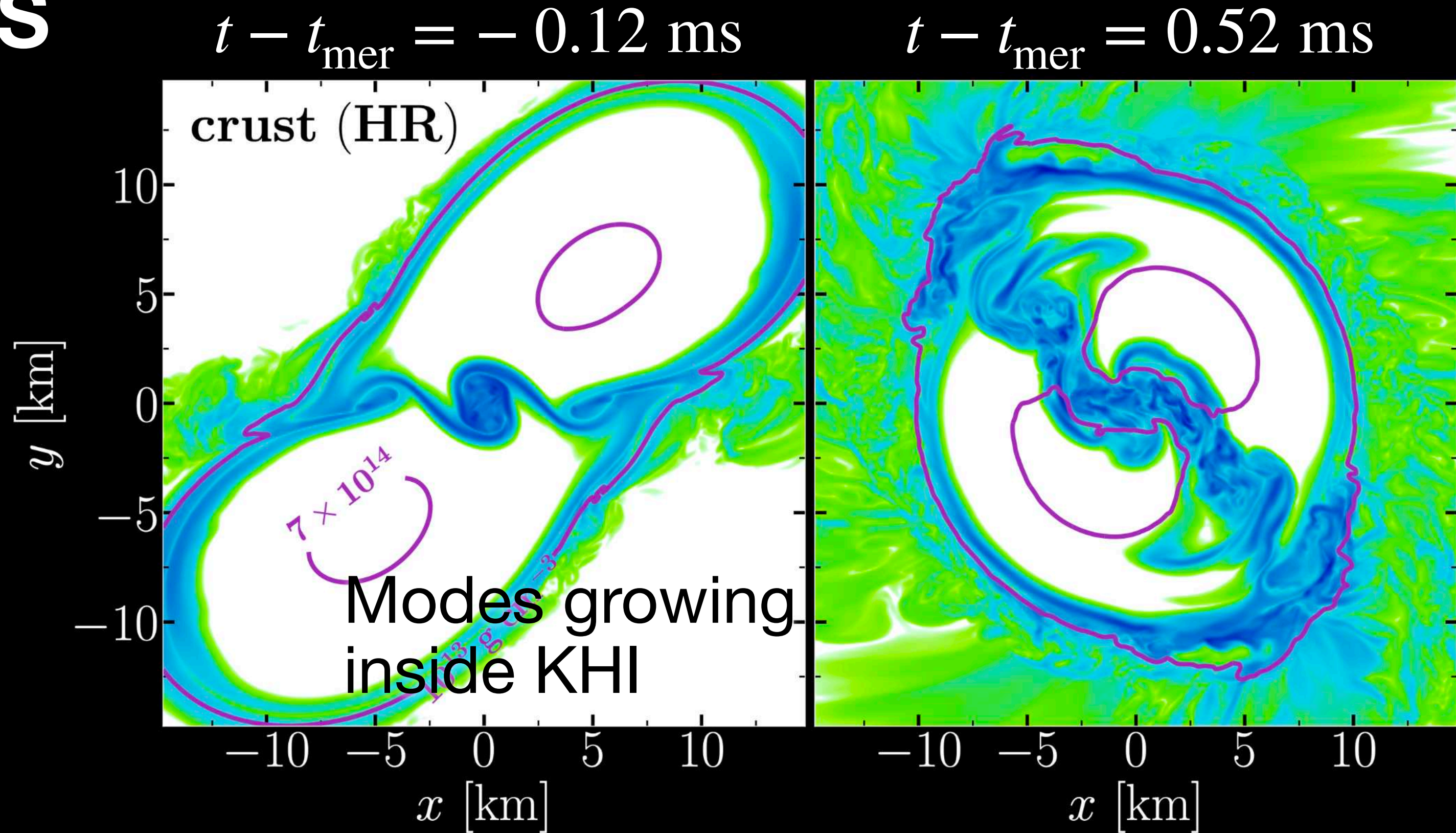
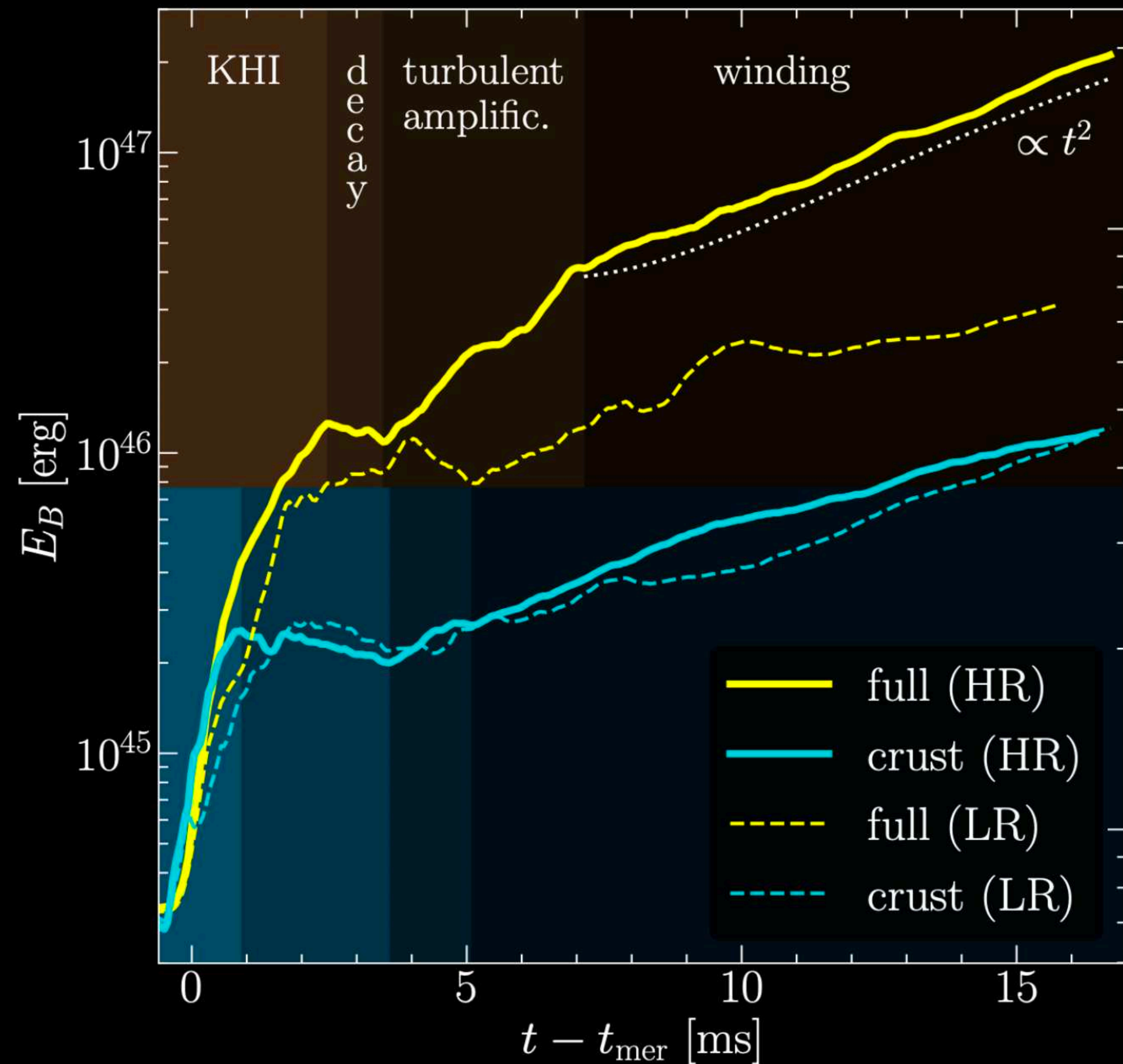
What is a turbulence, magnetic dynamo?

Saturation



Examples of turbulent dynamos

KHI instabilities in merging NS



exponential growth $\longrightarrow$ pause
exponential growth $\longleftarrow$ LSD

Chabanov+ (2023) *ILES*
FIL

Examples of turbulent dynamos

Milky Way-type galaxies in cosmological sims

exponential growth



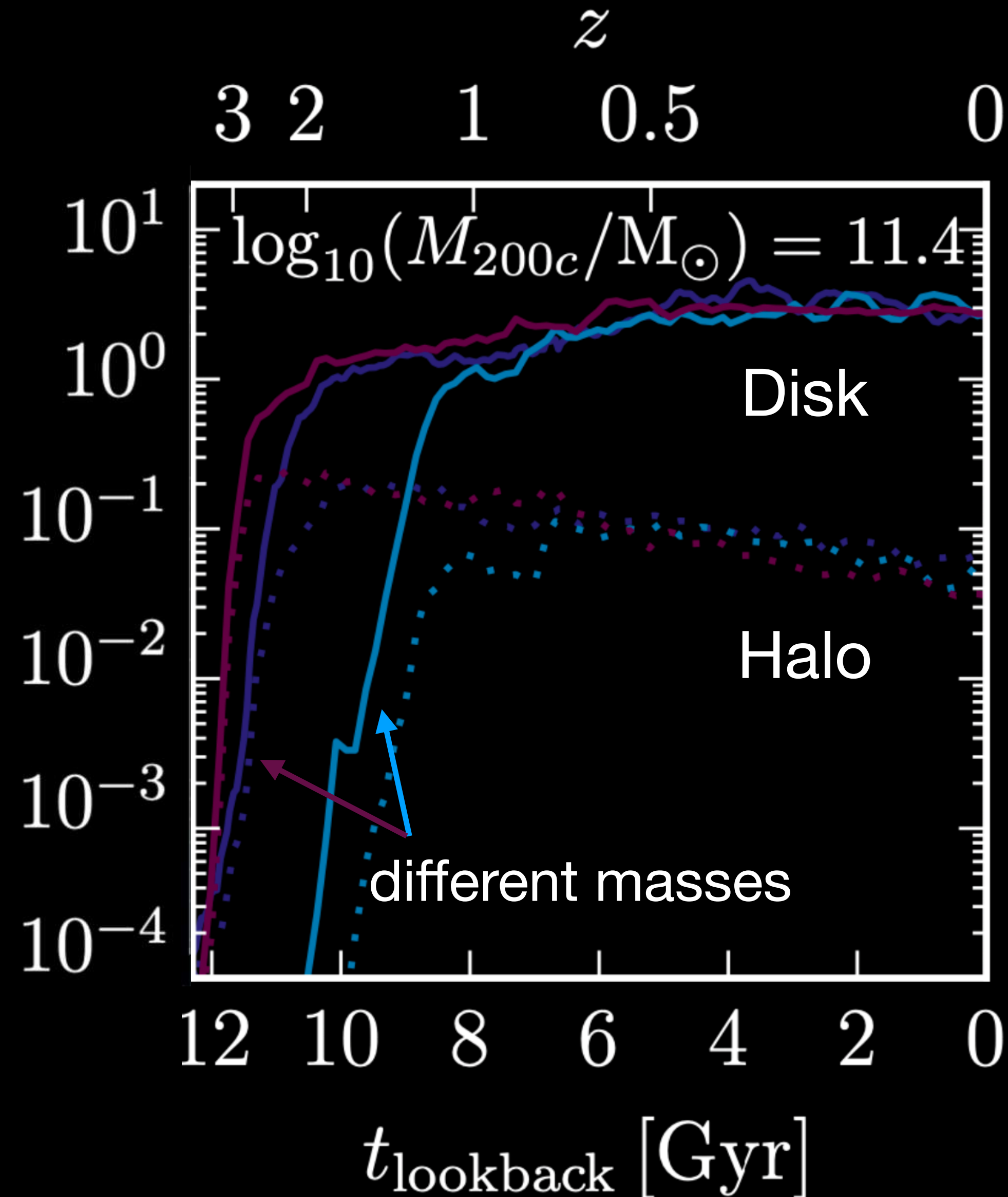
linear growth



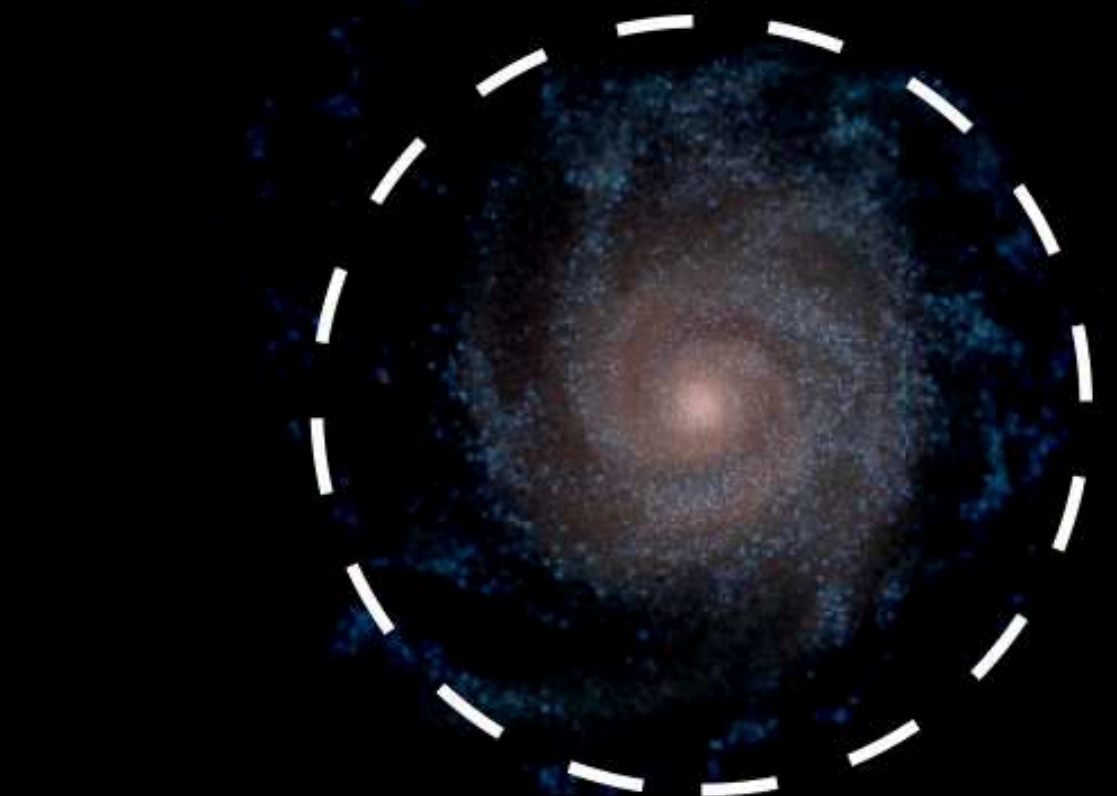
saturation

$$\mathcal{E}_{\text{kin}} \sim \mathcal{E}_{\text{mag}}$$

B [μG]



$$\log_{10}(M_{200c}/M_{\odot}) = 11.4$$



5kpc

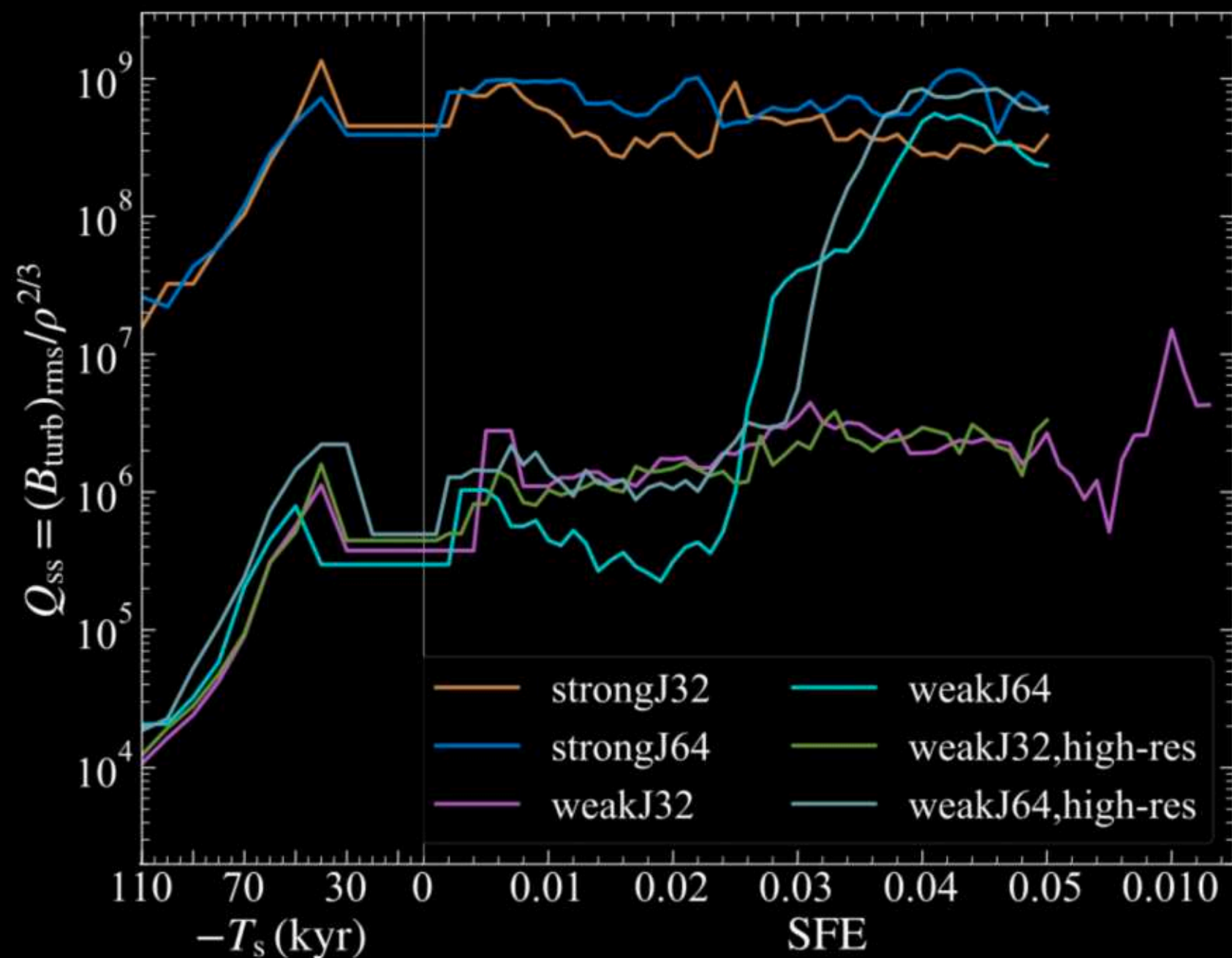


Pakmor+ (2024) *AREPO*

Examples of turbulent dynamos

There are many, across all scales (all MHD)

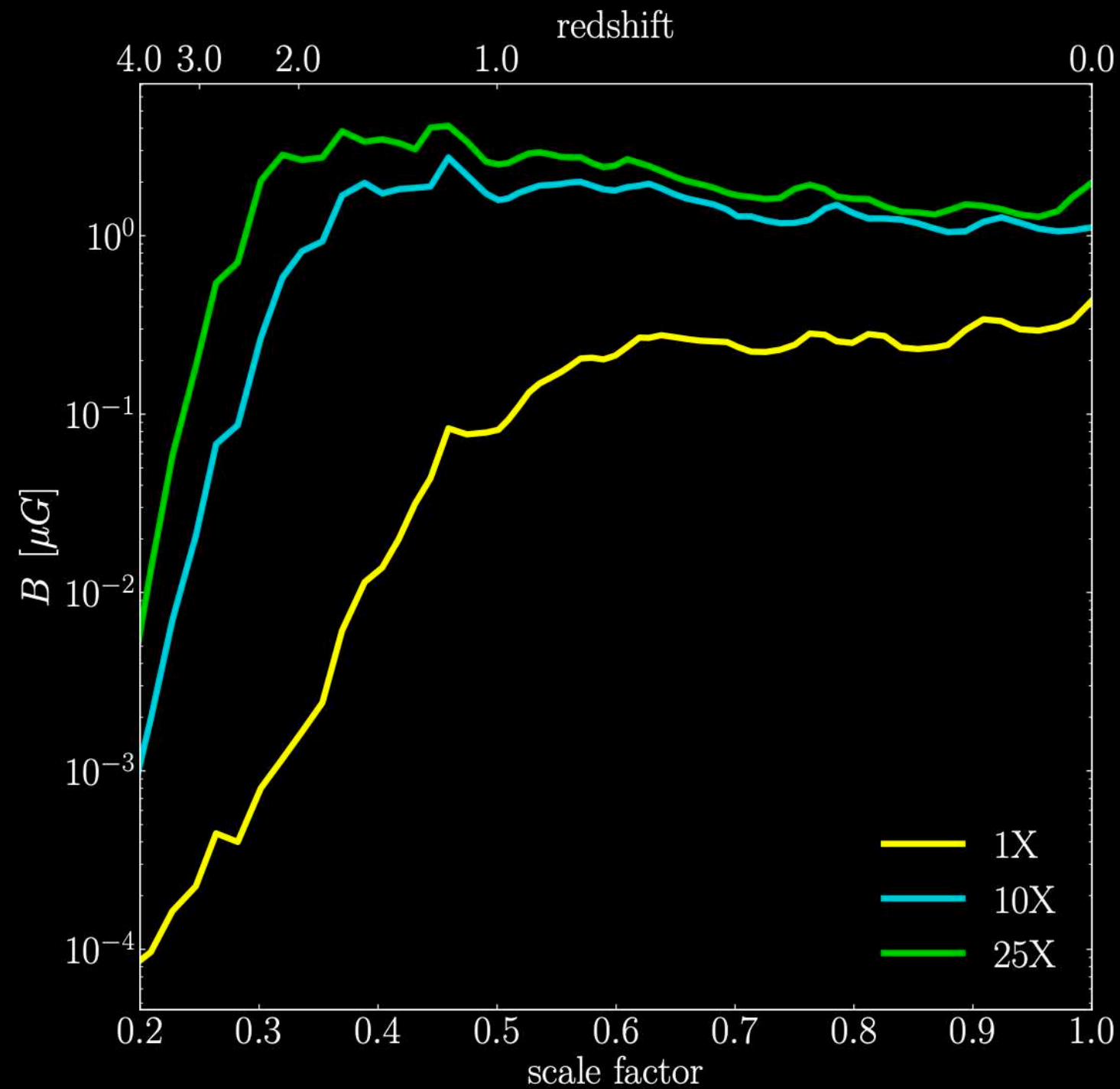
Sharda+2021



Molecular clouds in first generation stars

FLASH

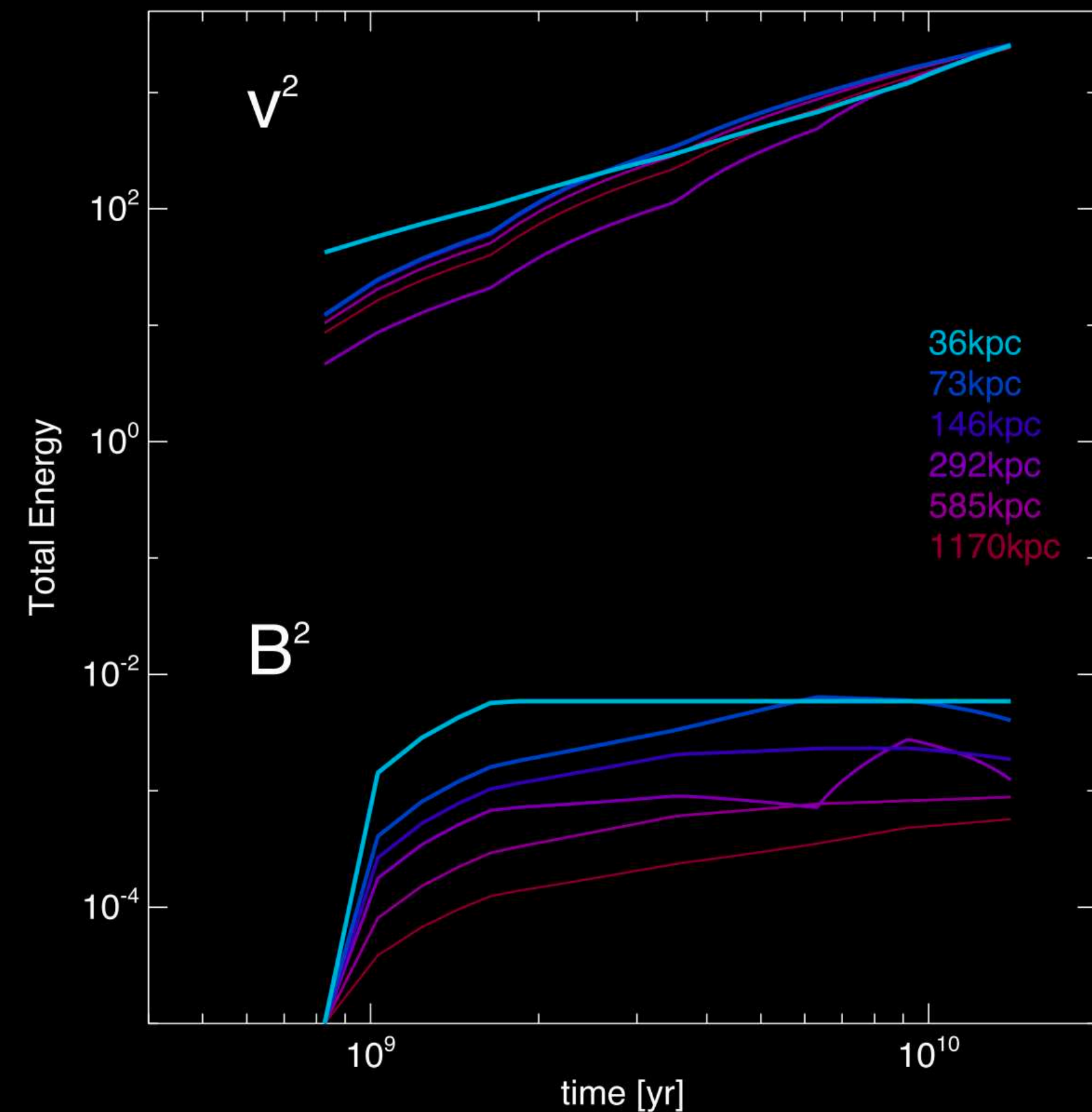
Steinwandel+2021



Intracluster medium

RAMSES

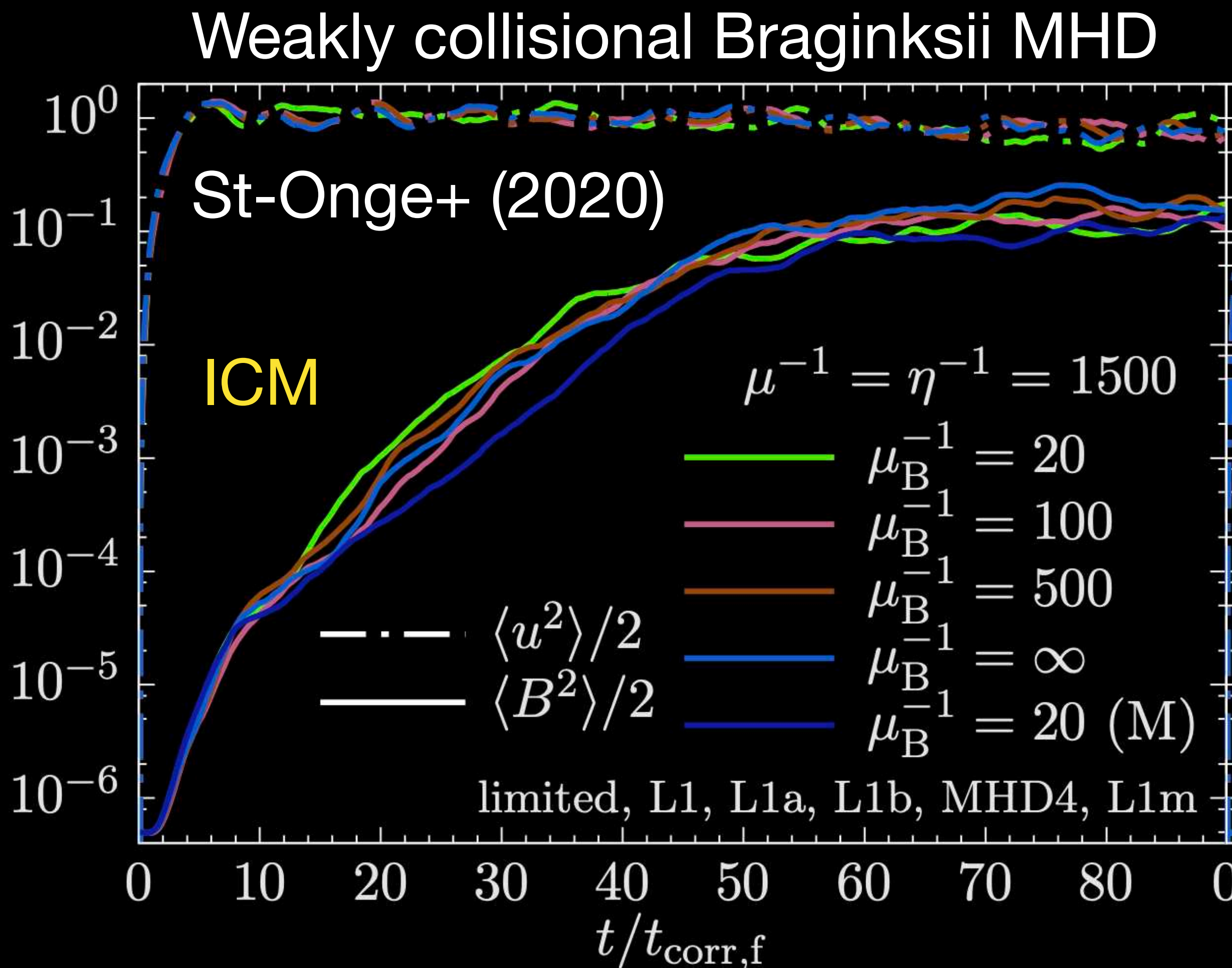
Vazza+2014



Cosmic filaments

ENZO

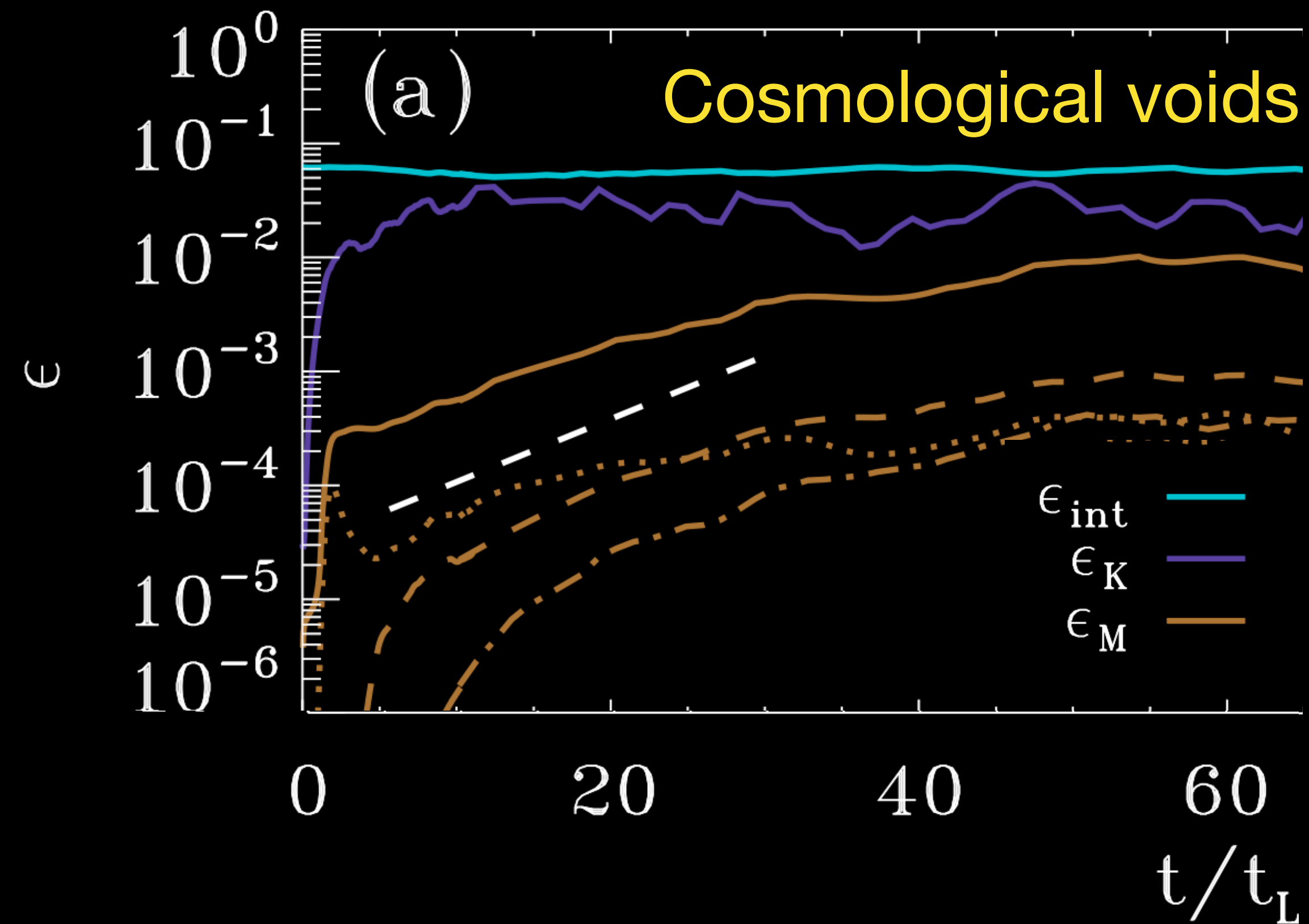
Examples of turbulence dynamos and plasma regimes



(added anisotropic viscous
Braginskii stress term into MHD)

$\nabla \cdot (\hat{\mathbf{b}} \otimes \hat{\mathbf{b}} (\hat{\mathbf{b}} \otimes \hat{\mathbf{b}} : \nabla \mathbf{v}))$ *Snoopy*

Collisionless plasma
magnetogenesis coupled to dynamo



Sironi+2023
(PIC: pair plasma)
TRISTAN-MP

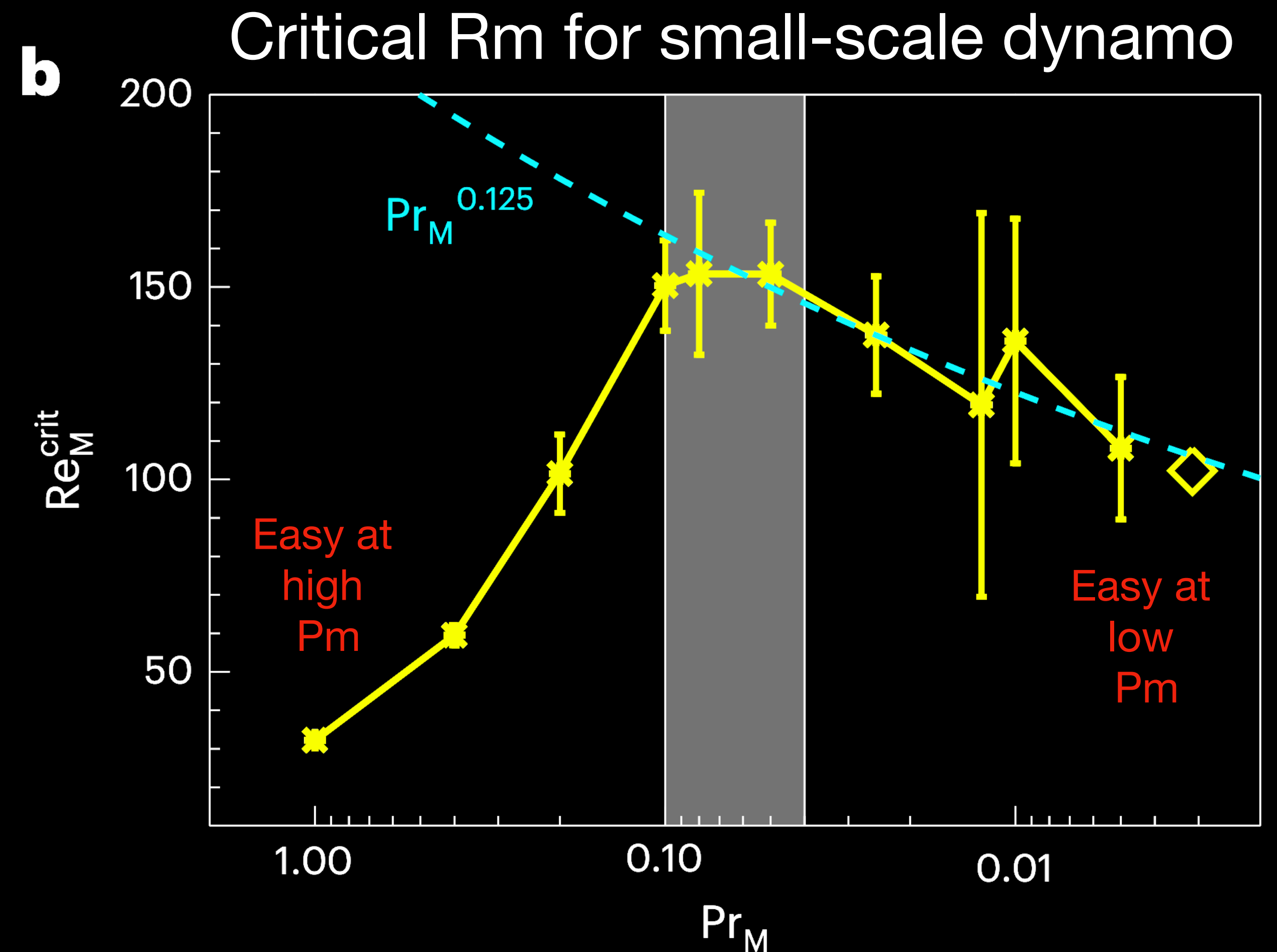
Turbulence dynamos

Ubiquitous across multiple phenomena / regimes

Extremely simple ingredients

1. $\nabla \otimes \mathbf{u}$
(dynamo is $B_j B_i \partial_i u_j$)

2.
$$\text{Rm} = \frac{U_0 L}{\eta} \gtrsim 100$$



Simulations in this talk: 100s of no net-flux boxes

- 400 or so **highly-modified version of finite volume code *FLASH***, second-order in space approximate Riemann solver with framework outlined in [Bouchut+ \(2010\)](#), tested in *FLASH* in [Waagen+ \(2011\)](#).
- Compressible non-helical, visco/resistive MHD turbulence driven with finite correlation time OU process on $L/2$.
- No net magnetic flux. Pure turbulent magnetic field

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$d_t(\rho \mathbf{u}) + \nabla \cdot \mathbb{F} = \frac{1}{\text{Re}} \nabla \cdot \sigma_{\text{viscous}} + \rho \mathbf{f}$$

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{u} \times \mathbf{B}) + \frac{1}{\text{Rm}} \nabla^2 \mathbf{B}$$

$$\nabla \cdot \mathbf{B} = 0$$

DB: EXTREME_Turb_hdf5_plt_cnt_0100
Cycle: 118366 Time: 1.25

Pseudocolor
Var: mag - Speed
Max: 62.41
Min: 0.2508

Volume
Var: dens
Max: 319.1
Min: 0.002483

Volume
Var: dens
Max: 319.1
Min: 0.0009338

mag. field
mass density

Visualisation: Salvatore Cielo (VisIt)

Simulation

• 400 or so highly space approxin (2010), tested in

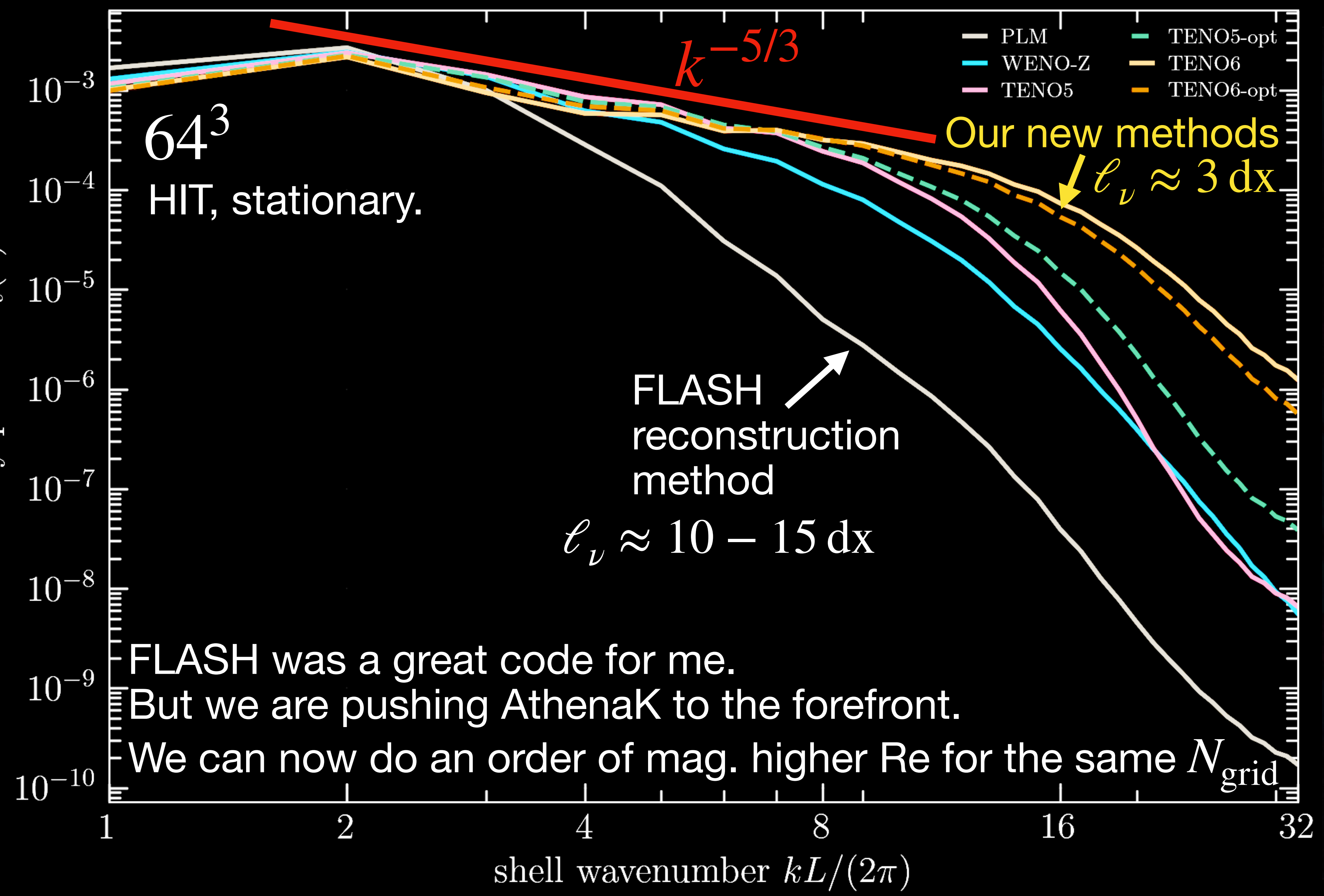
- Compressible r correlation time
- No net magneti

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\partial_t (\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p$$

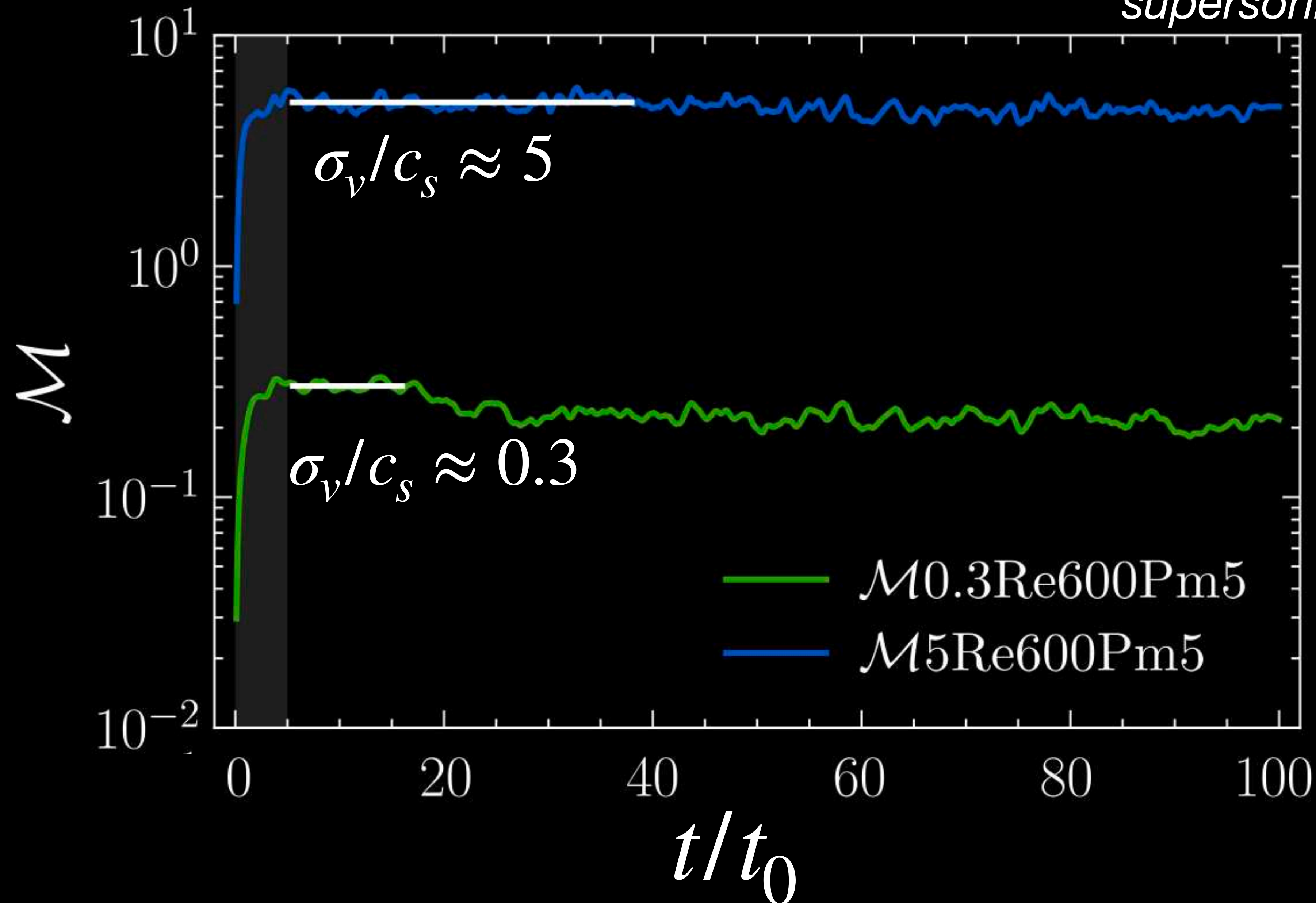
$$\partial_t \mathbf{B} = \nabla \times \mathbf{E}$$

$$\nabla \cdot \mathbf{B} = 0$$



Simulations in this talk

Kriel, Beattie+ (2025). *Fundamental scales II: the kinematic stage of the supersonic dynamo*



$$\text{Pm} = \frac{\nu}{\eta} \propto \frac{T^4}{n_e}$$

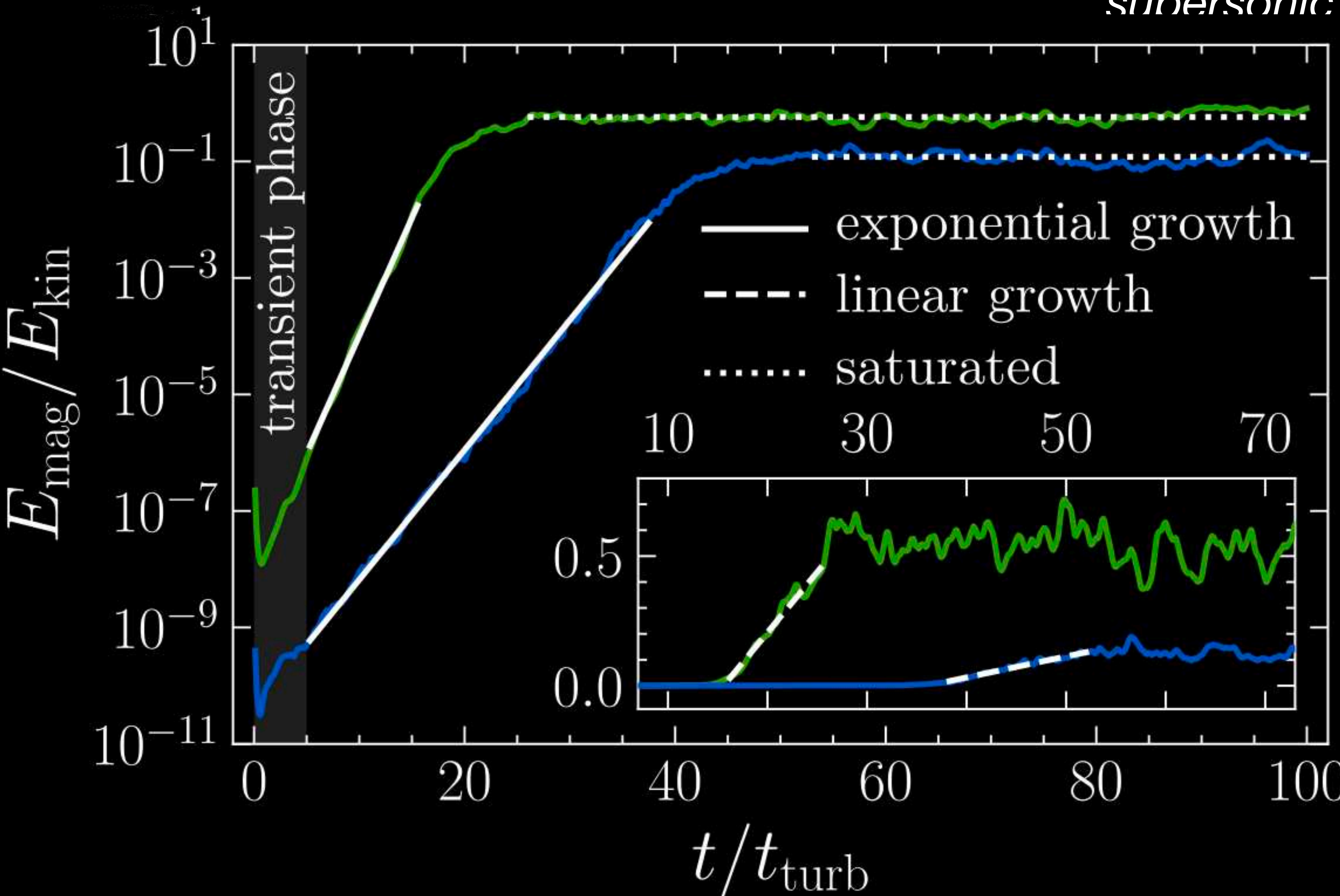
$$\text{Rm} = \frac{U_0 L}{\eta}$$

$$\text{Re} = \frac{U_0 L}{\nu}$$

$$\mathcal{M} = \frac{\langle u^2 \rangle^{1/2}}{\langle c_s \rangle}$$

Simulations in this talk

Kriel, Beattie+ (2025). *Fundamental scales II: the kinematic stage of the supersonic dynamo*



$$\text{Pr}_m = \frac{\nu}{\eta} \propto \frac{T^4}{n_e}$$

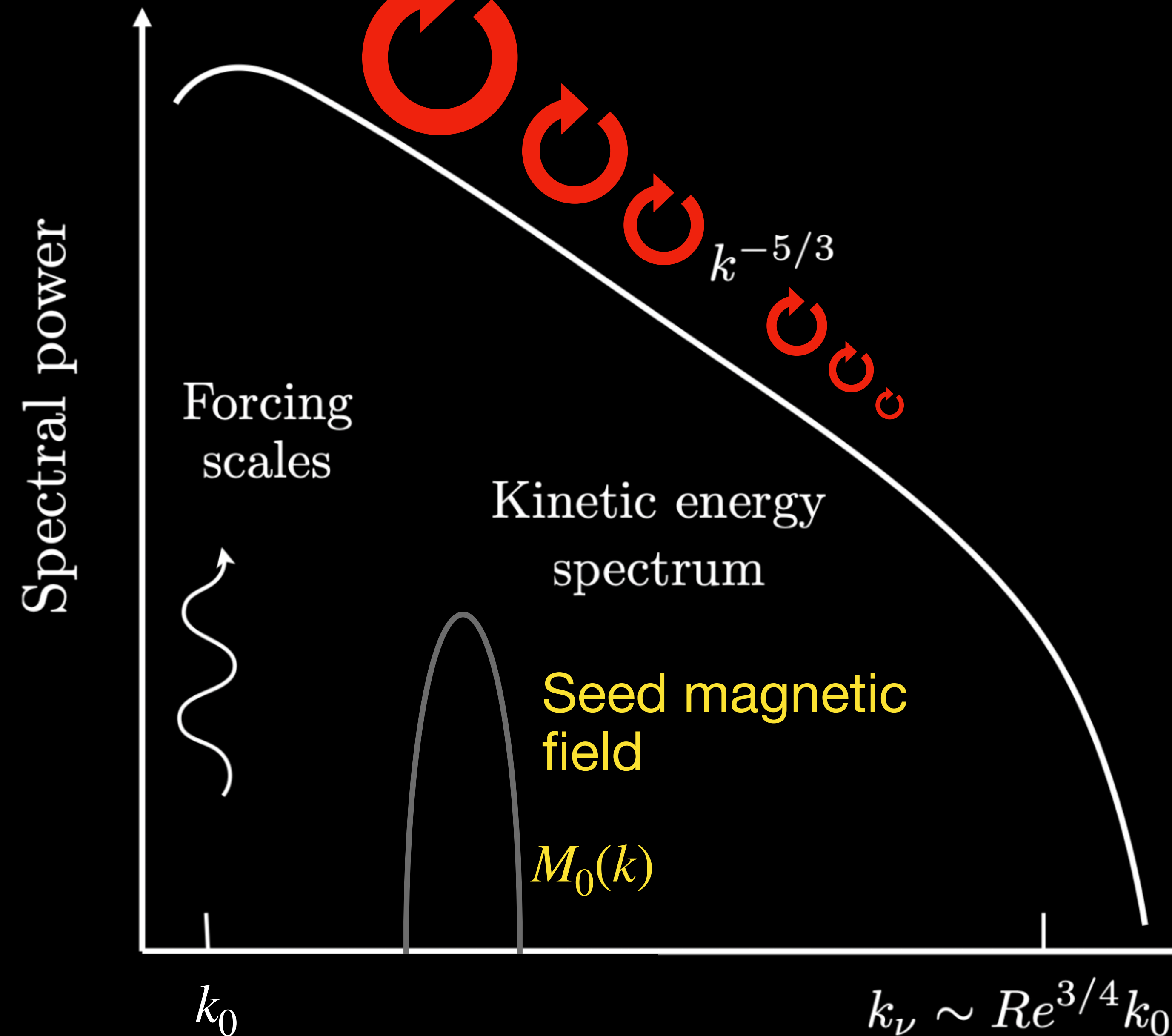
$$\text{Rm} = \frac{U_0 L}{\eta}$$

$$\text{Re} = \frac{U_0 L}{\nu}$$

$$\mathcal{M} = \frac{\langle u^2 \rangle^{1/2}}{\langle c_s \rangle}$$

Turbulence dynamo

Modified from
Rincon (2019)



$$Pm = \frac{\nu}{\eta} \gg 1$$
$$k_\eta \gg k_\nu$$

Do we need to worry about
the seed field in turbulent
dynamamos?

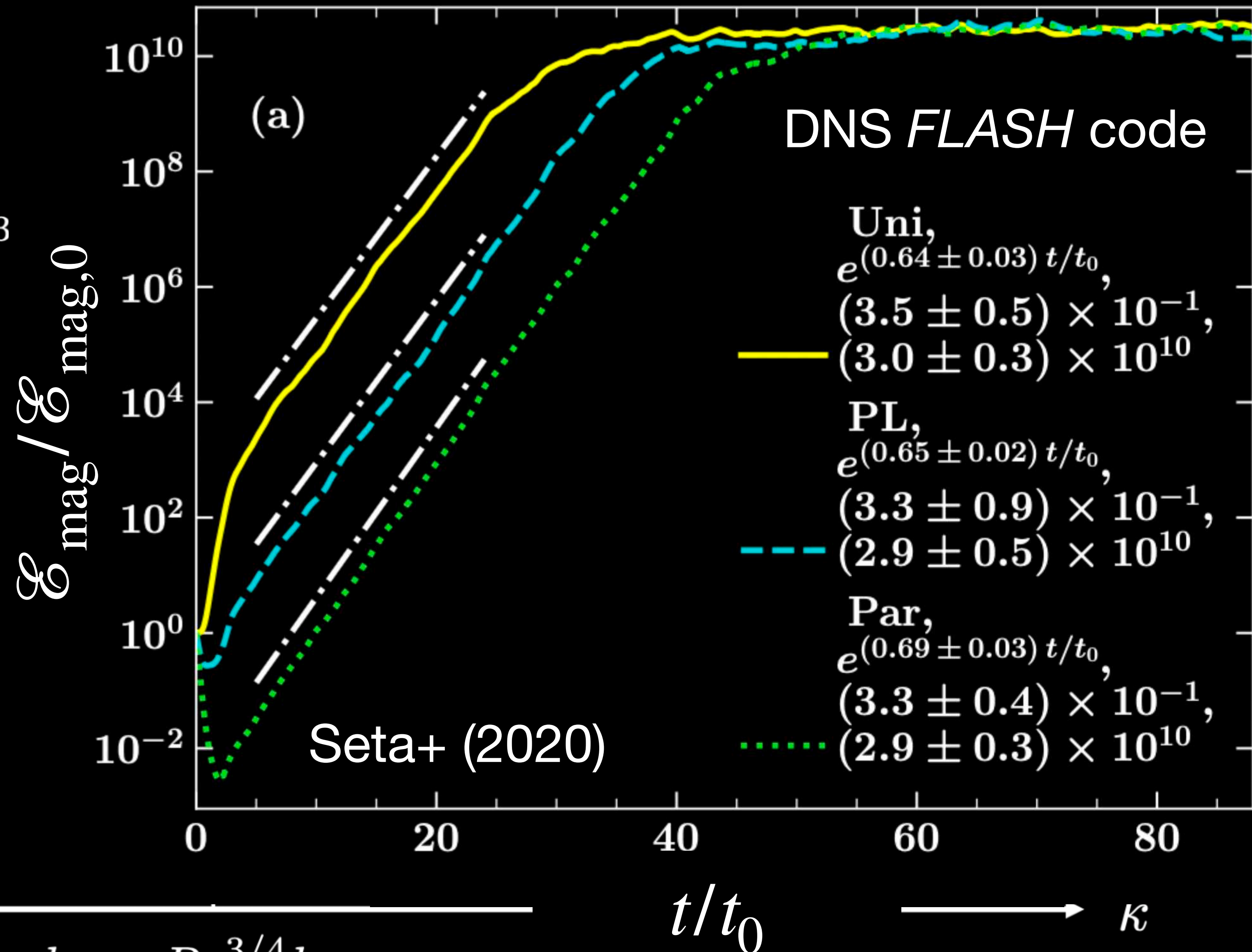
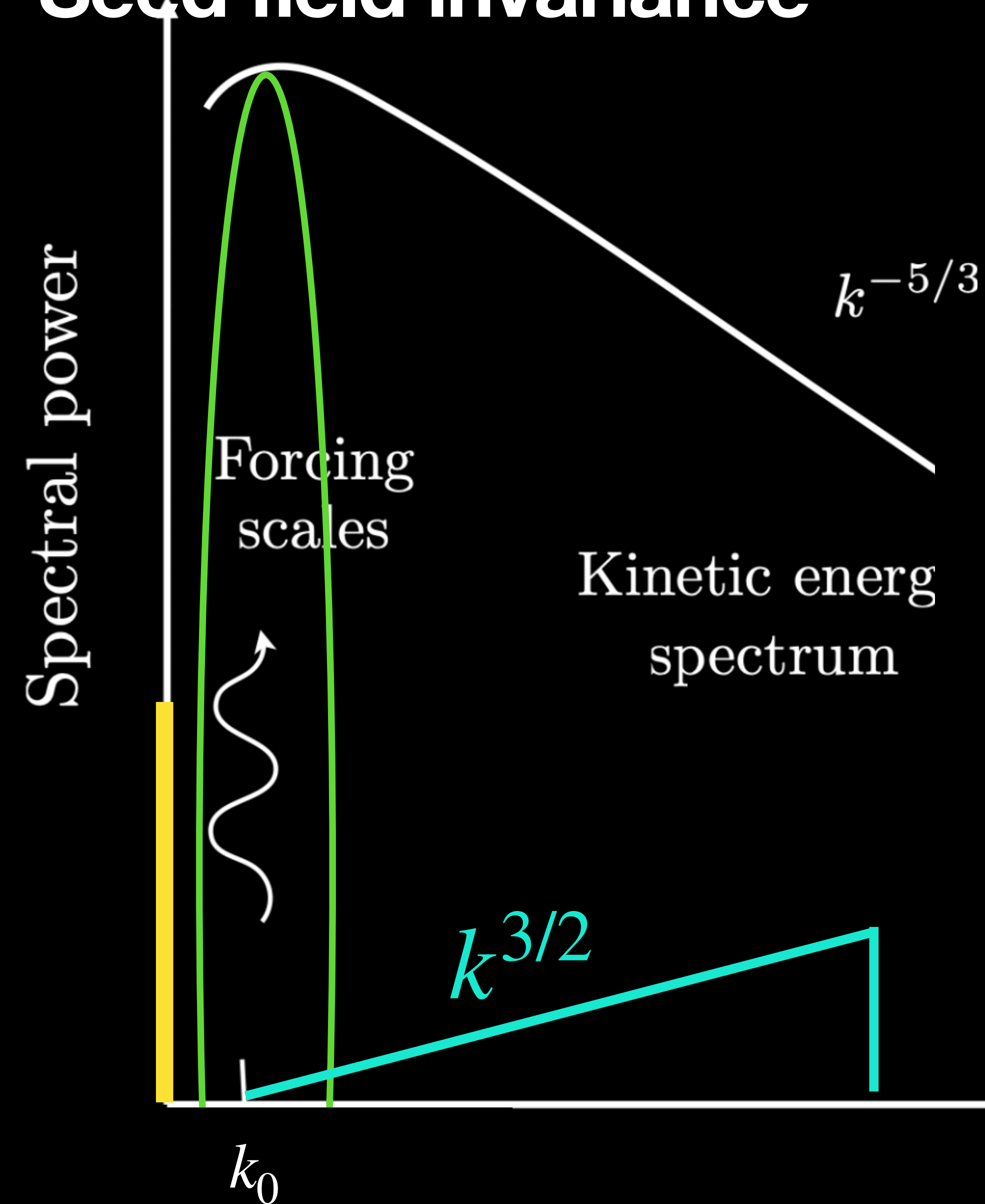
i.e., does the initial state
influence the final state?

Turbulence dynamo

Seed field invariance

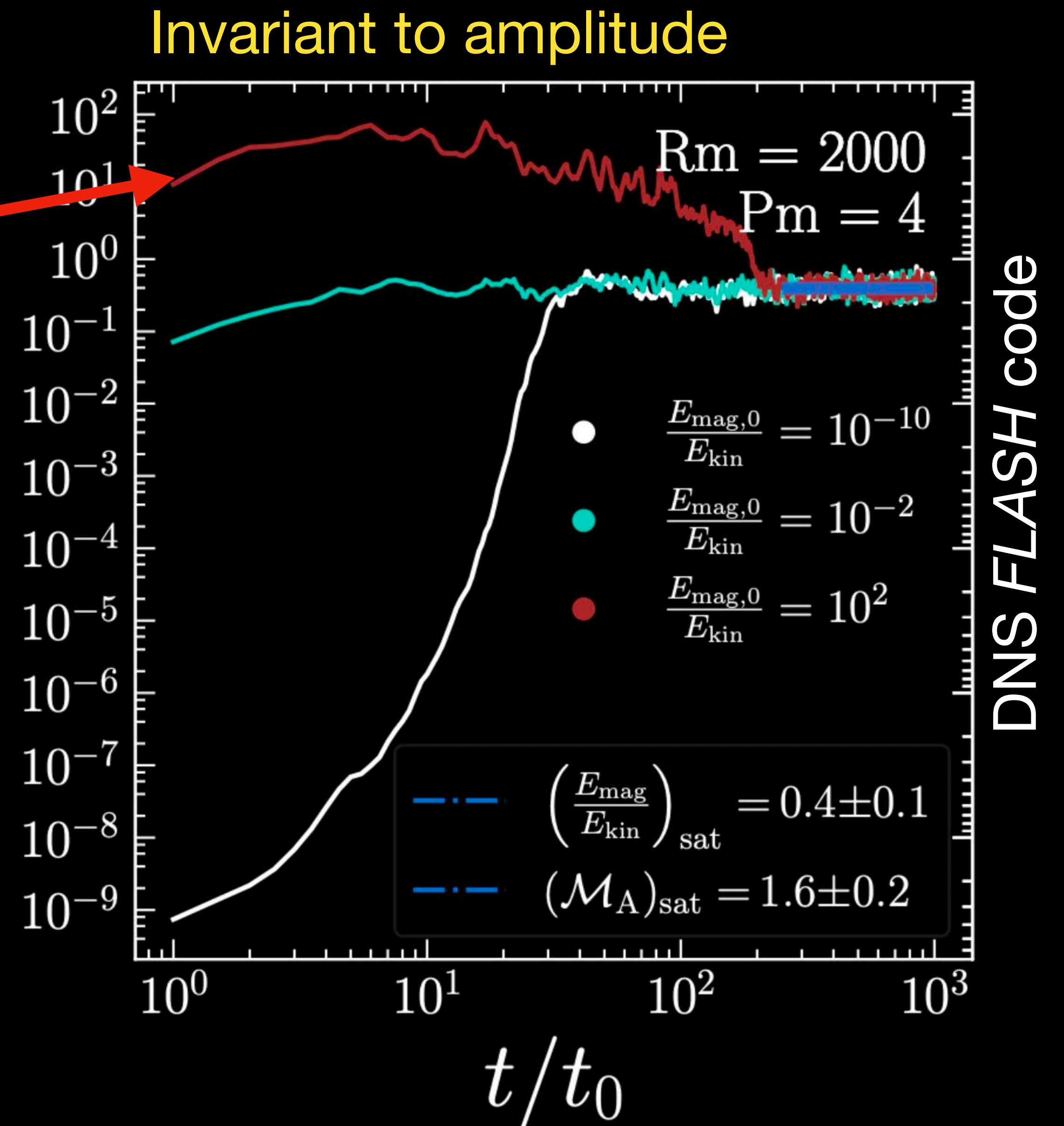
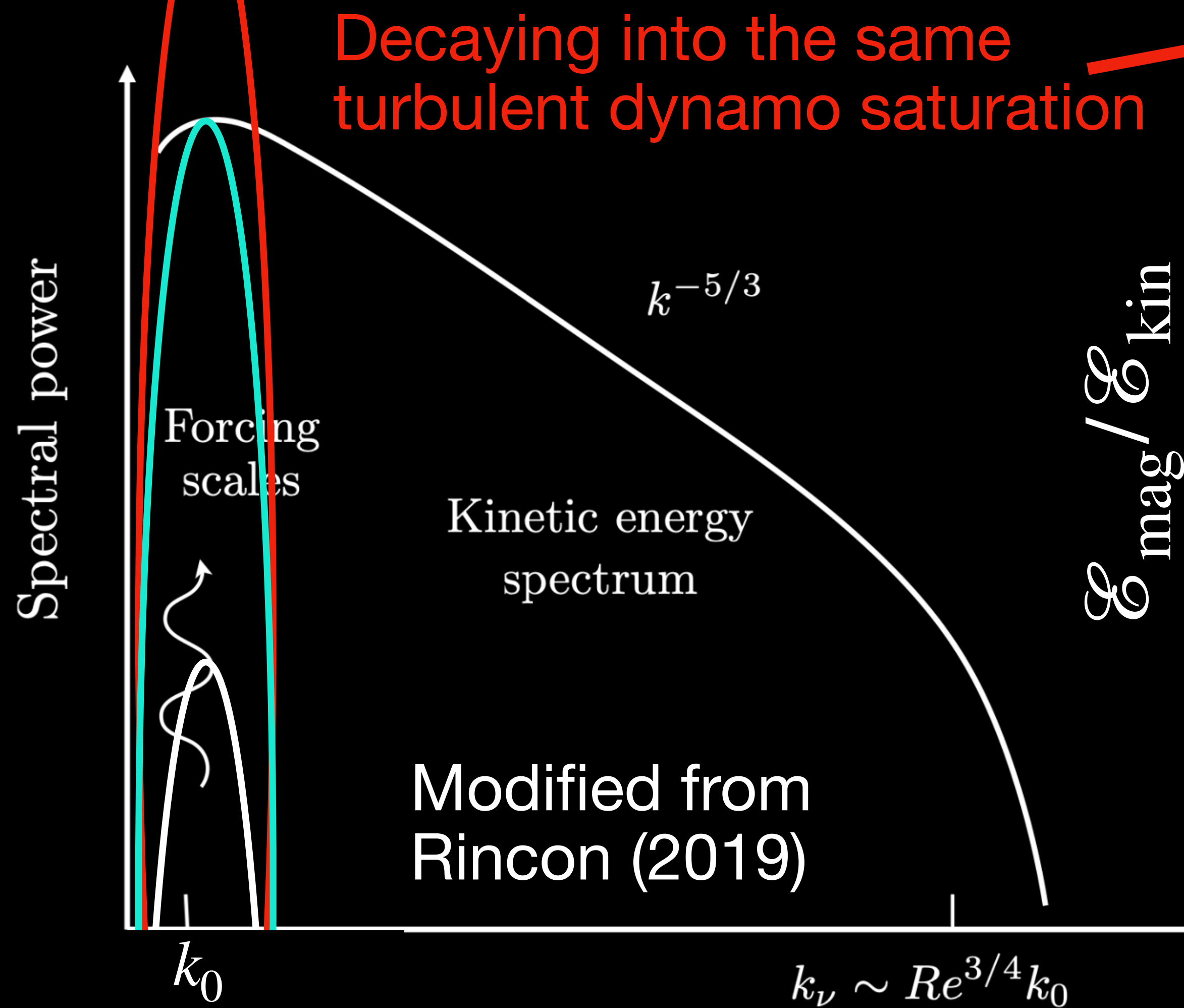
Modified from
Rincon (2019)

Invariant to structure



Turbulence dynamo

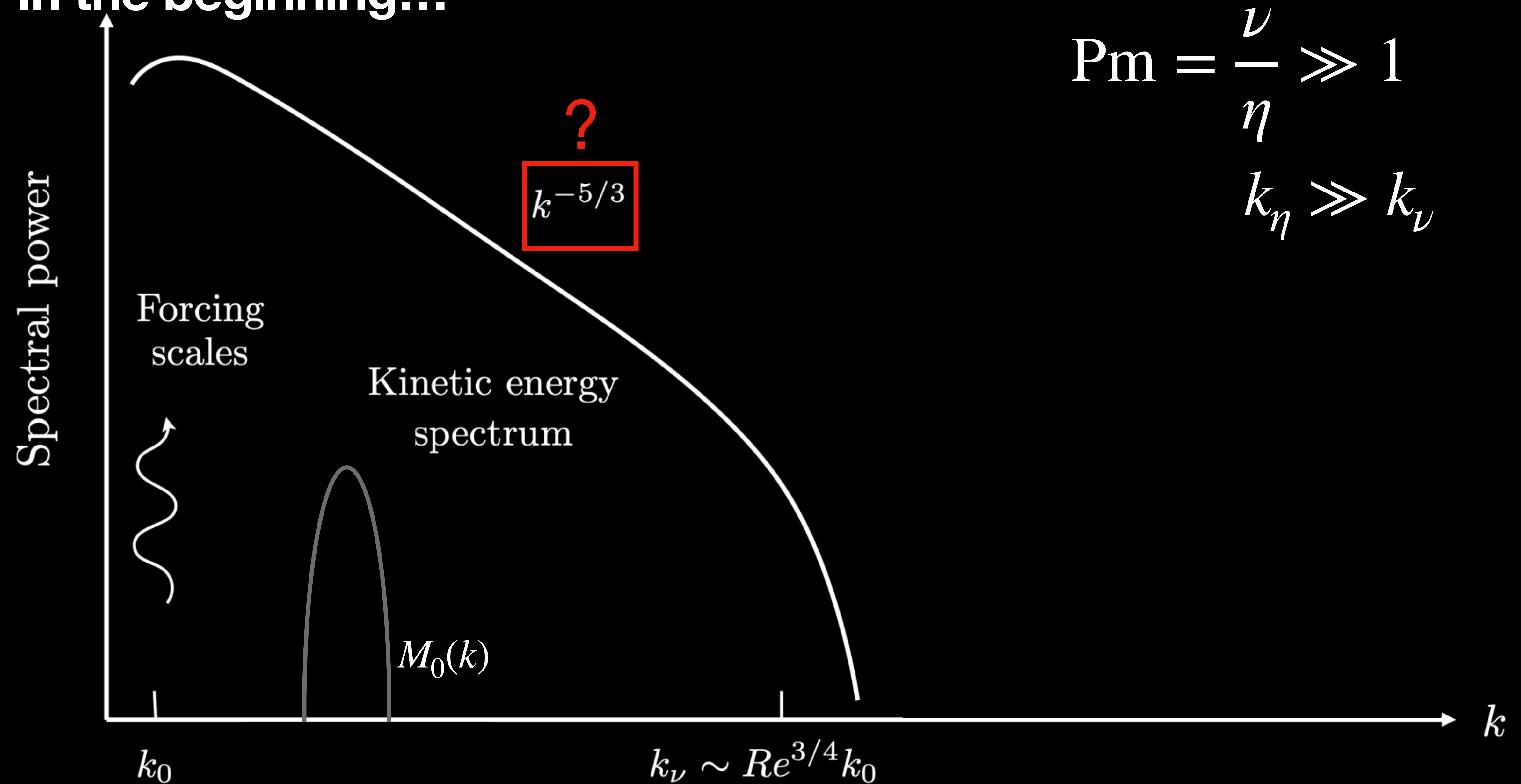
Seed field invariance



Turbulence dynamo

In the beginning...

Modified from
Rincon (2019)



$$Pm = \frac{\nu}{\eta} \gg 1$$
$$k_\eta \gg k_\nu$$

Turbulence dynamo

Kinematic regime: the sub-viscous range

Derived from $k^{-5/3}$ velocity spectrum

$$\mathcal{M} \ll 1 \quad k_\nu \sim \text{Re}^{3/4}$$

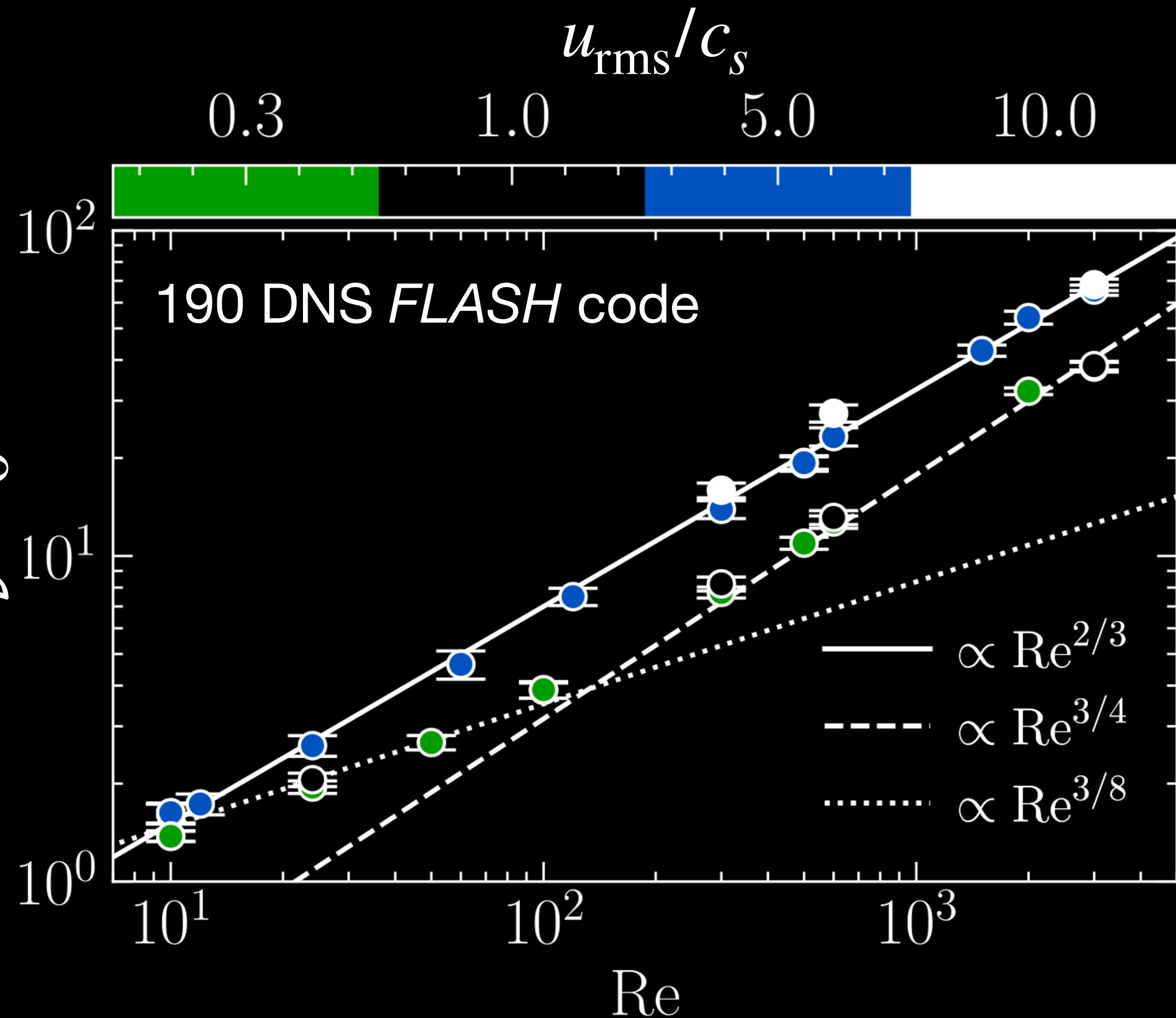
Kolmogorov41

Derived from k^{-2} velocity spectrum

$$\mathcal{M} \gg 1 \quad k_\nu \sim \text{Re}^{2/3}$$

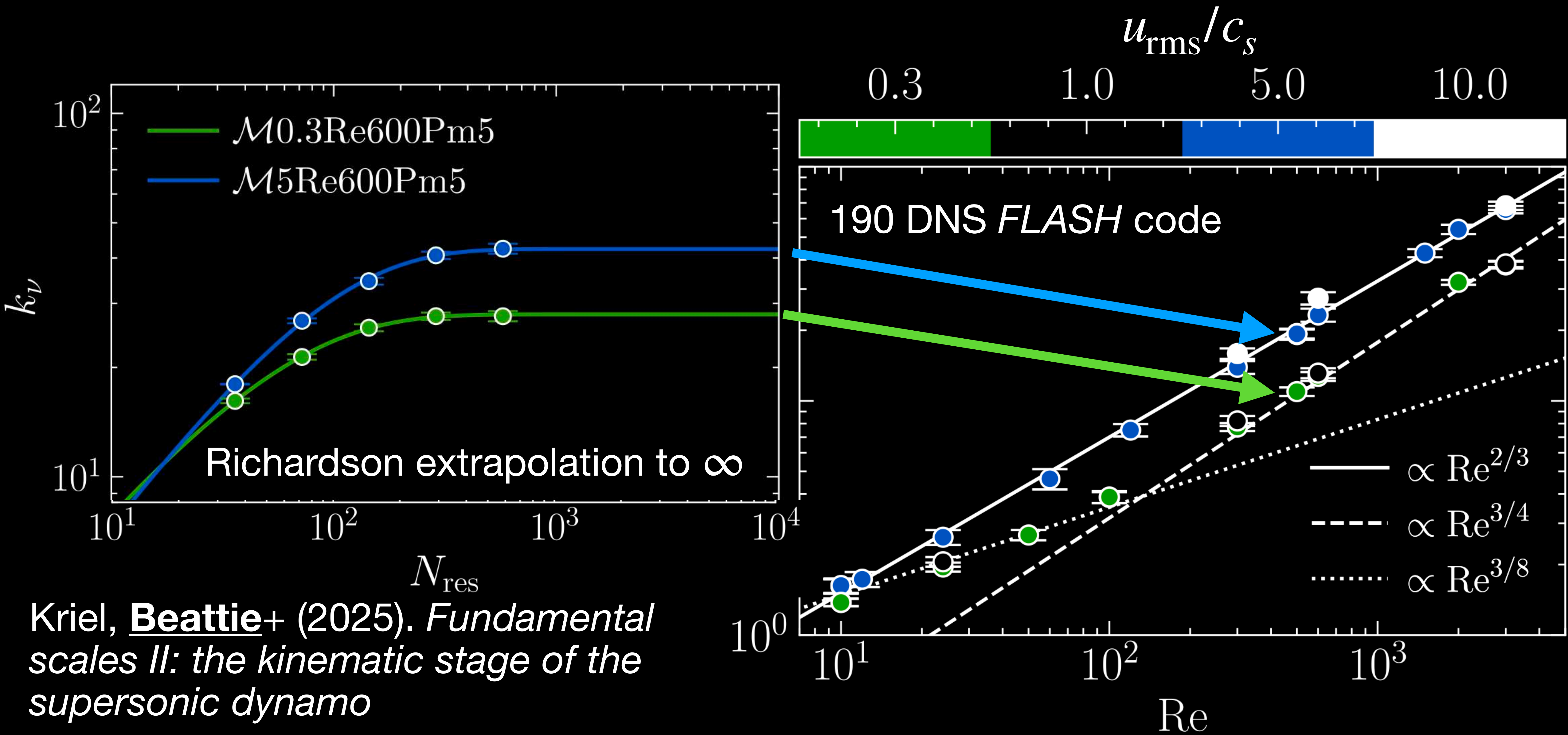
Schober+(2015)
Beattie+(2025)

Kriel, **Beattie**+ (2025). *Fundamental scales II: the kinematic stage of the supersonic dynamo*



Turbulence dynamo

Kinematic regime: the sub-viscous range

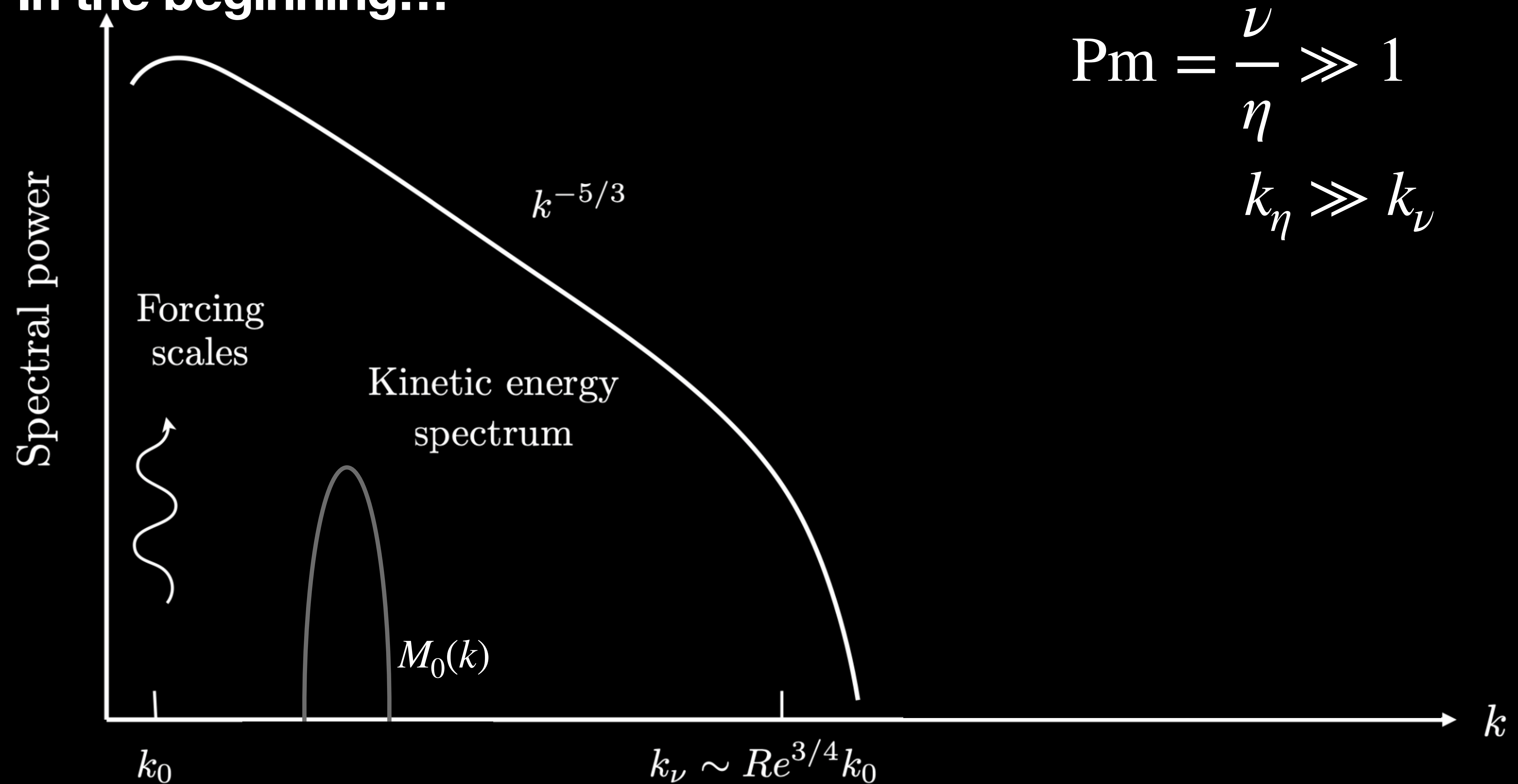


Kriel, **Beattie**+ (2025). *Fundamental scales II: the kinematic stage of the supersonic dynamo*

Turbulence dynamo

In the beginning...

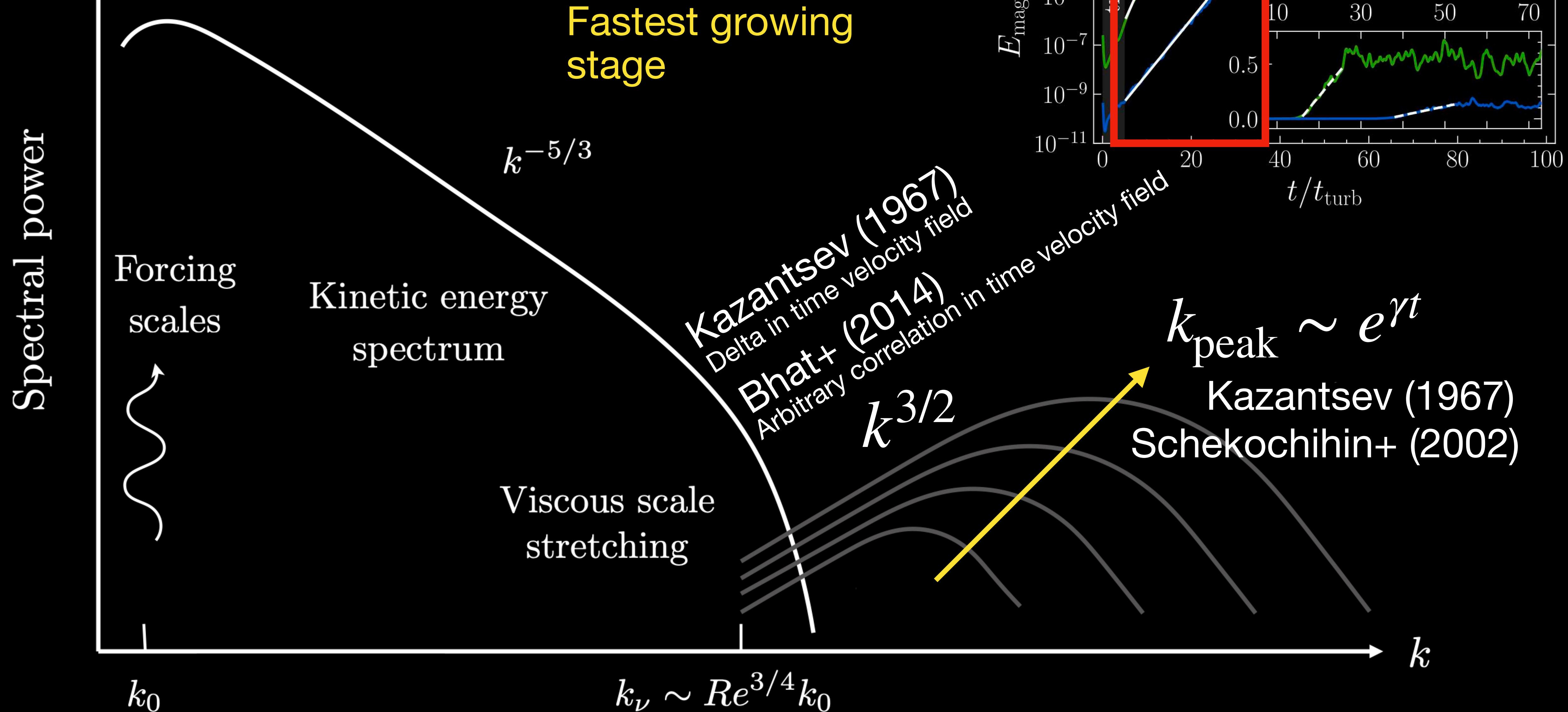
Modified from
Rincon (2019)



$$Pm = \frac{\nu}{\eta} \gg 1$$
$$k_\eta \gg k_\nu$$

Turbulence dynamo

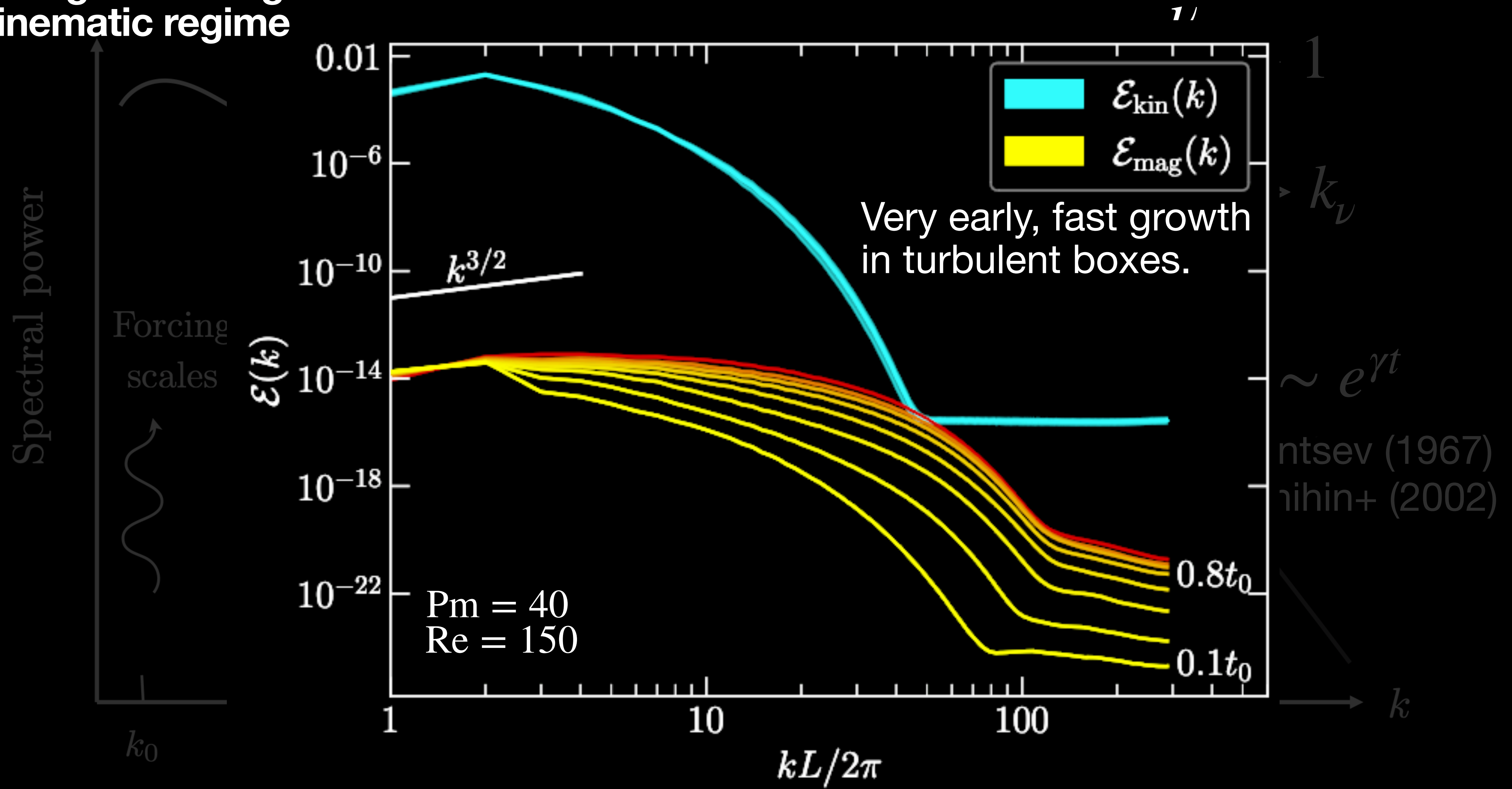
First growth stage: diffusion-free
kinematic regime



Turbulence dynamo

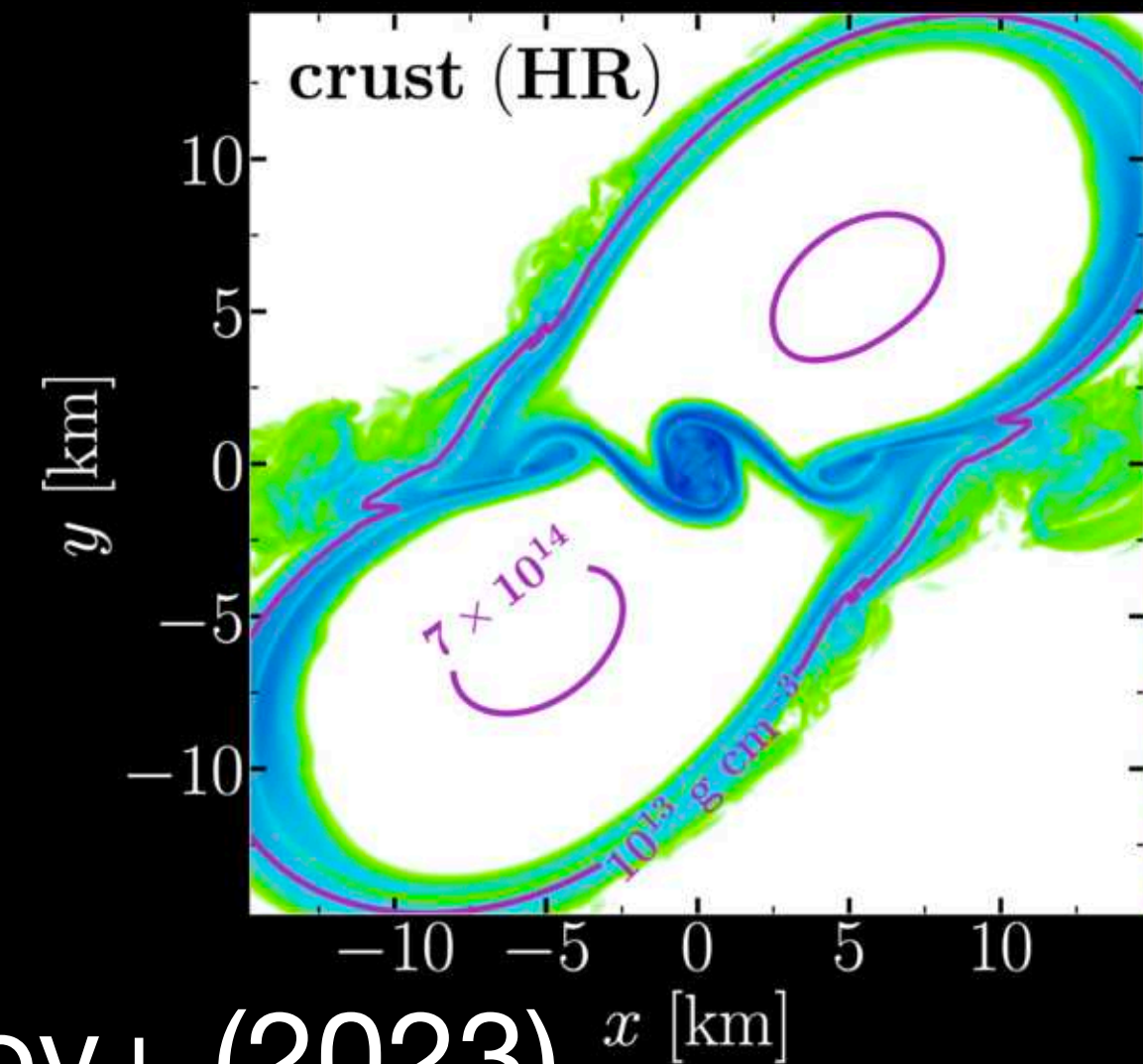
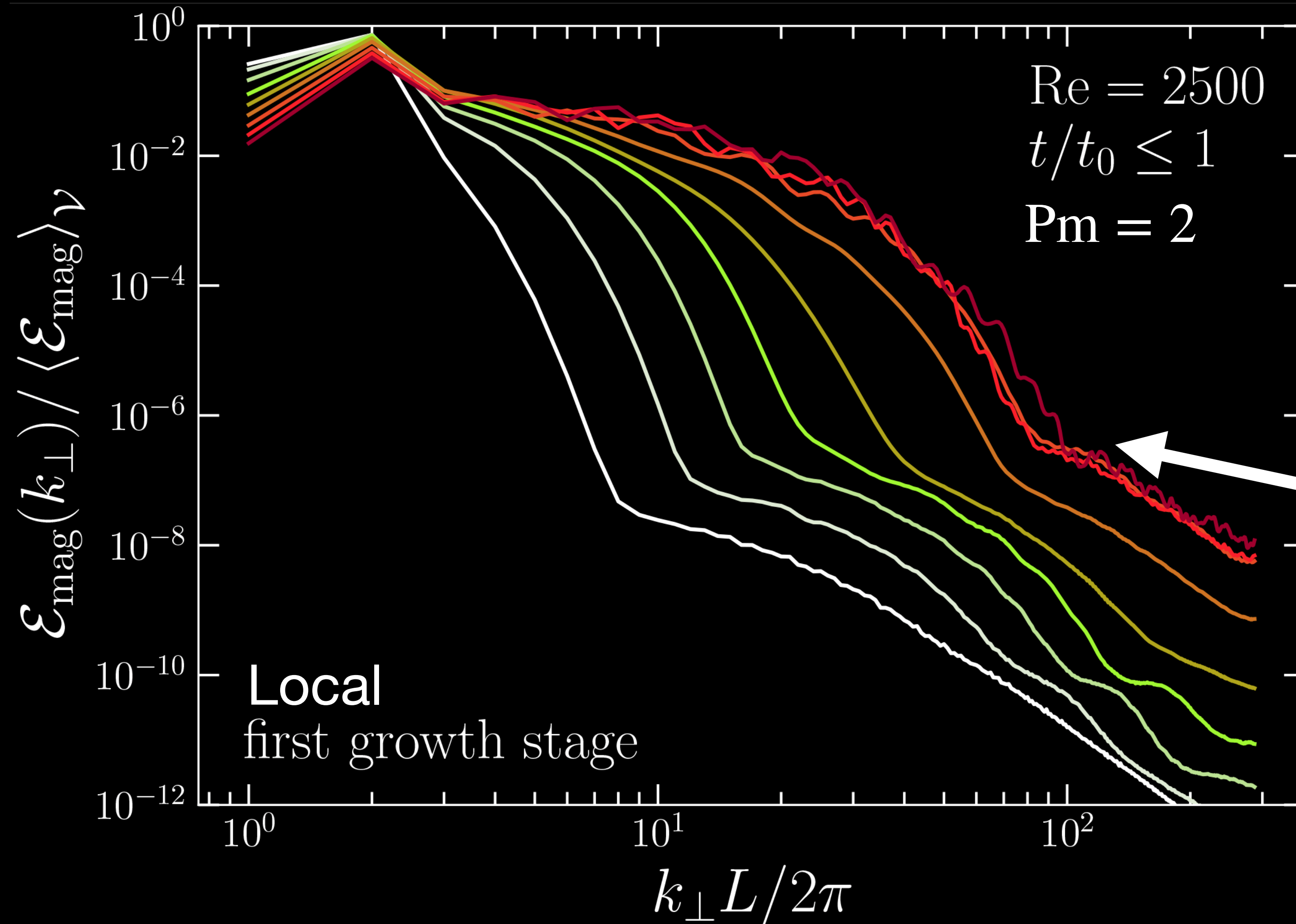
First growth stage: diffusion-free
kinematic regime

Modified from
Rincon (2019)

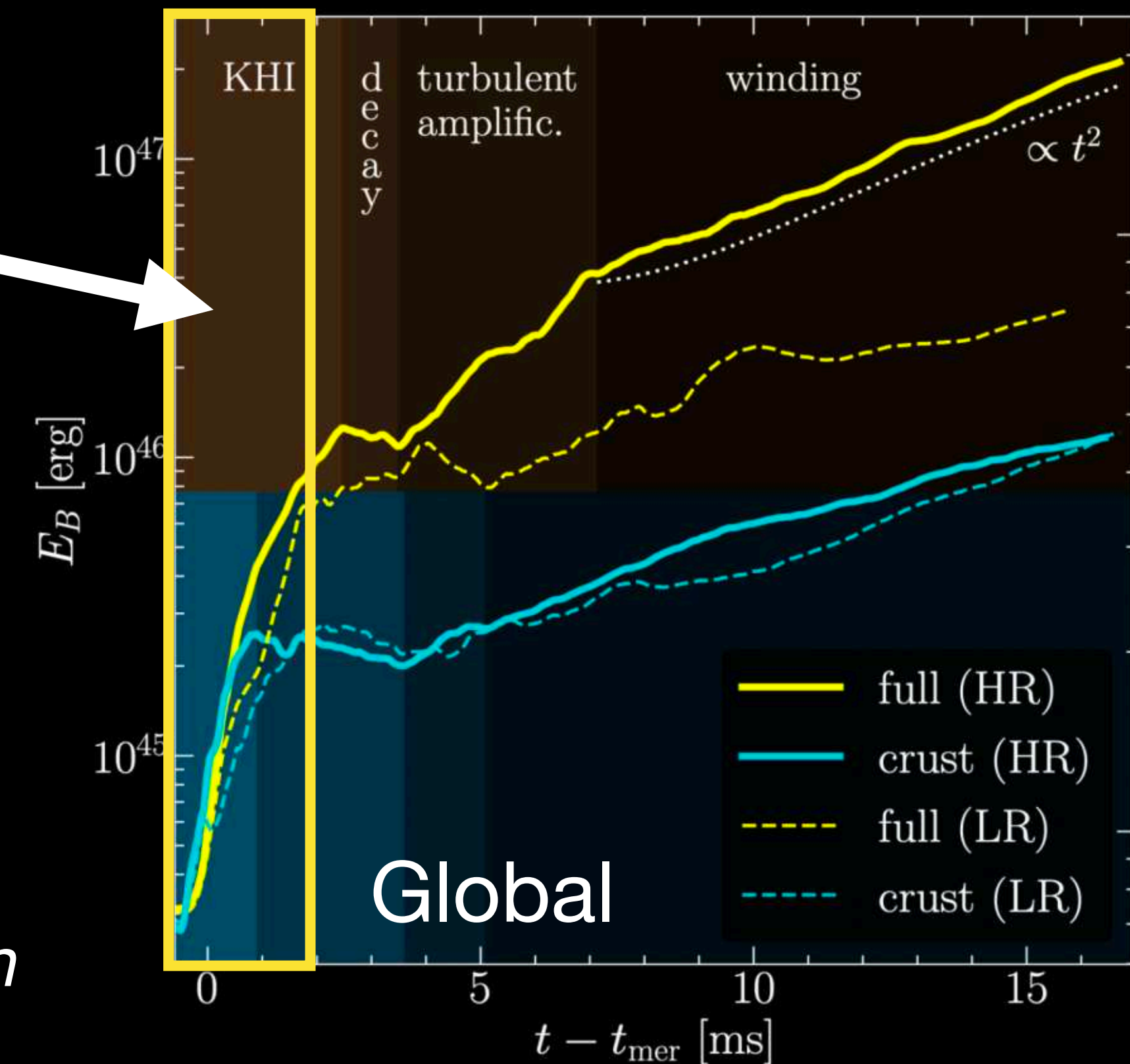


Turbulence dynamo

Local KHI dynamo simulations



Chabanov+ (2023)



Beattie+ (in prep.) *The Kinematic Kelvin-Helmholtz Dynamo Between Merging Compact Objects.*

Turbulence dynamo

The engine of the kinematic dynamo

Growth rate dominated by the smallest possible scales of the flow gradients

$$\gamma = \langle \hat{B}_i \hat{B}_j \partial_j u_i \rangle \sim u_\nu / \ell_\nu \sim 1/t_\nu, \quad \begin{aligned} u_\ell / \ell &\sim \varepsilon^{1/3} \ell^{-2/3}, \\ t_\ell &\sim \ell^{2/3}. \end{aligned}$$

put in units of outer scale turnover time $t_0 = \ell_0 / u_0$

$$t_0 \gamma \sim t_0 / t_\nu,$$

and to summarise,

K41 spectrum (incompressible)

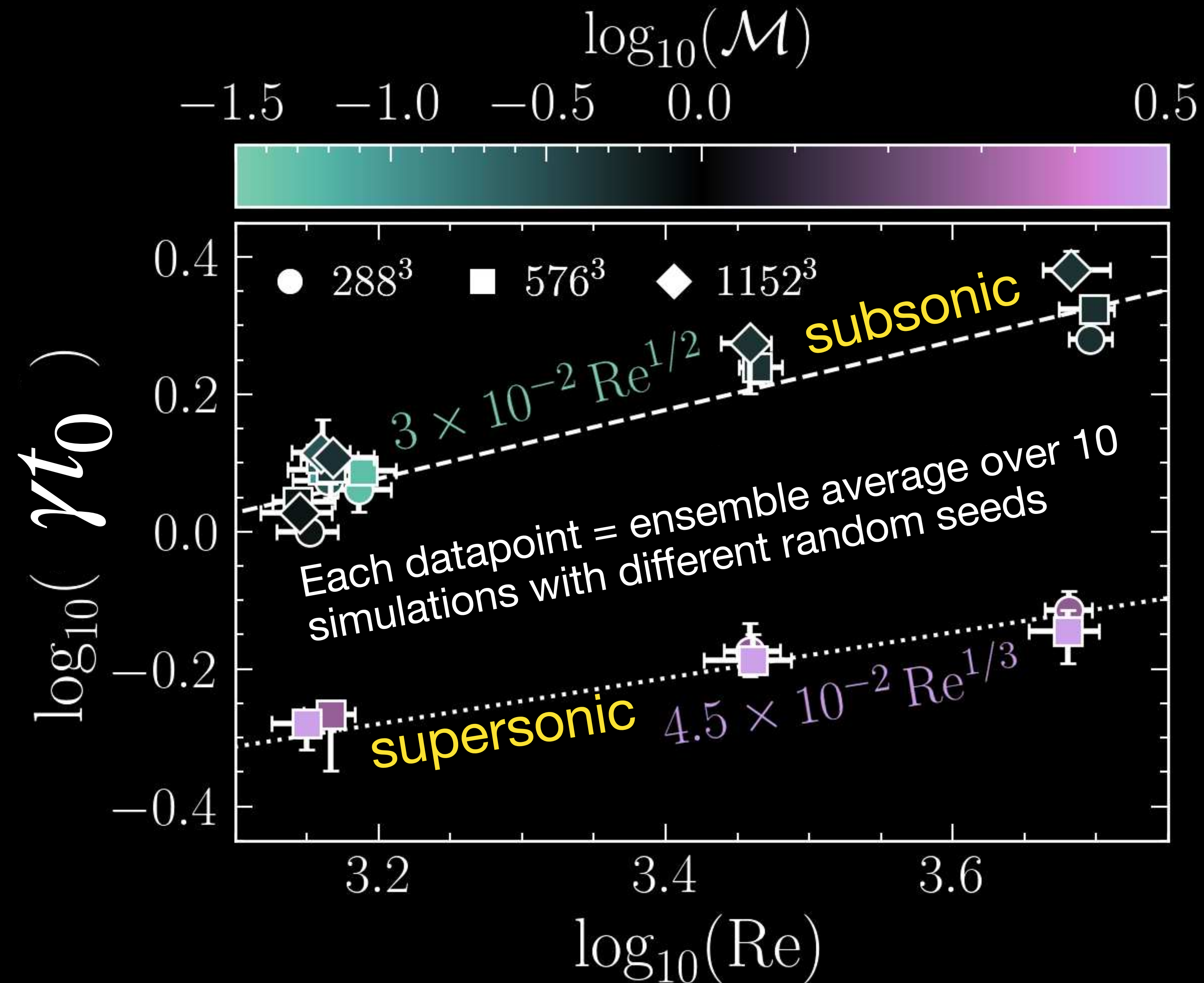
$$t_0 \gamma \sim (\ell_0 / \ell_\nu)^{2/3} \sim (\text{Re}^{3/4})^{2/3} \sim \text{Re}^{1/2},$$

Burgers spectrum (supersonic)

$$t_0 \gamma \sim (\ell_0 / \ell_\nu)^{1/2} \sim (\text{Re}^{2/3})^{1/2} \sim \text{Re}^{1/3}.$$

Turbulence dynamo

Confronting γ with data



The engine of the
kinematic dynamo is k_ν

K41 spectrum (incompressible)

$$t_0 \gamma \sim \text{Re}^{1/2}, \quad \checkmark$$

Burgers spectrum (supersonic)

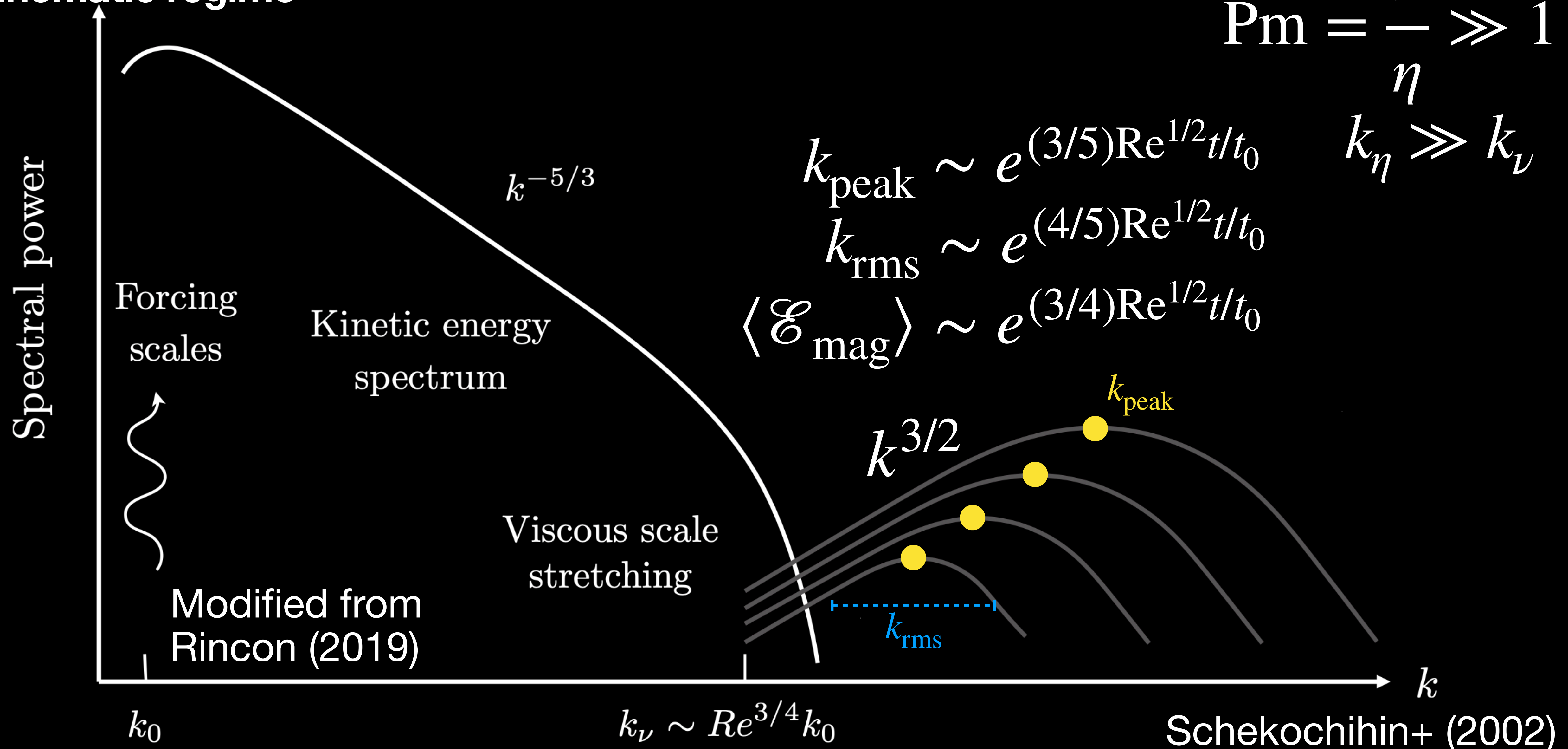
$$t_0 \gamma \sim \text{Re}^{1/3}. \quad \checkmark$$

Turbulence dynamo

First growth stage: diffusion-free
kinematic regime

$$\text{Pm} = \frac{\nu}{\eta} \gg 1$$

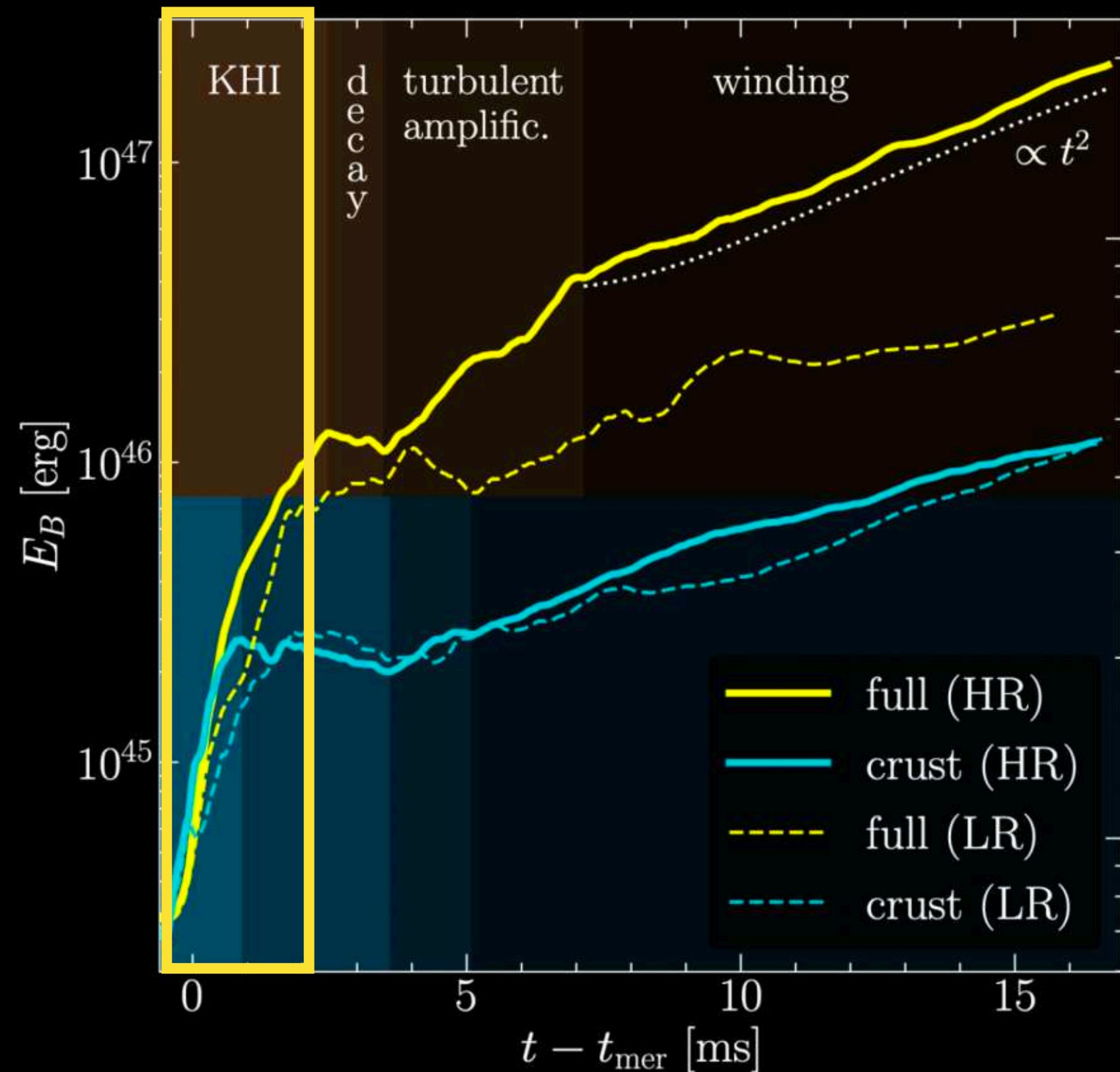
$$k_\eta \gg k_\nu$$



Turbulence dynamo

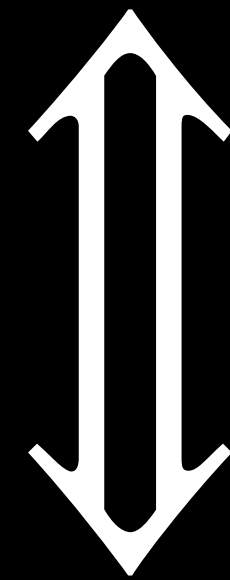
Local KHI dynamo simulations

Chabanov+ (2023)



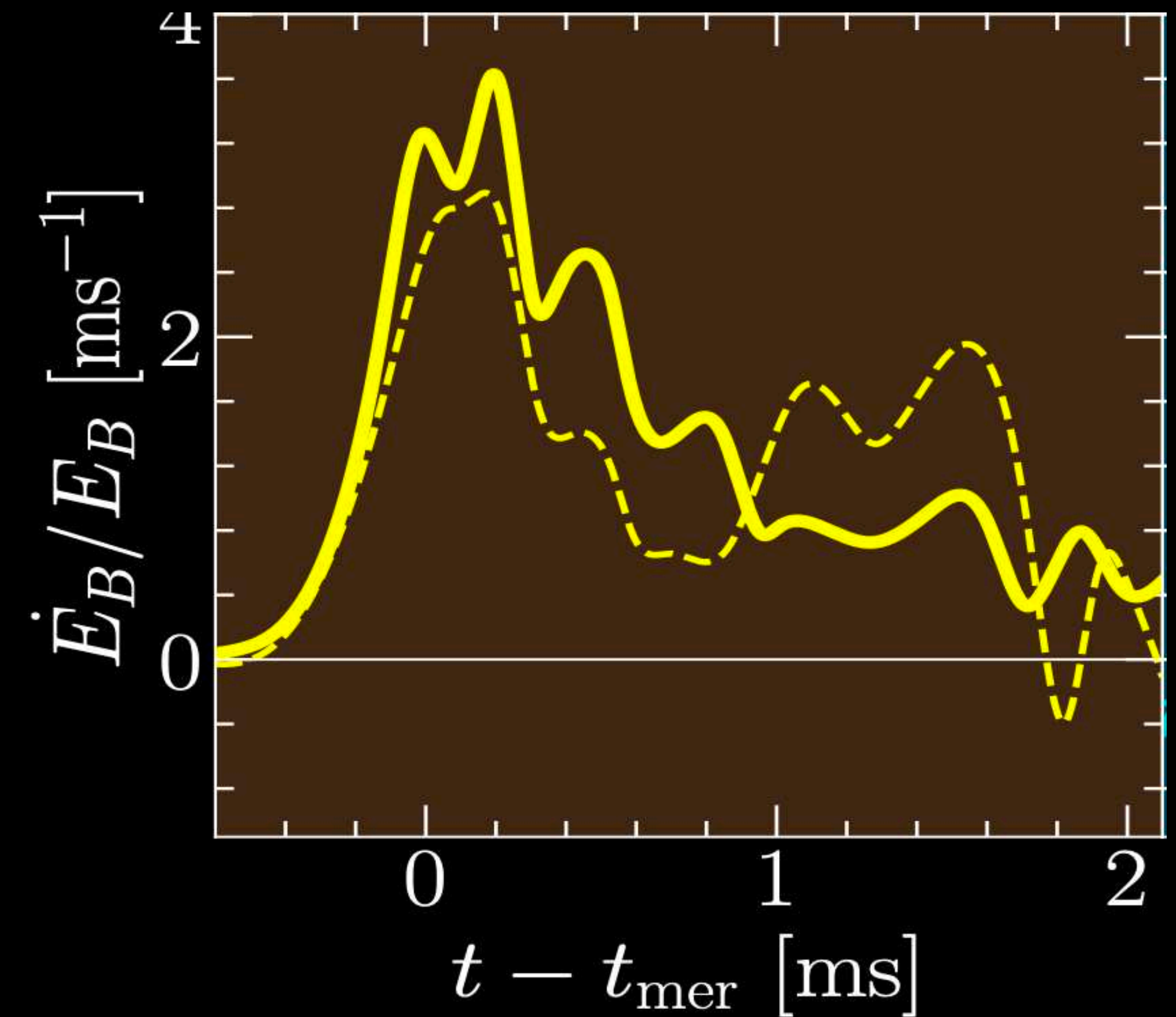
Neutrinos create a bulk viscosity

$$\text{Re} \sim 10^6+$$



$$\gamma \sim 10^3 (\text{ms})^{-1}$$

An extreme challenge for global sims



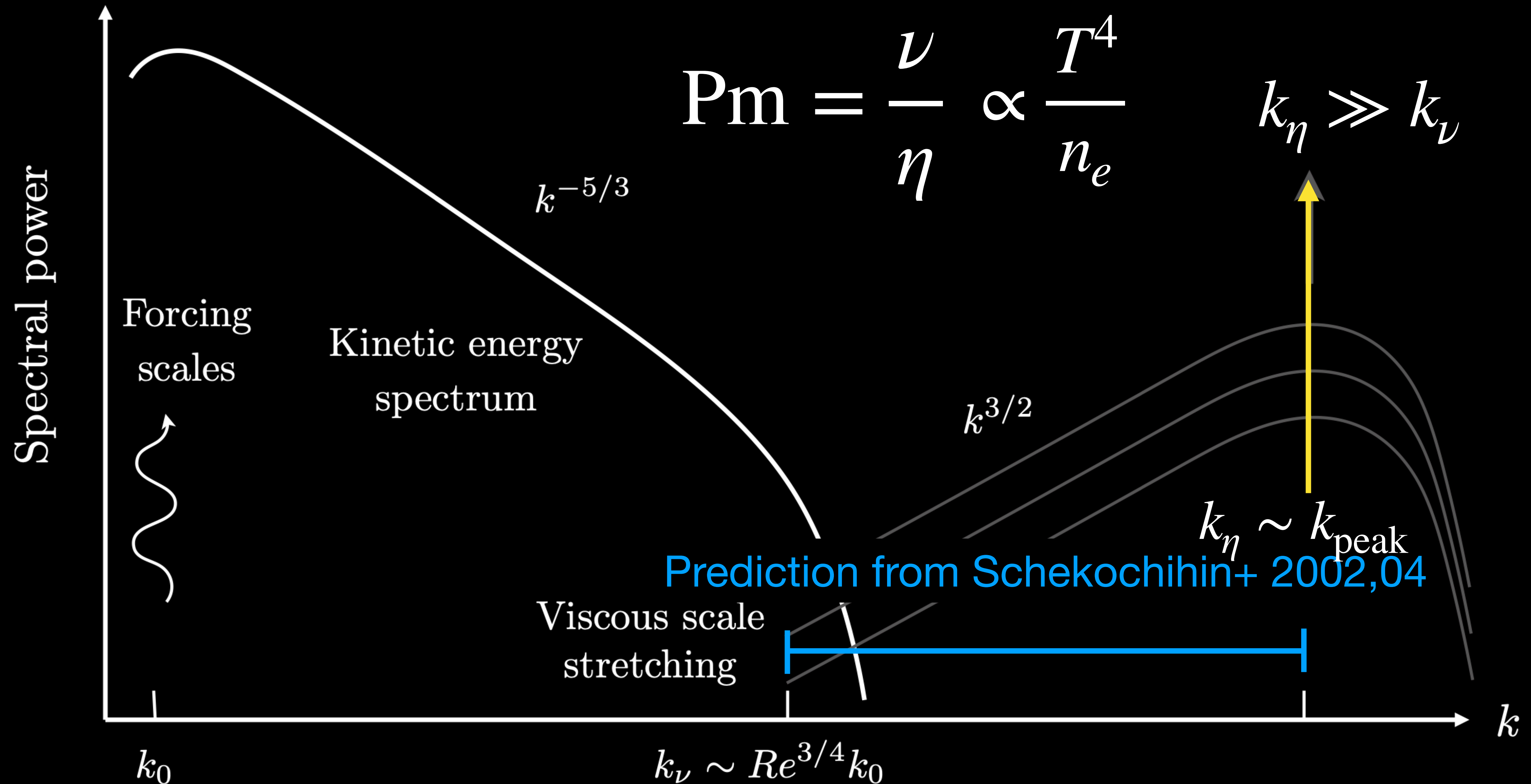
(bulk viscosity can act as an effective shear viscosity for the turbulent dynamo)

Beattie+ (2025, MNRAS) Taking control of compressible modes: bulk viscosity and the turbulent dynamo

Turbulence dynamo

First growth stage: kinematic regime & the sub-viscous range

Modified from Rincon (2019)



Turbulence dynamo

Kinematic regime: the sub-viscous range

Modified from
Rincon (2019)

$$\text{Pm} = \frac{\nu}{\eta} \gg 1$$
$$k_\eta \gg k_\nu$$

stretching at the viscous scale

$$\frac{u_\nu}{\ell_\nu} \sim \frac{\eta}{\ell_\eta^2}$$

$$k^{-5/3}$$

dissipation at the resistive scale

$$\ell_\eta \sim \left(\frac{\ell_\nu \eta}{u_\nu} \right)^{1/2} \sim \left(\frac{\nu \ell_\nu}{u_\nu} \right)^{1/2} \text{Pm}^{-1/2} \sim \ell_\nu \text{Pm}^{-1/2}$$

$$k^{3/2}$$

another prediction... independent of cascade

Prediction from Schekochihin+ 2002,04

Viscous scale
stretching

$$k_0$$

$$k_\nu \sim \text{Re}^{3/4} k_0$$

k

Spectral power

Turbulence dynamo

Most of the energy is in the sub-viscous modes!

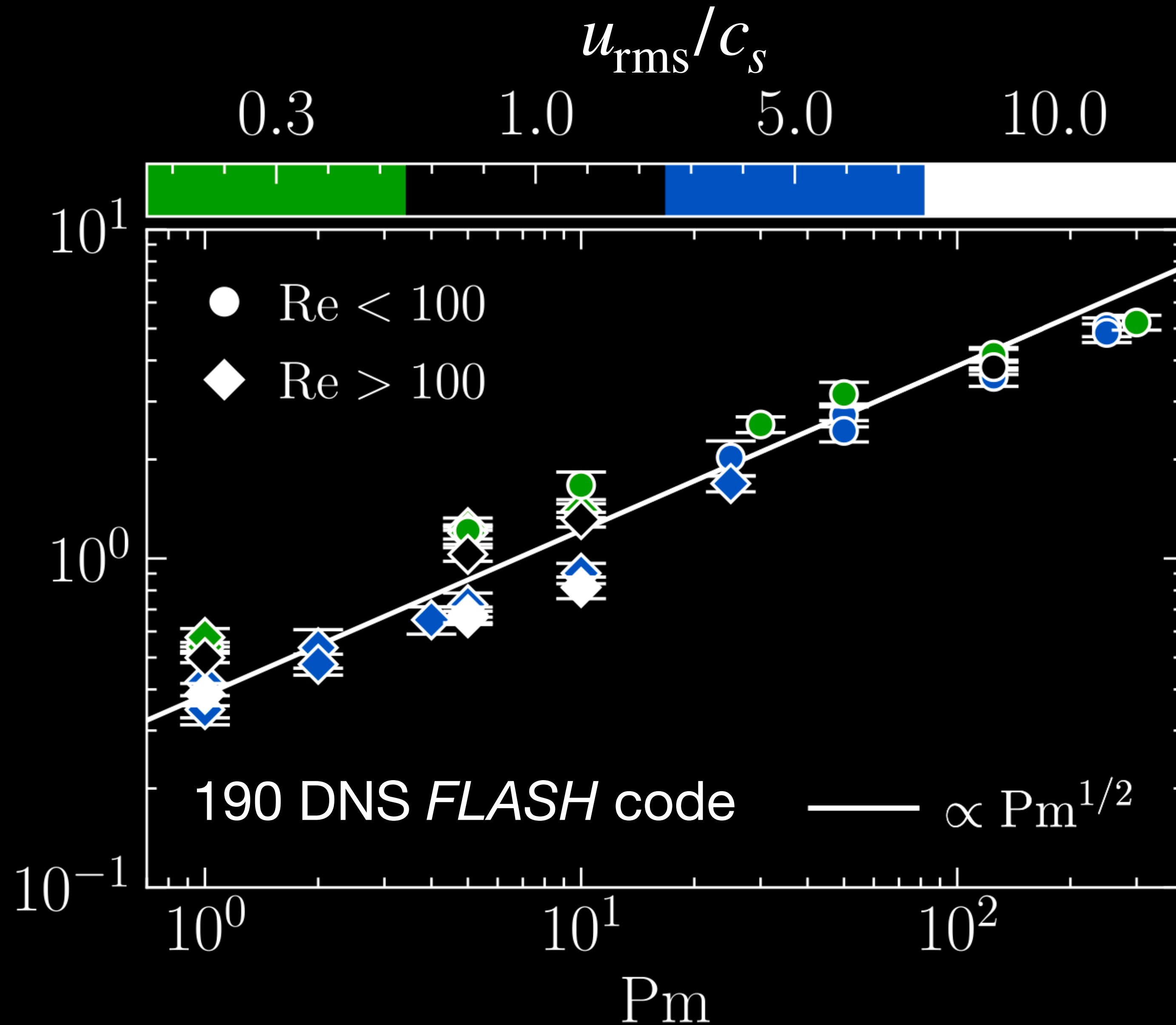
stretching at the viscous scale

$$\frac{u_\nu}{\ell_\nu} \sim \frac{\eta}{\ell_\eta^2}$$

dissipation at the resistive scale

$$\frac{k_\eta}{k_\nu} \sim \text{Pm}^{1/2}$$

Kriel, Beattie+ (2025). *Fundamental scales II: the kinematic stage of the supersonic dynamo*



How fast do the sub-viscous modes populate?

(Assuming $M_0(k) = M_0\delta(k - k_0)$)

Cold ISM

$$\begin{aligned} \text{CNM, } \text{Re} \sim 10^{10}, \text{Pm} \sim 10^4, t_0 \sim 10 \text{ pc} / 3 \text{ km s}^{-1} \sim \text{Myr} \quad \Delta t \ll t_0 \\ \Delta t \sim \ln(\text{Re}^{3/4} \text{Pm}^{1/2}) \gamma^{-1} \implies \Delta t \sim 20 \gamma^{-1} \implies \Delta t \sim 200 \text{ years,} \\ \sim 8 \text{ orders of mag.} \end{aligned}$$

Hot ISM and ICM

$$\begin{aligned} \text{HIM, } \text{Re} \sim 10^2, \text{Pm} \sim 10^{20}, t_0 \sim 30 \text{ pc} / 10 \text{ km s}^{-1} \sim \text{Myr} \\ \Delta t \sim 25 \gamma^{-1} \implies \Delta t \sim \text{Myr}, \Delta t \sim t_0 \\ \sim 11 \text{ orders of mag.} \end{aligned}$$

$$\begin{aligned} \text{ICM, } \text{Re} \sim 10^2, \text{Pm} \sim 10^{29}, t_0 \sim 100 \text{ kpc} / 200 \text{ km s}^{-1} \sim 10^2 \text{ Myr} \\ \Delta t \sim 40 \gamma^{-1} \implies \Delta t \sim 400 \text{ Myr}, \Delta t \sim t_0 \\ \sim 17 \text{ orders of mag.} \end{aligned}$$

Turbulence dynamo

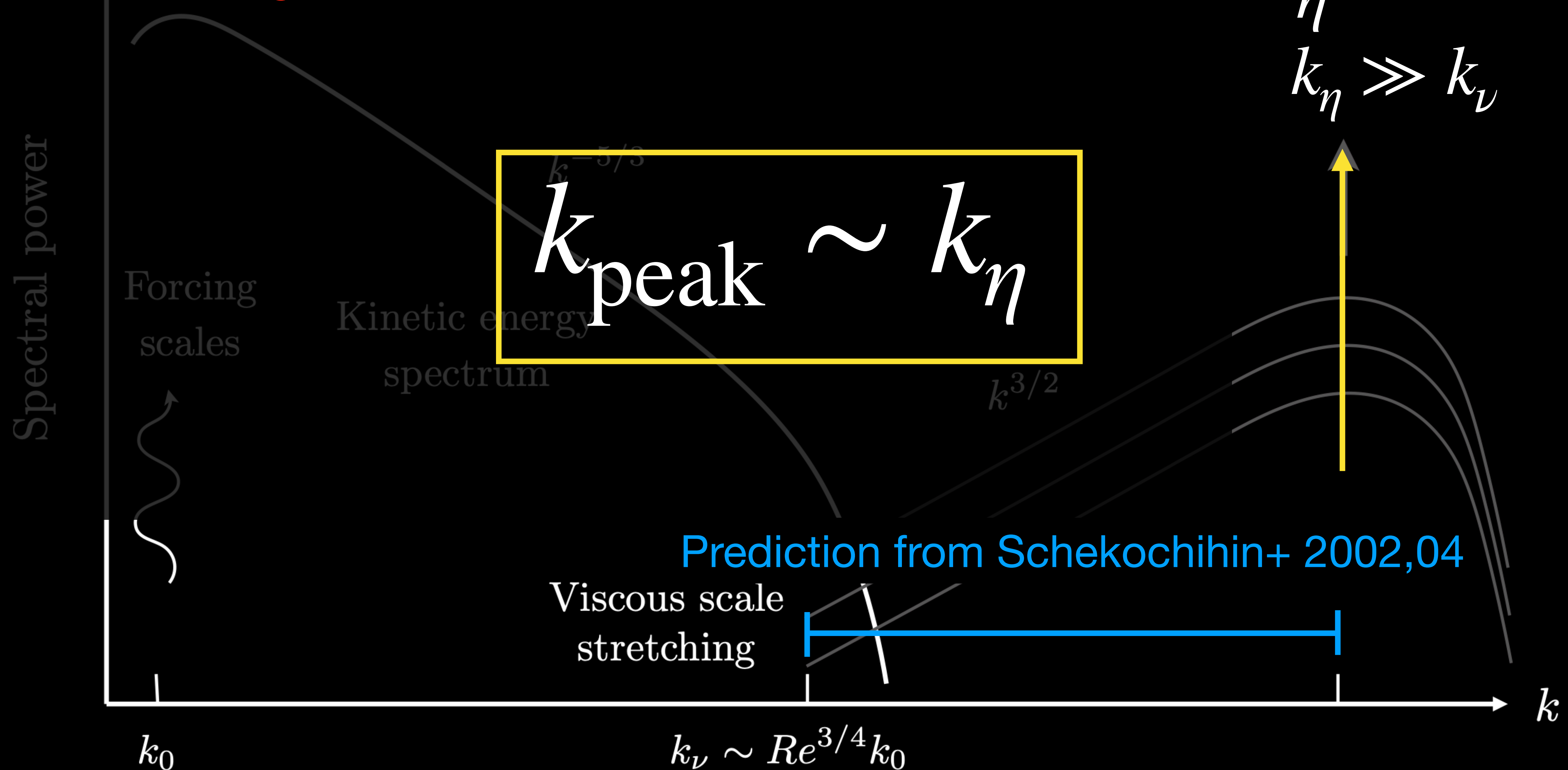
Kinematic regime: the sub-viscous range

Modified from
Rincon (2019)

$$Pm = \frac{\nu}{\eta} \gg 1$$

$$k_\eta \gg k_\nu$$

stretching at the viscous scale

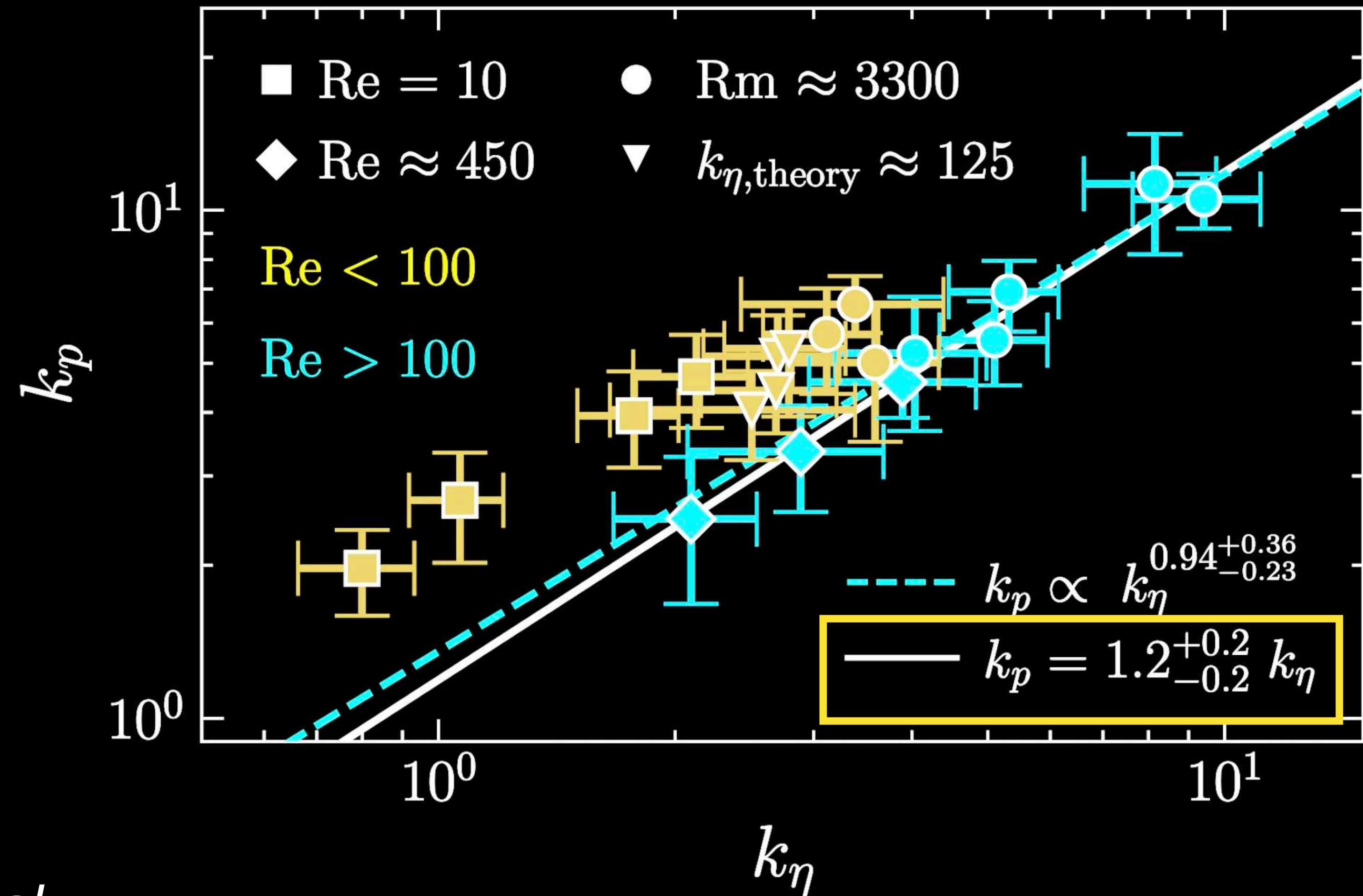


Turbulence dynamo

Most of the energy on the resistive scale!

The magnetic energy is concentrated at the smallest scales!

$$k_{\text{peak}} \sim k_{\eta} \quad \checkmark$$



Kriel, Beattie+ (2025). *Fundamental scales I: the kinematic stage of the subsonic dynamo*

Turbulence dynamo

Viscosity and resistivity matter!

$$\text{Pm} = \frac{\nu}{\eta} \gg 1$$

An extreme challenge for
global simulators using
ideal MHD for dynamo!

peak energy (depends on Rm)

$$k_\eta \sim \text{Pm}^{1/2} k_\nu$$

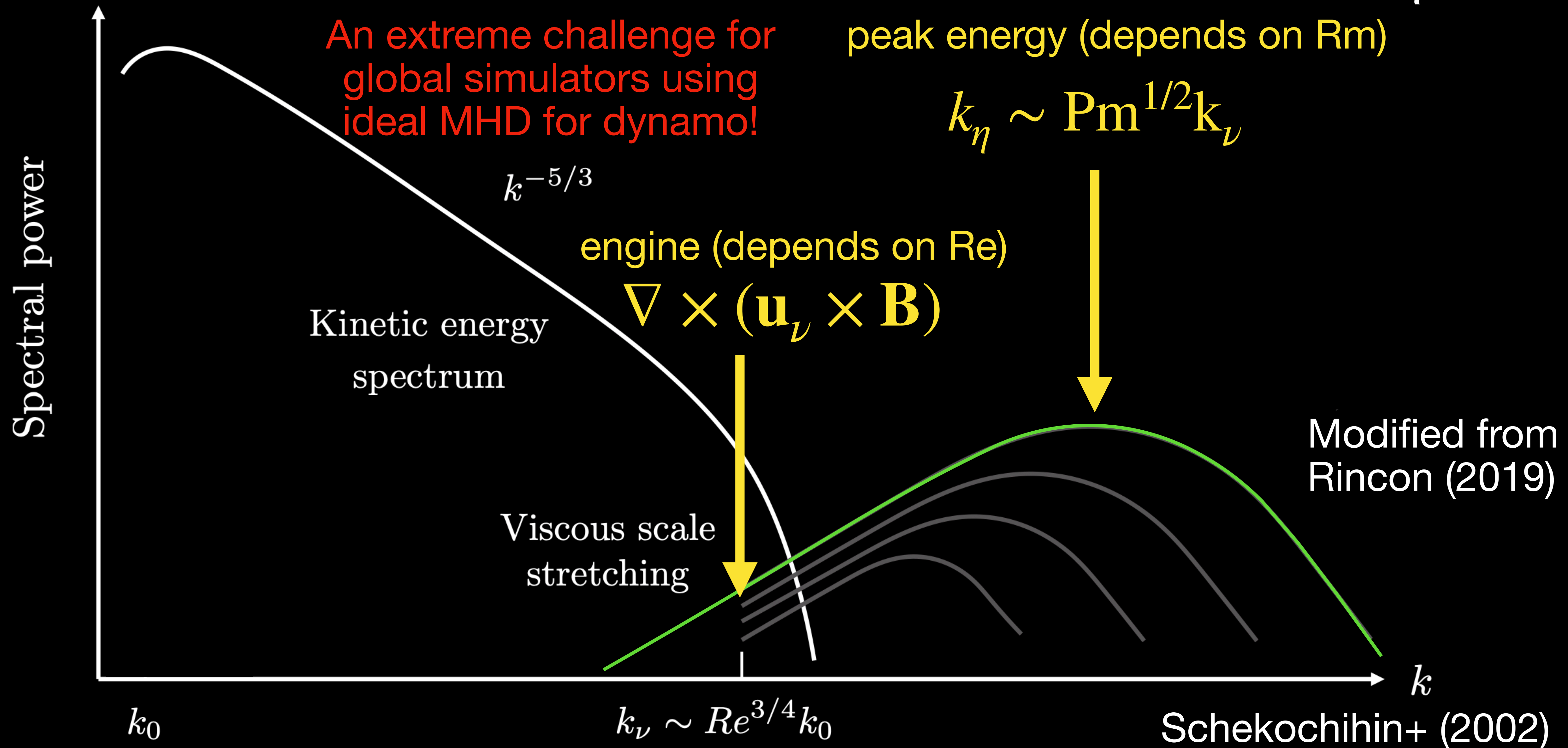
engine (depends on Re)

$$\nabla \times (\mathbf{u}_\nu \times \mathbf{B})$$

Kinetic energy
spectrum

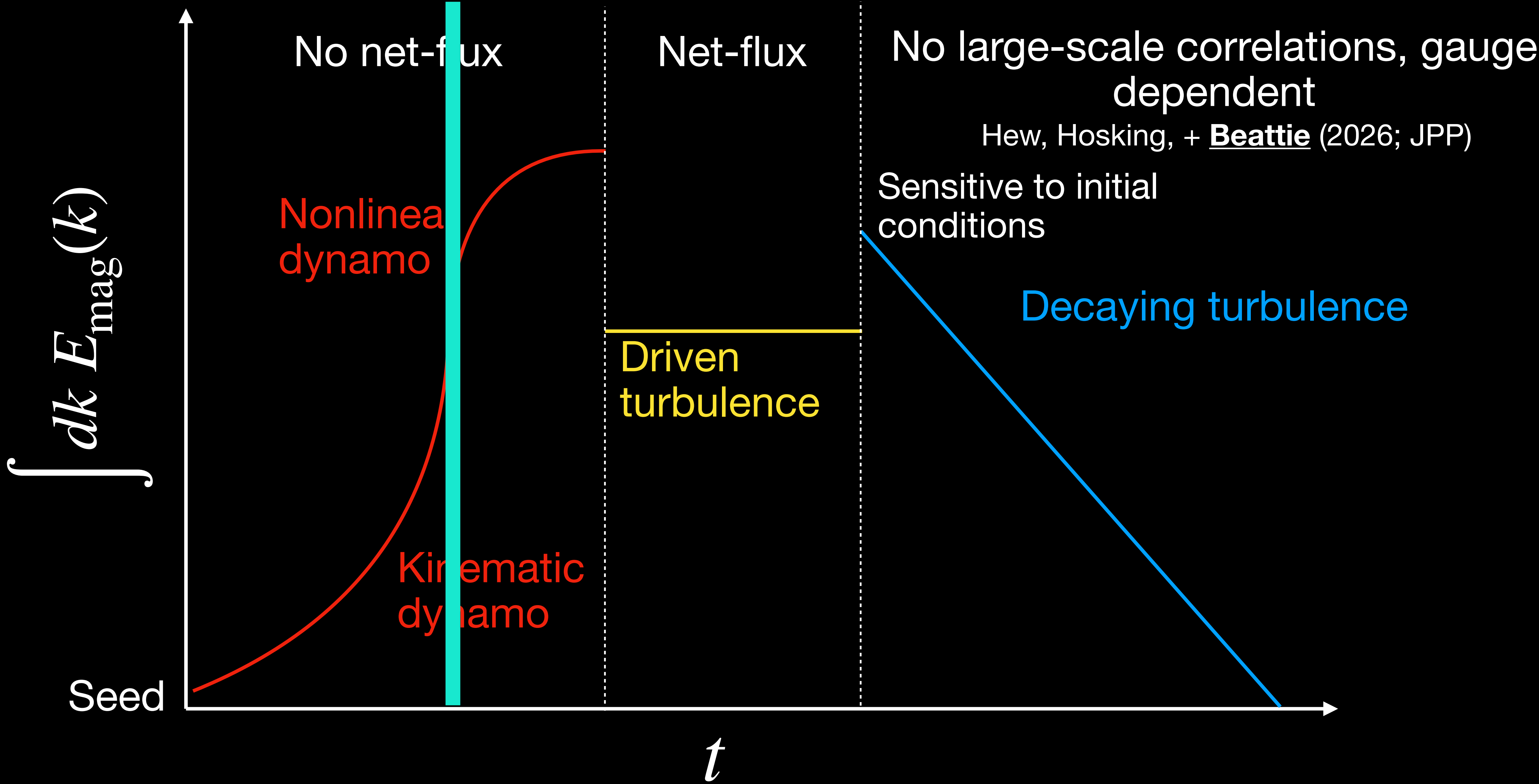
Viscous scale
stretching

Modified from
Rincon (2019)



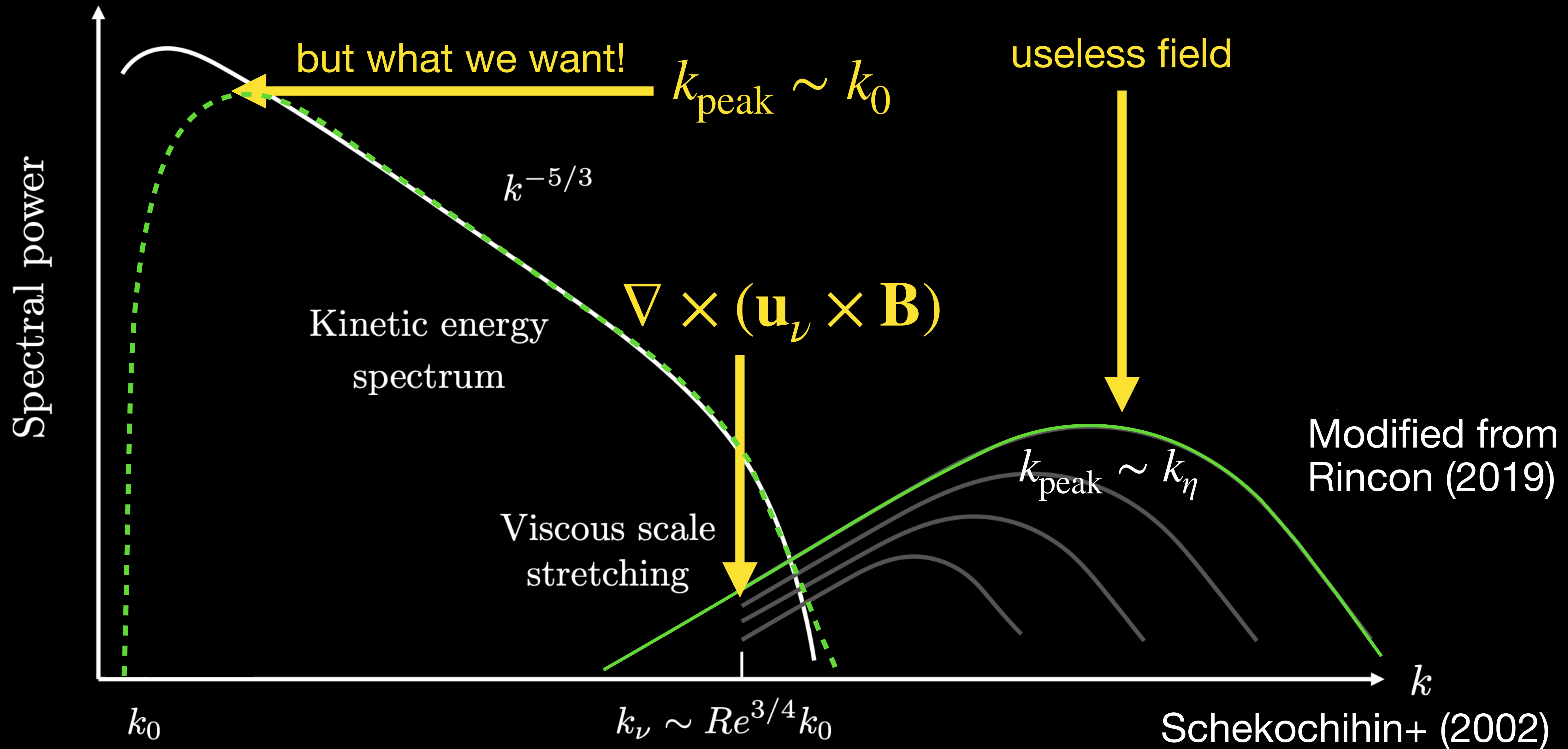
The lifecycle of a high-k mode spans across a patchwork quilt of theories

This afternoon we're here



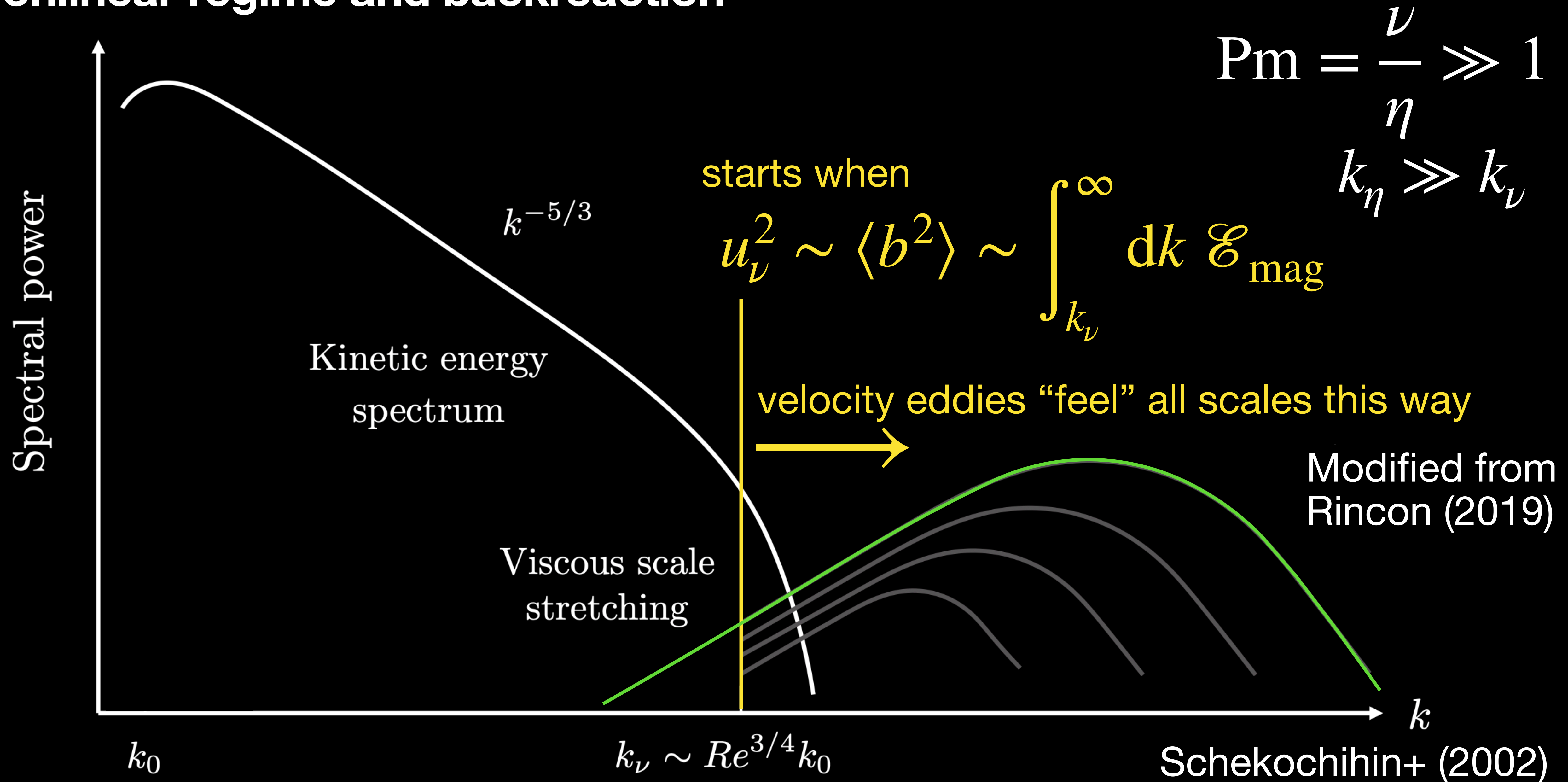
Turbulence dynamo

If the turbulence dynamo only generates small-scale fields it's useless



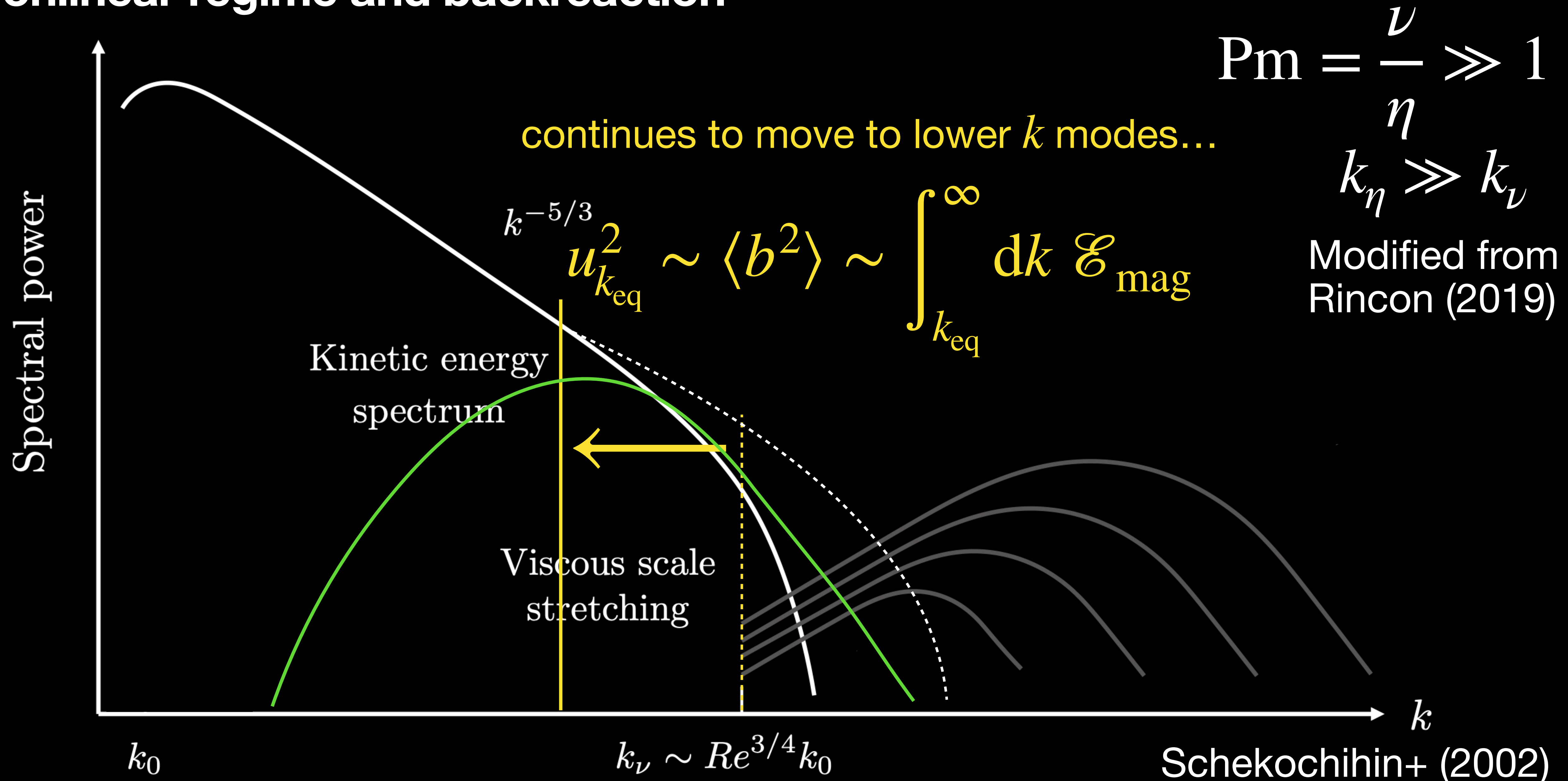
Turbulence dynamo

Nonlinear regime and backreaction



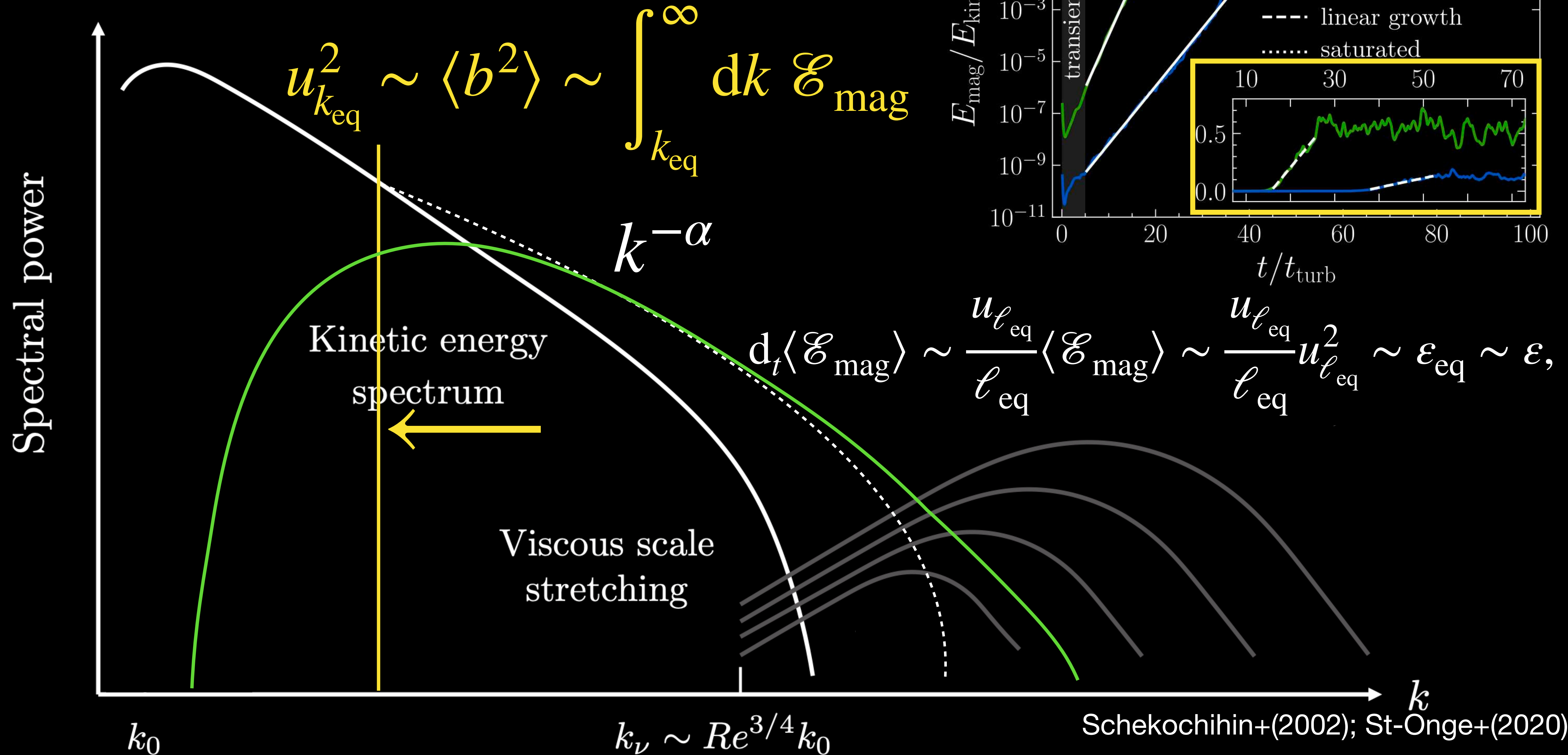
Turbulence dynamo

Nonlinear regime and backreaction



Turbulence dynamo

Nonlinear regime and backreaction



Schekochihin+(2002); St-Onge+(2020)

Galishnikova+(2023); **Beattie**+(2025)

Turbulence dynamo

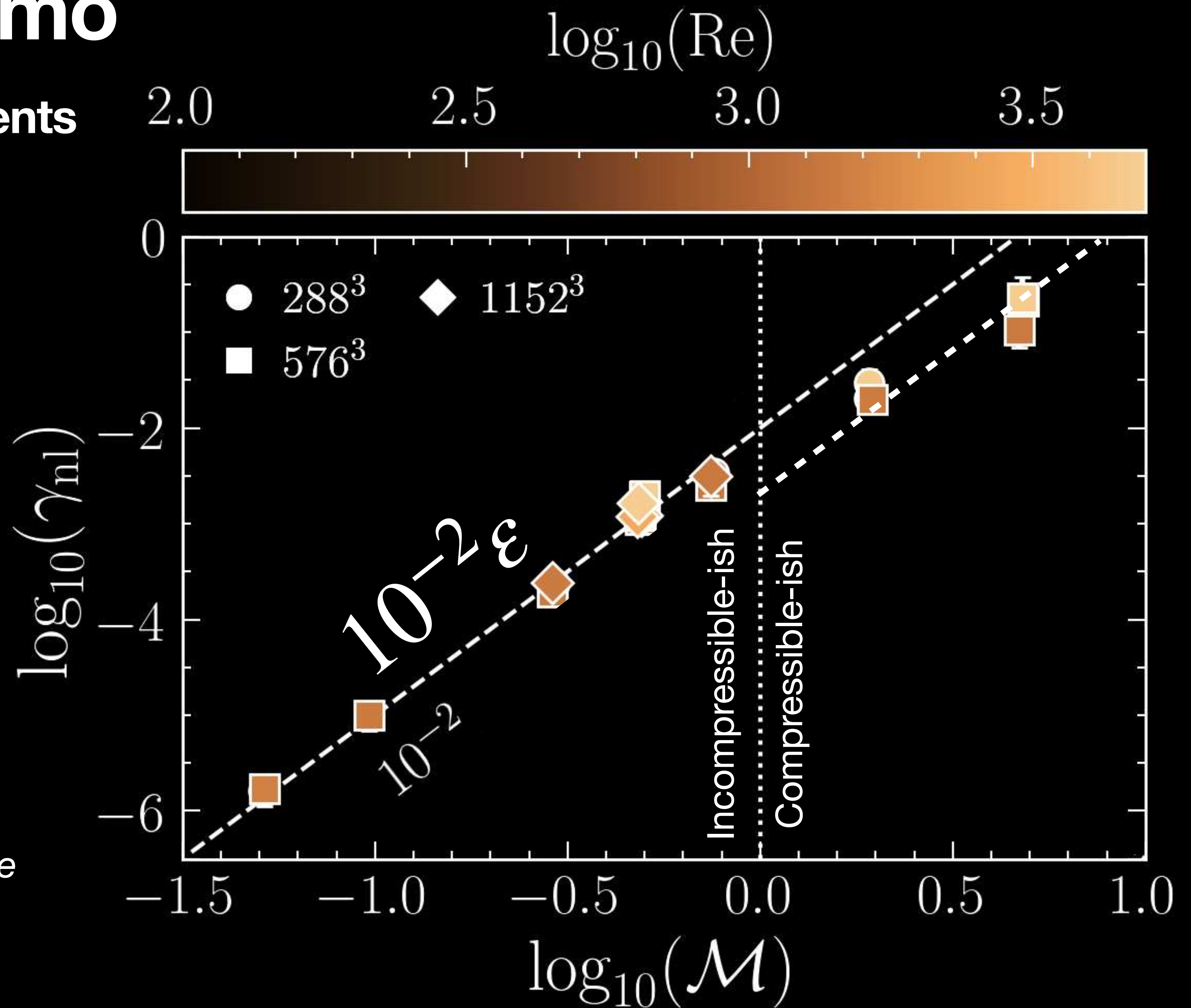
Nonlinear regime:
fed by flux, not velocity gradients

The engine of the
nonlinear dynamo is the
energy flux of the
cascade

$$\langle \mathcal{E}_{\text{mag}} \rangle \sim \varepsilon t$$



Kriel, **Beattie**+ (2026). *The universal growth of magnetic energy during the nonlinear phase of subsonic and supersonic small-scale dynamos*



Turbulence dynamo

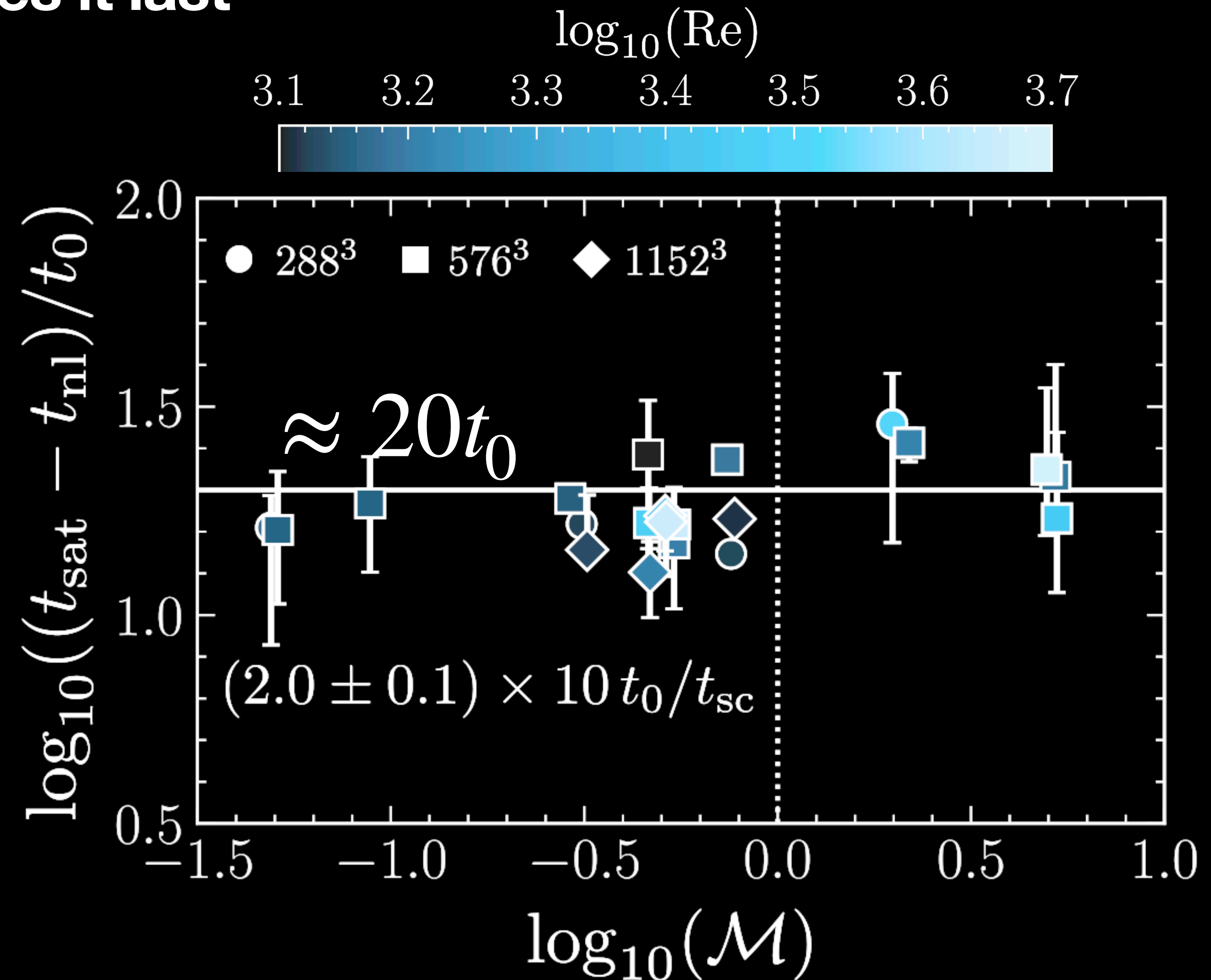
Nonlinear regime: how long does it last

$$\Delta t_{\text{nl}} \approx 20 t_0$$

universally

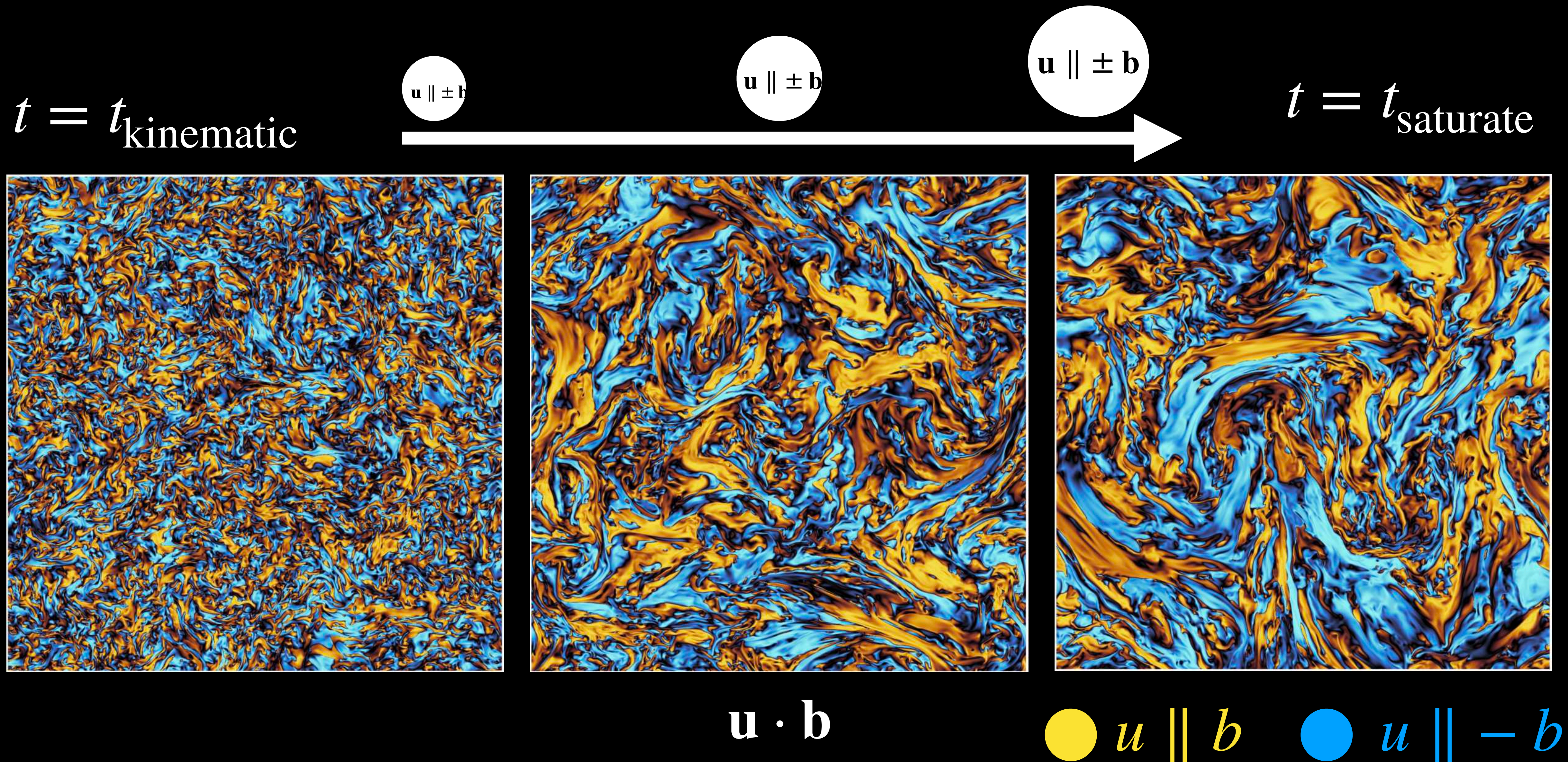
Somewhat mysterious to us at the moment. Implies efficiency is universal for super or subsonic nonlinear regime.

Kriel, **Beattie**+ (2026). *The universal growth of magnetic energy during the nonlinear phase of subsonic and supersonic small-scale dynamos*



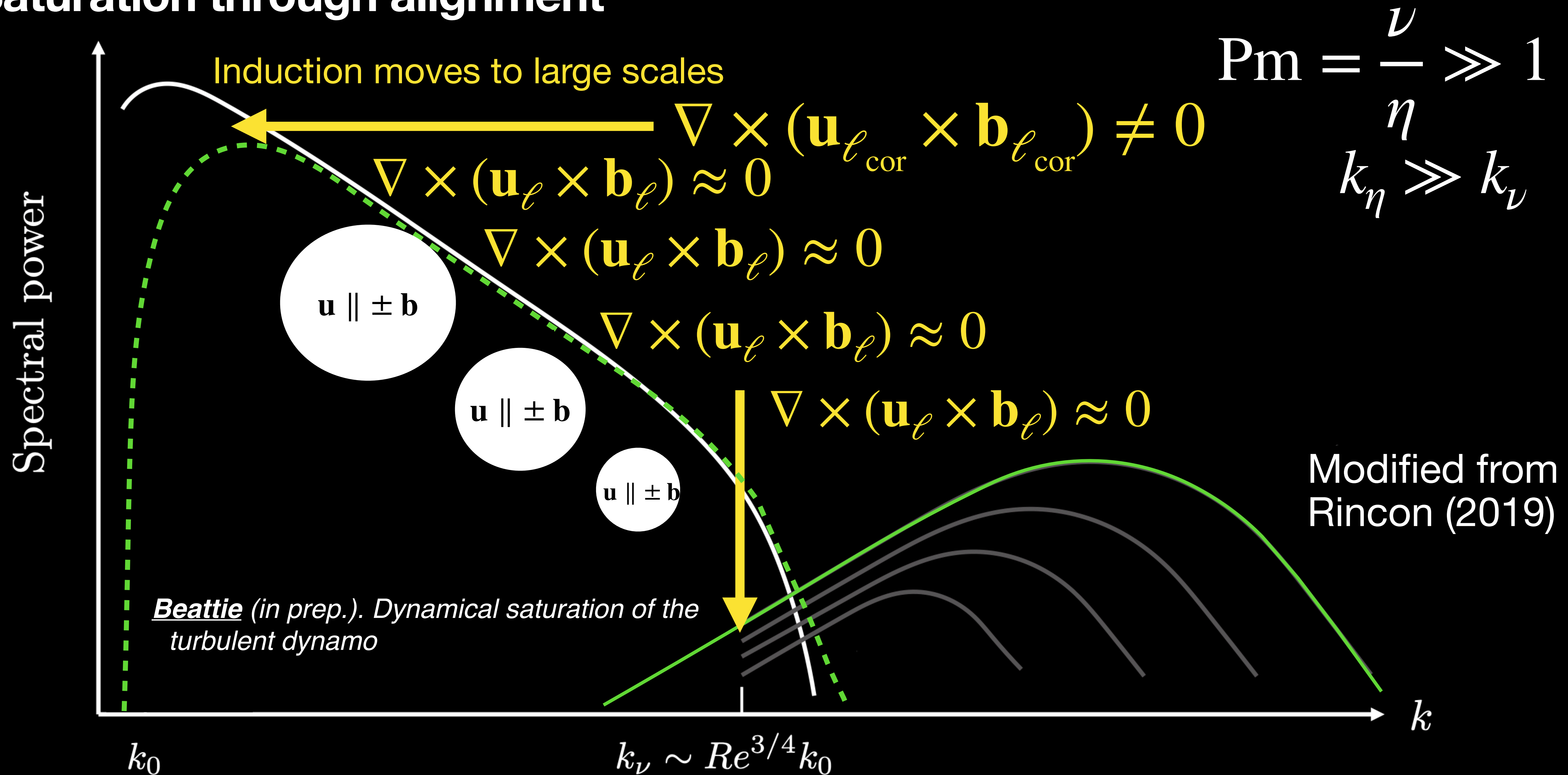
Turbulence dynamo

Nonlinear regime: what does the backreaction look like?



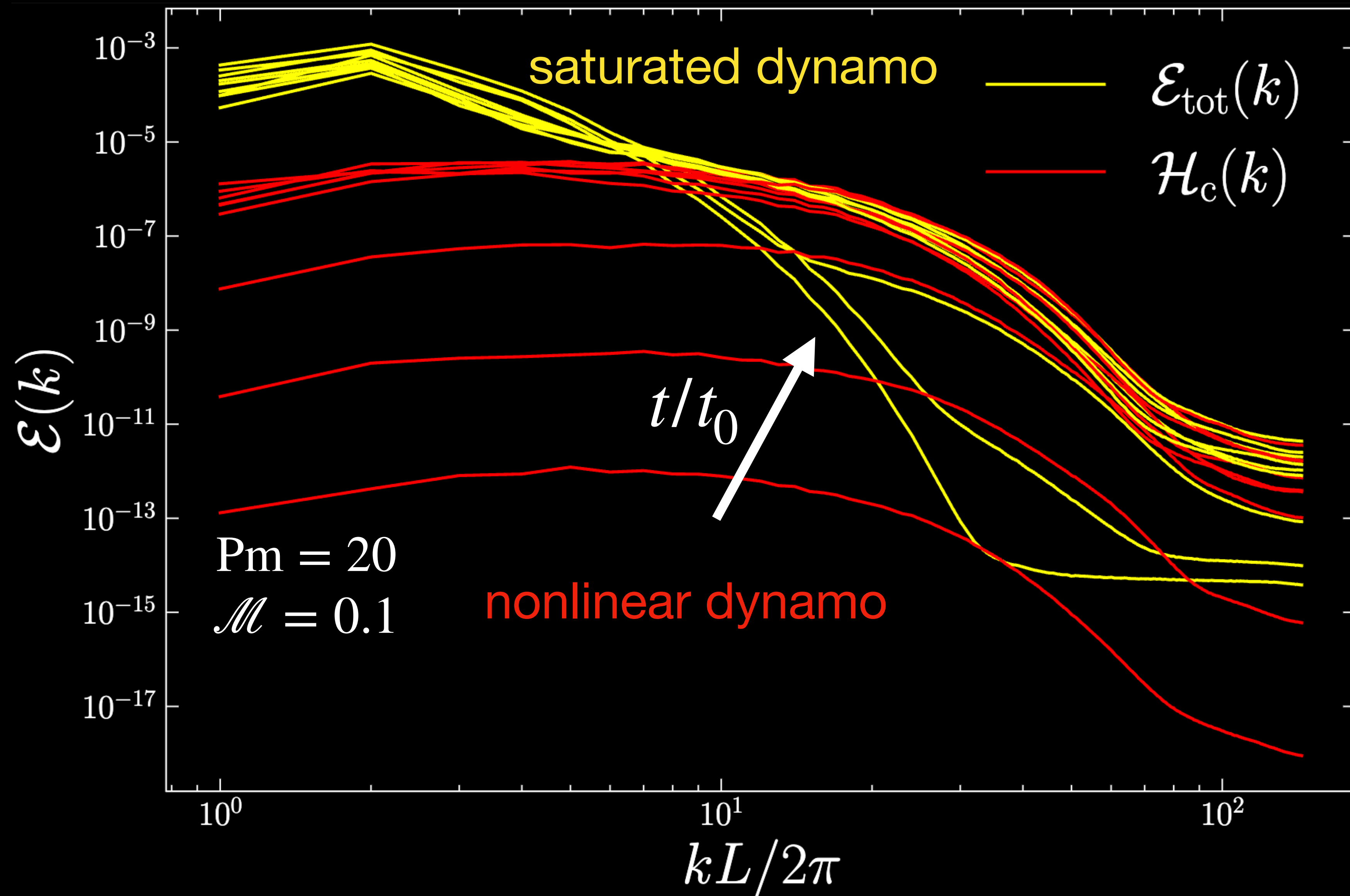
Turbulence dynamo

Saturation through alignment



Turbulence dynamo

Saturation through alignment



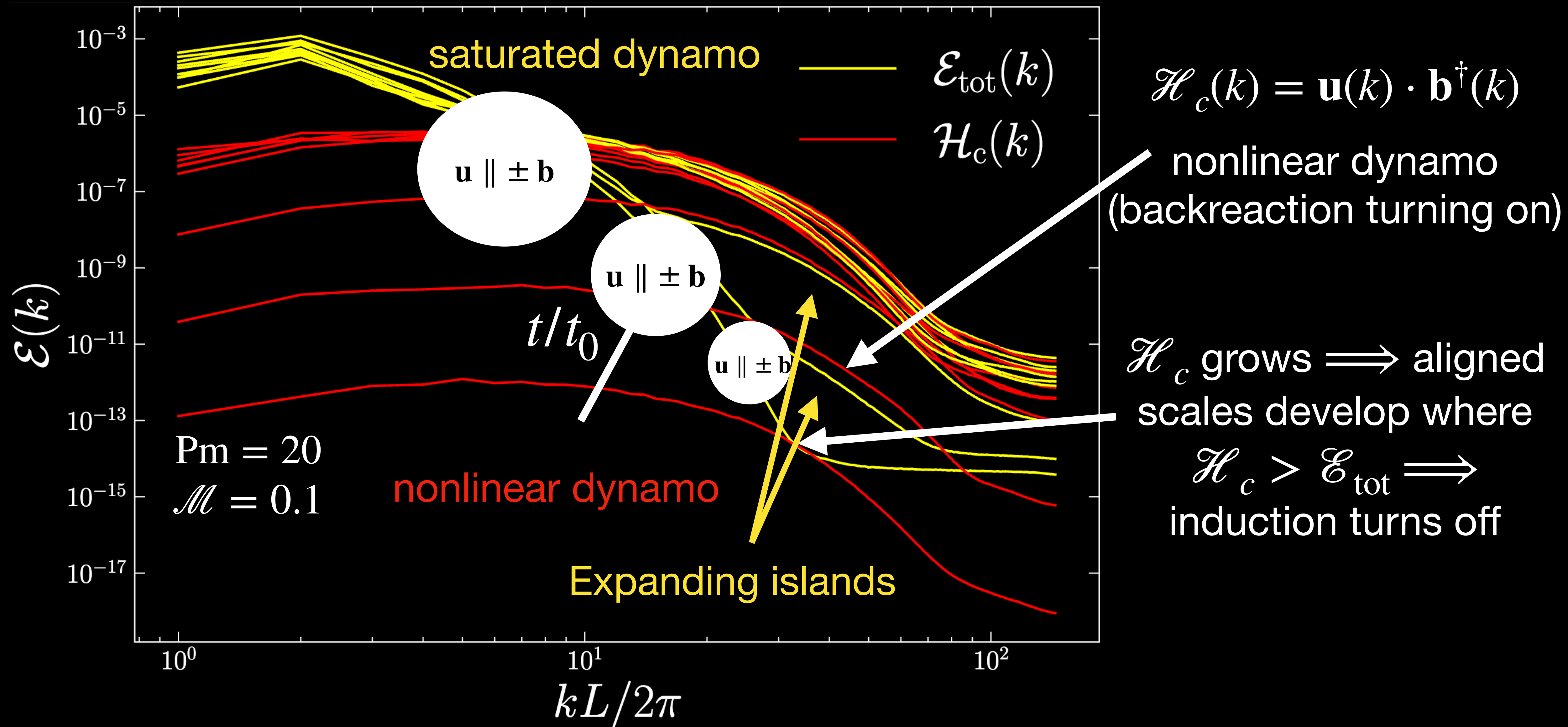
$$\mathcal{H}_c(k) = \mathbf{u}(k) \cdot \mathbf{b}^\dagger(k)$$

$\mathcal{H}_c$ grows $\implies$ aligned scales develop where

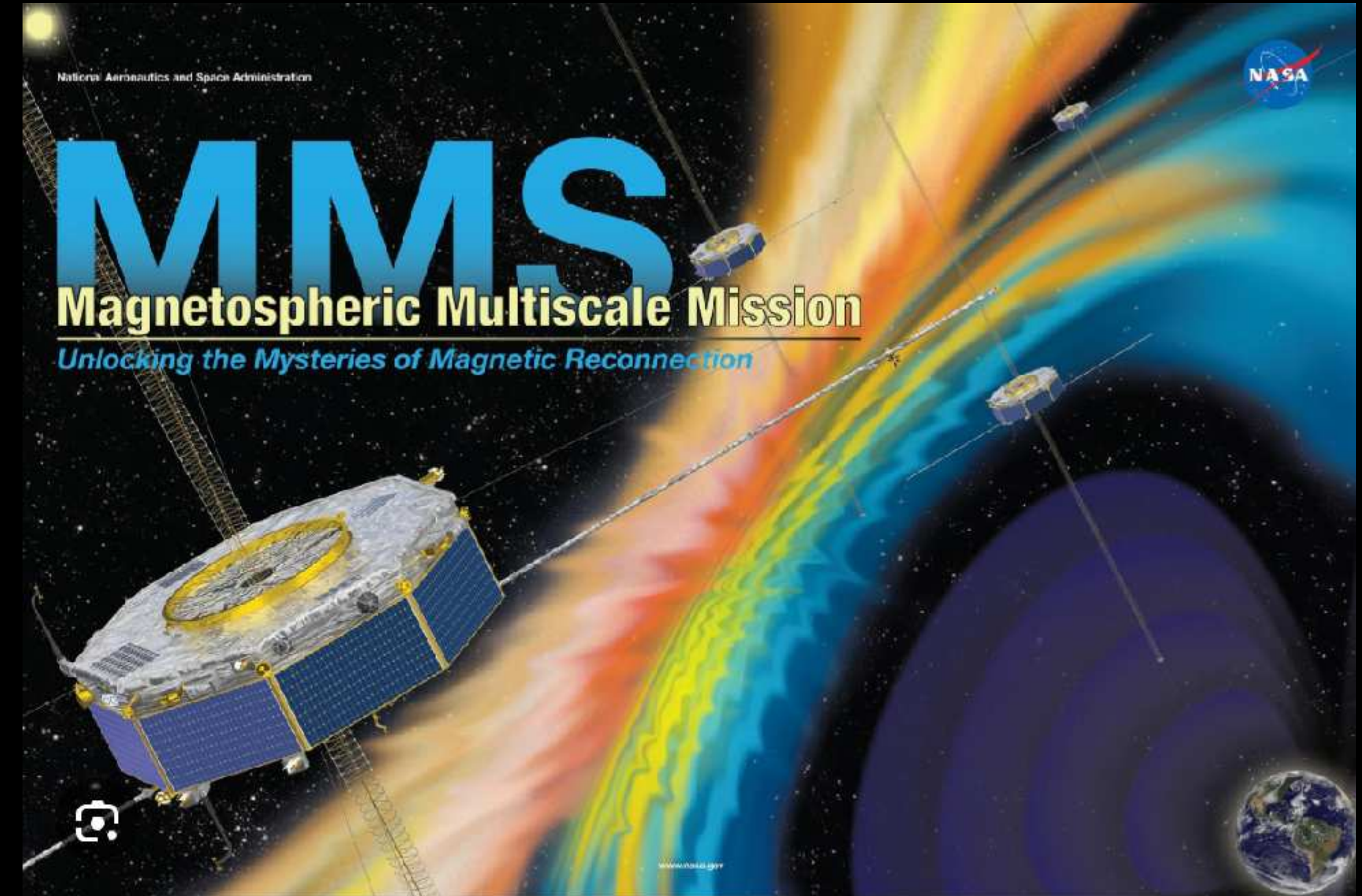
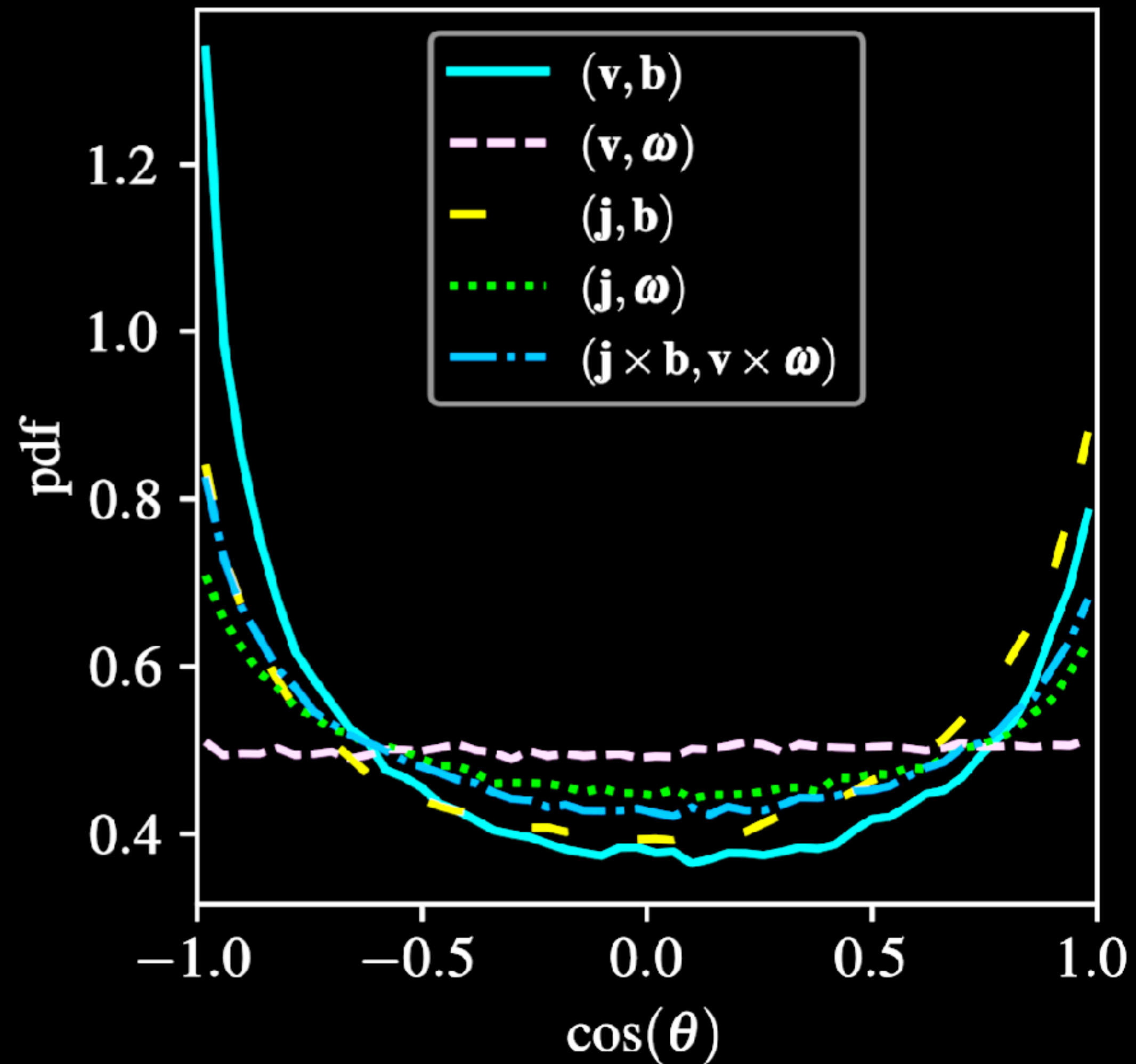
$\mathcal{H}_c > \mathcal{E}_{\text{tot}} \implies$
induction turns off

The dynamical saturation of the dynamo

Saturation through alignment



Patchy helicities in Earth's magnetosheath



Typical turbulent magnetosheath

$$\mathcal{M}_{\text{turb}} \sim 0.2 \quad \beta \sim 10$$

10,080³ magnetized supersonic turbulence simulation

Beattie, Federrath, Klessen, Cielo & Bhattacharjee

1. What is the scaling of the energy cascade in compressible MHD turbulence with no net flux?
2. How are the characteristic scales organized in the compressible supersonic turbulence?
3. What are the saturation physics of the compressible turbulent dynamo?

PI of a three total 190million core-hour projects on superMUC-NG

ILES of compressible MHD turbulence

Turbulence: $\sigma_V/c_s \approx 4$, $\ell_0 = L/2$

Magnetic fields: $B = b_{\text{turb}}$, $\mathcal{M}_A \approx 2$

Three experiments for convergence tests:

- LOW-RES: 2,520³ (0.3Mcore-h, 8,640cores)
- MID-RES: 5,040³ (4.0Mcore-h, 34,560cores)
- HIGH-RES: 10,080³ (80.0Mcore-h, 148,240cores)

6 PB in data products $R_m \sim R_e \gtrsim 10^6$, $P_m \sim 1 - 2$



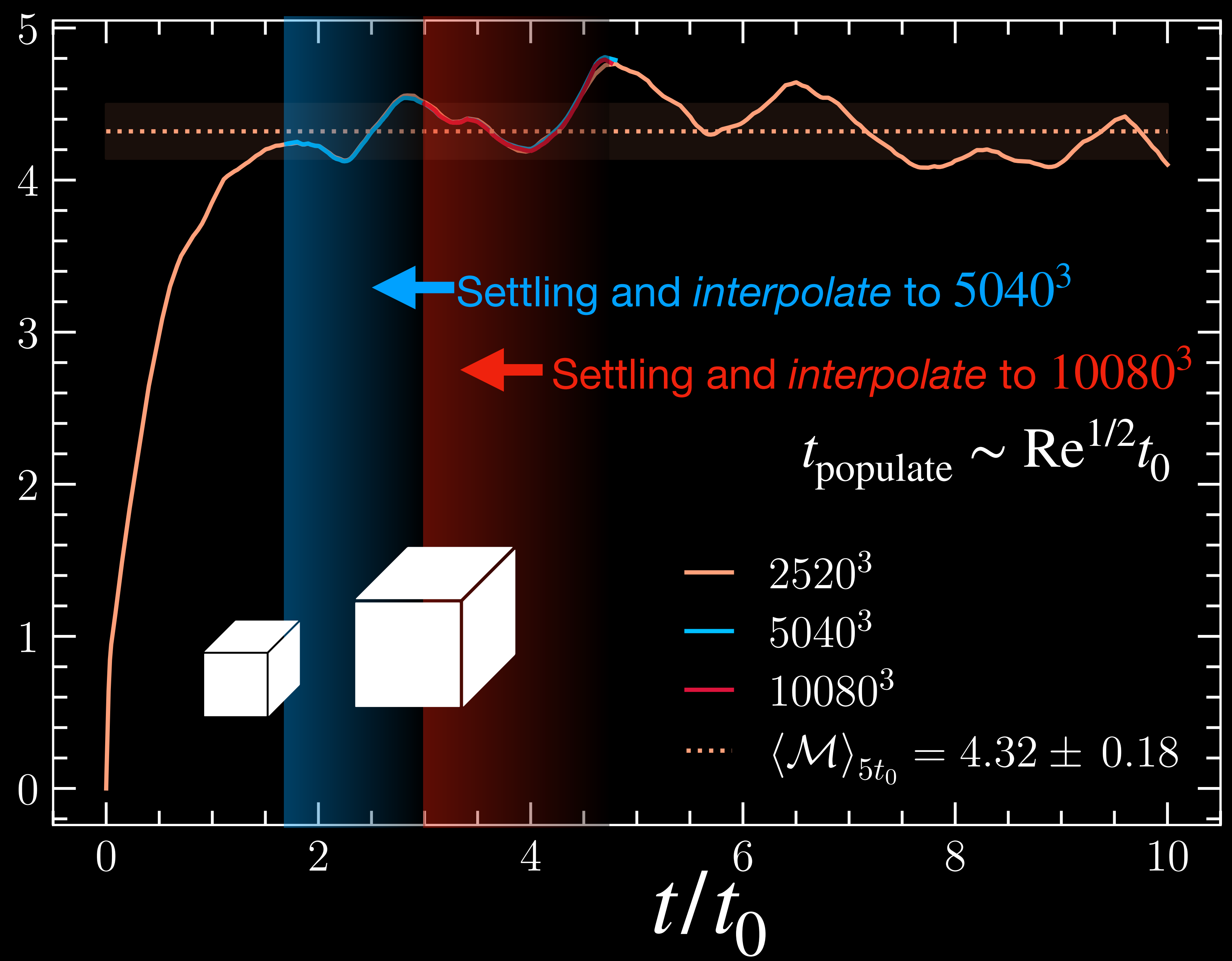
Beattie+ 2025 (*Nature Astronomy*). The spectrum of magnetized turbulence in the interstellar medium

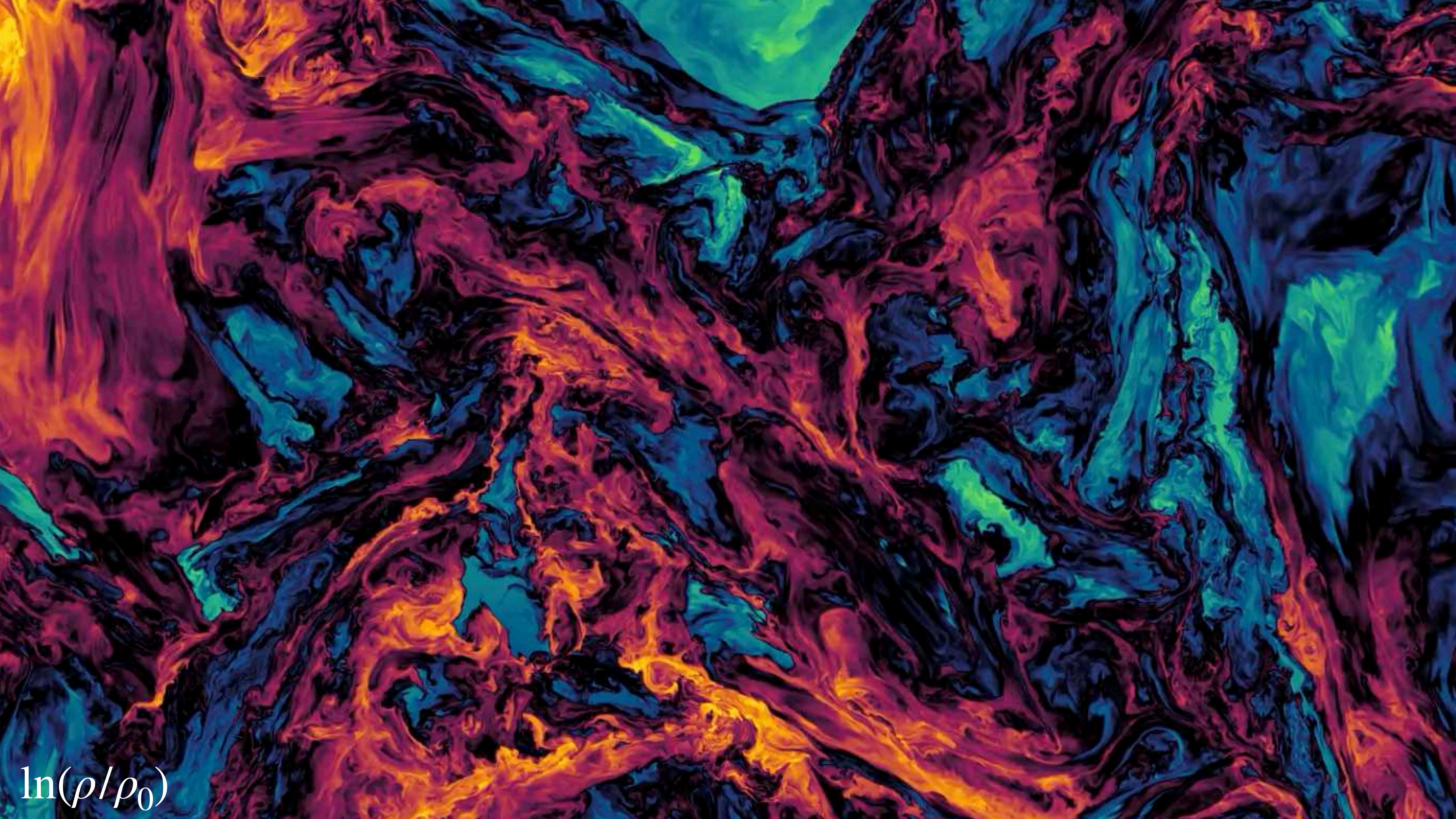
Beattie & Bhattacharjee, 2026 (PRL). Local relaxation and scale-dependent alignment in compressible, magnetized turbulence

Volume integral
Quantities

$$\mathcal{M} = \left\langle \frac{u^2}{c_s^2} \right\rangle_{\mathcal{V}}^{1/2}$$

Interpolation to
generate ICs for
successively higher
resolution
experiments

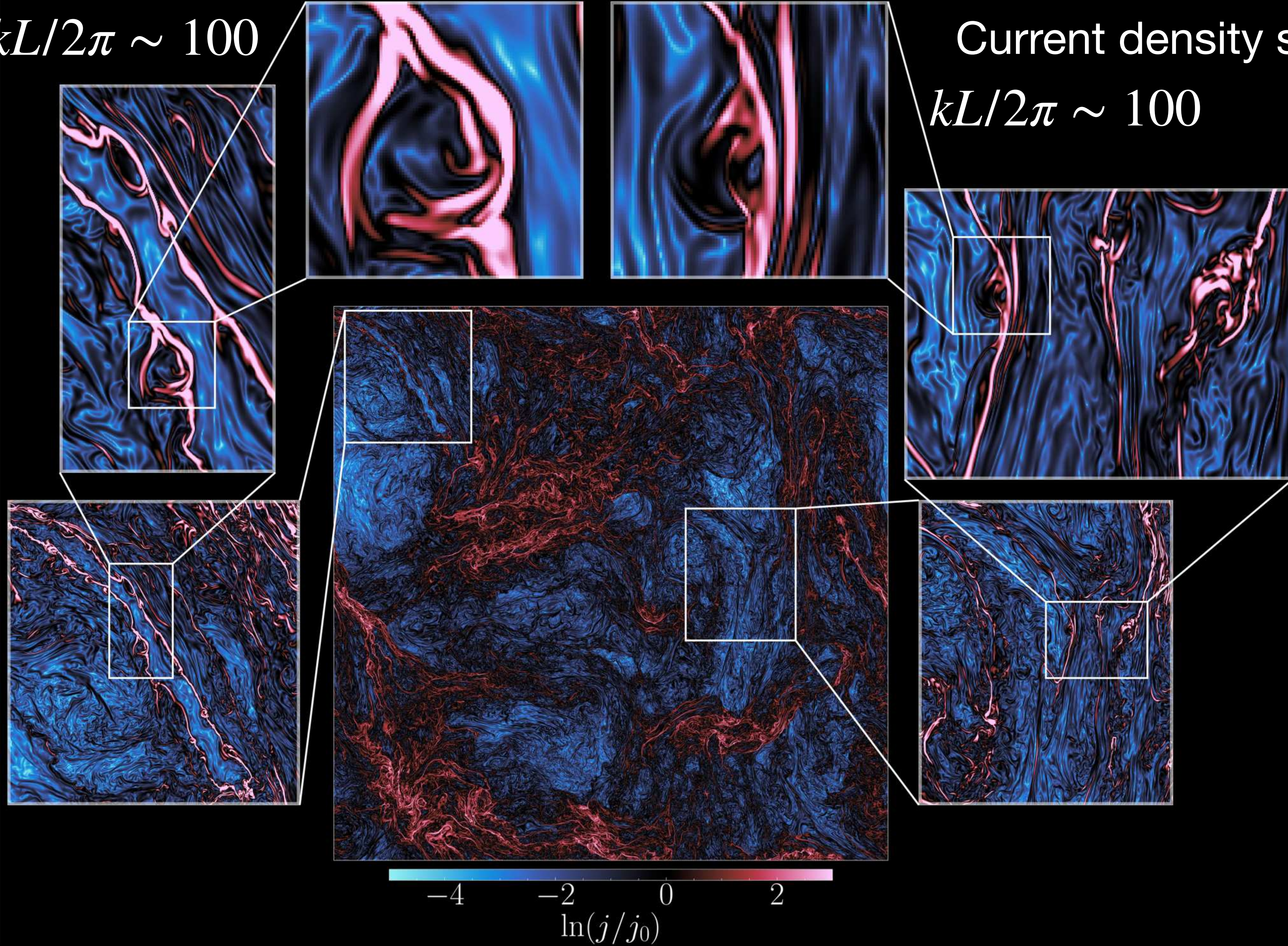




$\ln(\rho/\rho_0)$

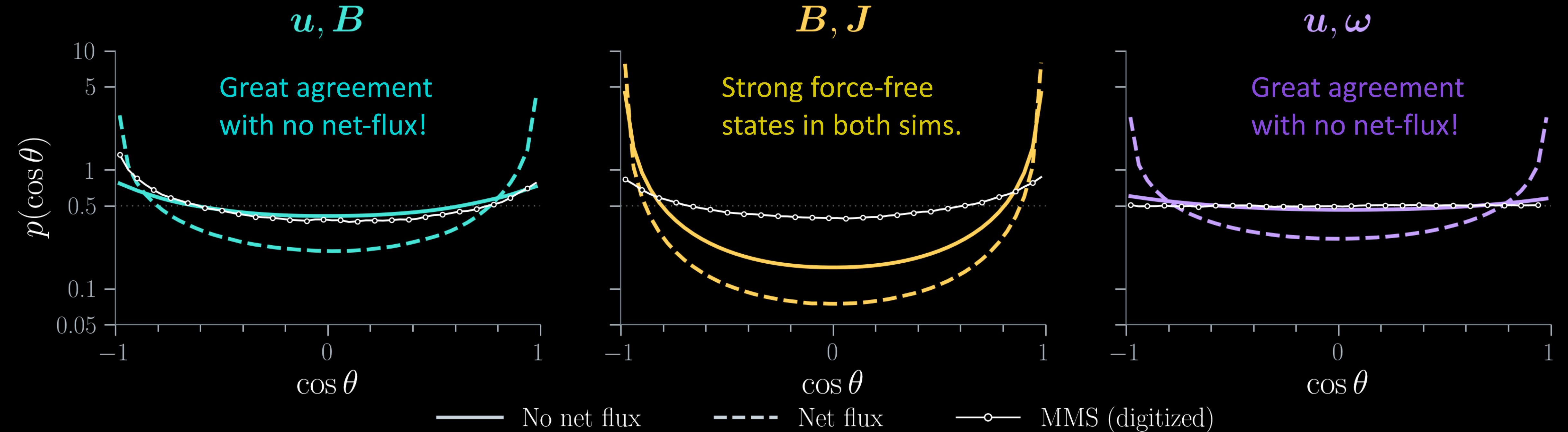
$kL/2\pi \sim 100$

Current density structures
 $kL/2\pi \sim 100$



Patchy helicities in simulation & magnetosheath

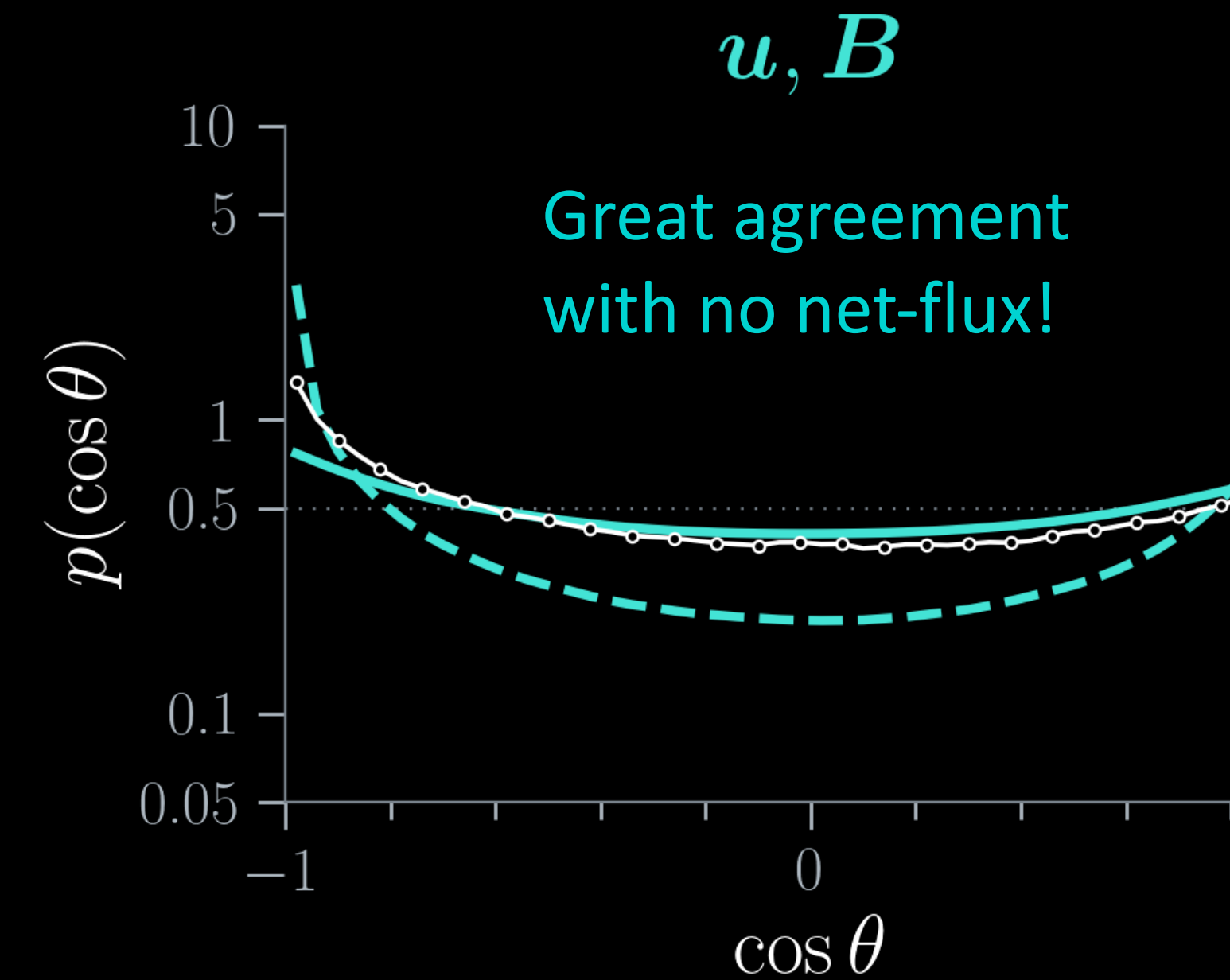
Pecora et al. (2023) for MMS data.



Patchy helicities in simulation & magnetosheath

Pecora et al. (2023) for MMS data.

Why?... have to invite me back :)



Physical Review Letters

ACCEPTED PAPER

Local relaxation and scale-dependent alignment in compressible, magnetized turbulence

James R. Beattie and Amitava Bhattacharjee

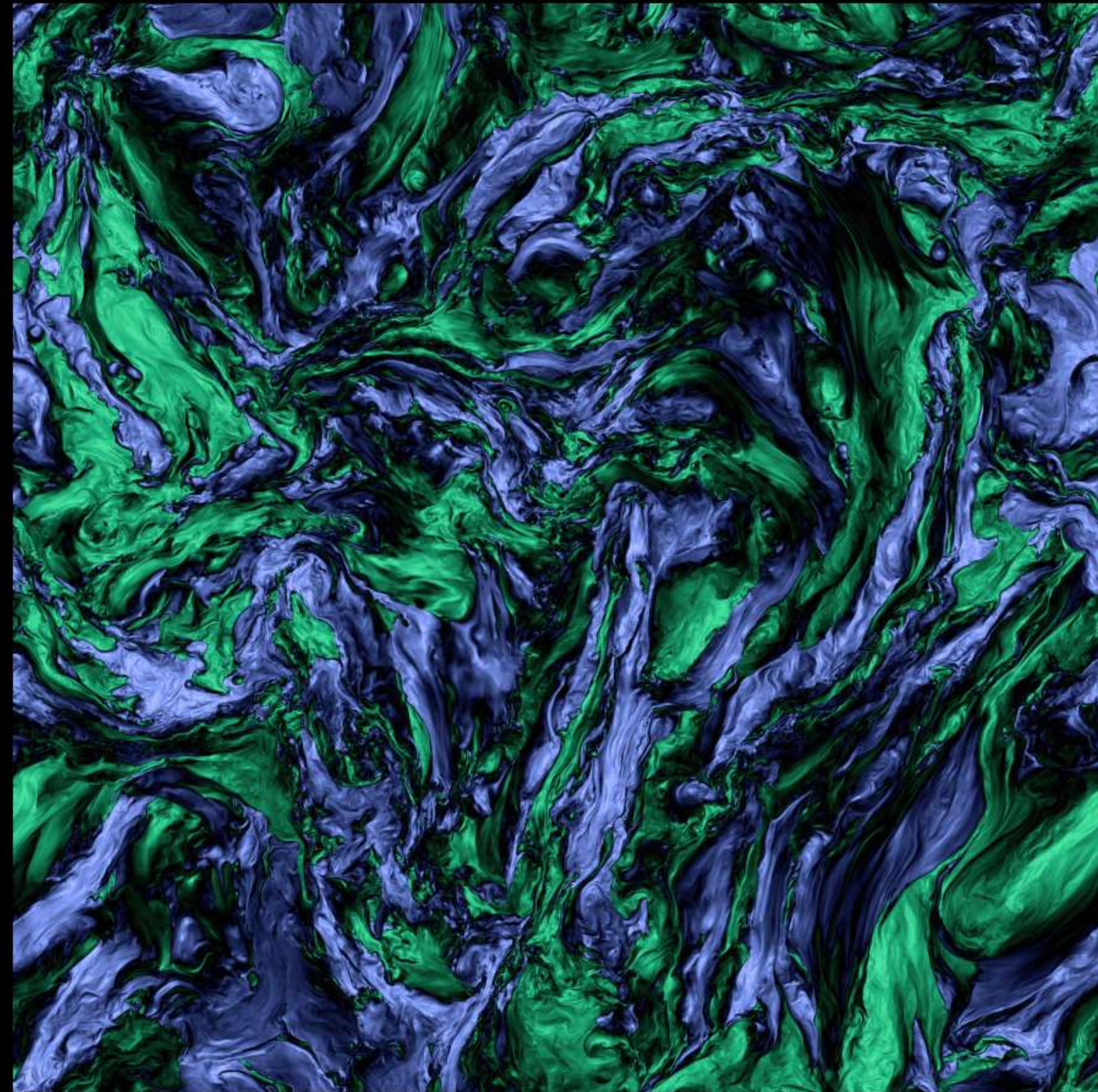
Phys. Rev. Lett. - **Accepted** 10 September, 2026

[Export Citation](#)

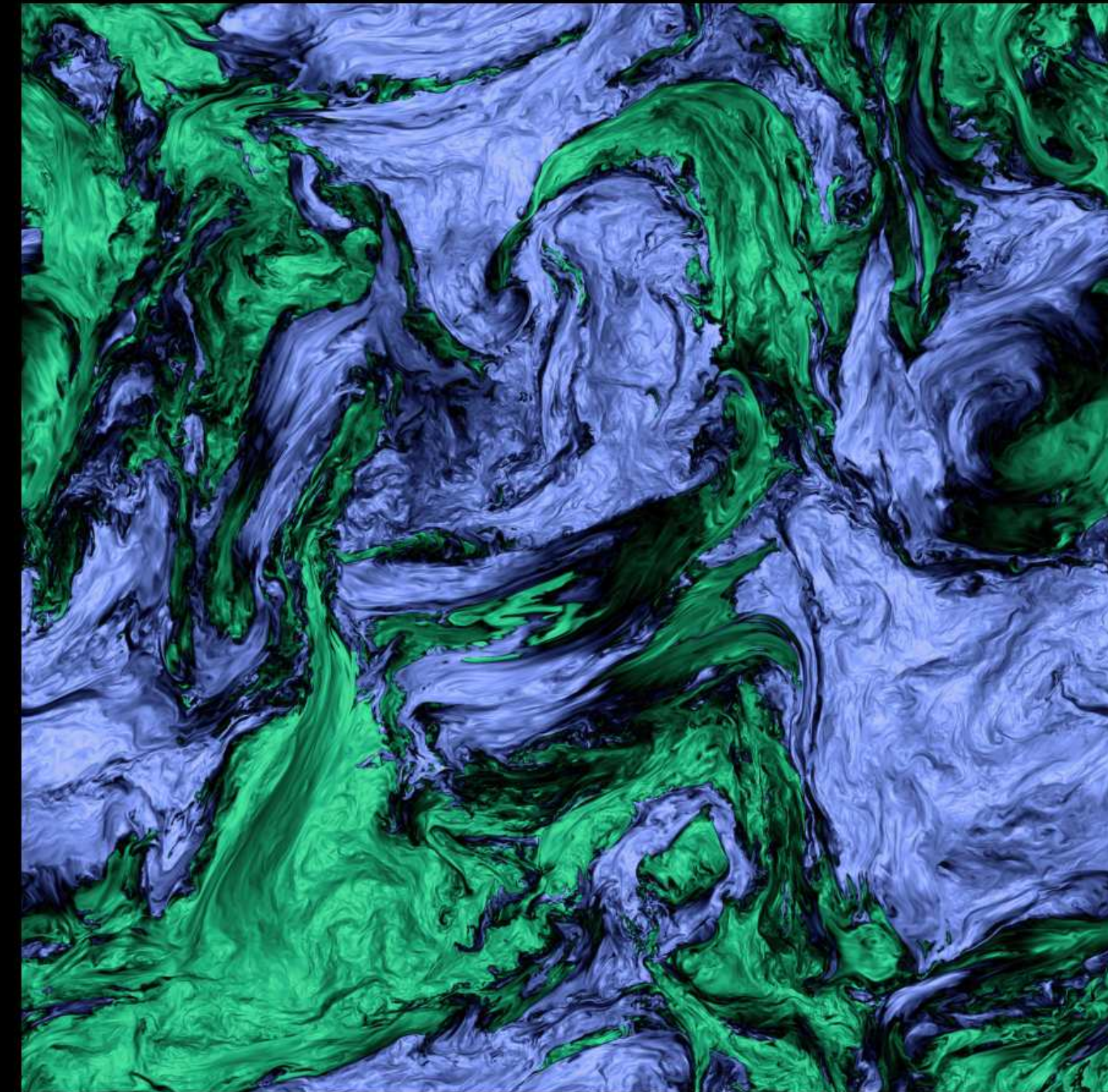
Conclusions for this afternoon talk

1. Kinematic and nonlinear theory at the level of integral energy work really well, with subtle but expected changes between the subsonic and supersonic regimes.
2. The alignment ends up playing an extremely important and generic role in saturating the turbulent dynamo...
 1. Potentially extremely universal, setting the saturation across all of the plasmas I introduced at the start of the talk!
 2. Come to Muni's plasma class to hear more about the simulation and saturation.

No net flux



Net flux



B_0
⊙

parallel

perpendicular

antiparallel

$$N = 10,080^3$$

$$\mathcal{M} \simeq 4$$

$$\mathcal{M}_A \simeq 2$$

$$\beta \simeq \frac{1}{2} \quad N = 10,368^3$$

160M core-hours · 6 PB · SuperMUC-NG · $\text{Re} \sim \text{Rm} \sim 10^6$ · ~150,000 cores

Classical relaxation: energy changes, helicity changes more slowly

Global constraints select aligned stationary states

$$E = \frac{1}{2} \int (u^2 + b^2) dV, \quad \mathbf{b} = \mathbf{B} / \sqrt{\mu_0 \rho_0}$$

Note that this kind of relaxation is FAST! $\mathcal{O}(t_A)$ (from 2D turbulence measurements)

Matthaeus+2021

Magnetic helicity

$$H_M = \int \mathbf{A} \cdot \mathbf{B} dV$$

$$\delta(E_M - \alpha H_M) = 0 \Rightarrow \mathbf{J} \parallel \mathbf{B}$$

Cross helicity

$$H_C = \int \mathbf{u} \cdot \mathbf{B} dV$$

$$\delta(E - \gamma H_C) = 0 \Rightarrow \mathbf{u} = \pm \mathbf{b}$$

Kinetic helicity (for incomp. hydro)

$$H_K = \int \mathbf{u} \cdot \boldsymbol{\omega} dV$$

$$\delta(E_K - \kappa H_K) = 0 \Rightarrow \mathbf{u} \parallel \boldsymbol{\omega}$$

Woltjer (1958) · Taylor (1974) · constraints reviewed in Banerjee et al. (2023)

Banerjee: follow the invariant budgets across scale

Forcing supplies · nonlinear transport redistributes · diffusion removes

Invariant correlation function

Transport fluxes

$$\partial_t R_I(\ell) = \langle F_I \rangle + \langle T_I \rangle + \langle D_I \rangle$$

Forcing fluxes

Diffusion fluxes

Forcing scales

Transport scales

Diffusion scales



Injection



Nonlinear exchange



Dissipation

large eddies



small eddies

Relaxed transport means frozen invariant fluxes

The principle of vanishing nonlinear transfers

$$\langle T_E \rangle = \langle T_{H_M} \rangle = \langle T_{H_C} \rangle = 0$$

The perfectly relaxed limit has no nonlinear interscale transfer (kinda duh).

What is interesting is that in this limit:

$$\boldsymbol{u} \times \boldsymbol{\omega} + \boldsymbol{j} \times \boldsymbol{b} = \nabla \Phi$$

$$\boldsymbol{u} \times \boldsymbol{b} = \nabla \Psi$$

Aligned states emerge in the appropriate gradient-free limits (which are generally set by the EOS and / different invariants).

Banerjee, Halder & Pan (2023) · pressure-balanced states can define Ψ , Φ

Thanks, questions?



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